THE CRYSTALLINE WORLDSHEET: A STRING THEORETICAL FRAMEWORK BASED ON φ AND π FOR THE DE SITTER OBSERVER PROBLEM

JORGE ARMANDO GONZÁLEZ GARCÍA, VÍCTOR MANUEL GONZÁLEZ GARCÍA, LUZ MARÍA GARCÍA ORDÓÑEZ, AND ITZEL MARION DRESSLER PÉREZ

Abstract. We propose that the obstruction to holography in de Sitter space is *epistemological*—a matter of representation—rather than *ontological*—a matter of which vacuum is realised. We trace it to the non-paradoxical self-reference of binary language, in the precise sense of the Lawvere fixed-point theorem and the Yanofsky diagonal $g(t) = \alpha(f(t,t))$, compounded by the dependence of quantum mechanics, as originally formulated, on Boolean logic. Resolving the obstruction at the intersection of set theory and intuitionism, we build a geometric framework carried on a single torus generated by $\varphi = 2\cos(\pi/5)$, whose modulus $|\Omega| = \frac{1}{2}$ we propose as the discrete microstate that low-dimensional holography could use. From this one microstate—without any ensemble average—we recover the AdS/CFT correspondence and, through the golden tower $z = \varphi^\sigma$, an M-theory type duality web, level by level. This construction happens to be inspired by Yuri Manin's hypothesised use of the same noncommutative tori that M-theory uses, but to approach the $\mathbb{F}_1$ program for the Riemann hypothesis. The observer enters not as a selector of the vacuum—the microstate is the unique self-consistent solution, so there is no landscape to choose from—but as the projection Π that participates by accumulating information: its quantum Fisher information, conjugate to the tower entropy and bounded by the ultraviolet cuts, builds the bulk geometry and, through the Landauer cost of each bit and the Jacobson–Verlinde passage from entropy to curvature, sources Einstein's equations. The objectivity threshold $f_{\text{crit}} = \frac{1}{2}$ coincides with the modulus of the microstate, and entanglement is geometry (ER = EPR-like) read from that modulus. The torus is proposed as the single microstate that low-dimensional holography and de Sitter appear to lack: the unique self-consistent solution its independent origins converge on, into which the AdS/CFT and M-theory corners project. Three companion papers develop, independently, the arithmetic, operator, and identity threads the construction invokes.

Contents

Date: **Received:** July 13, 2026; **Last modified:** July 30, 2026.
2020 *Mathematics Subject Classification.* Primary 81T30, 14G99, 11M26; Secondary 68V20.
Key words and phrases. Holography; de Sitter; M-theory; string theory; field with one element ($\mathbb{F}_1$); λ-rings; golden ratio; moduli spaces; formal verification.
Zenodo All Versions DOI: 10.5281/zenodo.21343602.

1. Introduction

1.1. **The problem: the observer in de Sitter.**

1.1.1. *The participatory observer.* The problem of holography in de Sitter space is, as Witten frames it in *Quantum Gravity in De Sitter Space*, that General Relativity "cannot be quantized—unless it is embedded in a more complete theory" [1]: a theory in which the observer is not introduced from outside but "described by the theory" [2]. In a cosmology with positive cosmological constant there is no asymptotic region to which an external observer may retreat; the observer is part of the system it measures and must therefore model the system that contains it. This is what makes the de Sitter case structurally different from the anti-de Sitter case, which has been the most successful approach to finding a UV complete description of quantum gravity, yet.

1.2. **Proposed solutions and their open issues.** In this section we review some of the most important proposed solutions to the problem, together with the issues that each one leaves unresolved.

1.2.1. *Strominger: dS/CFT and non-unitarity.* In 2001 Strominger proposed the dS/CFT correspondence [3]: quantum gravity in a D-dimensional de Sitter space is equivalent to a Euclidean conformal field theory living at future (or past) infinity, conceptualised as a $(D-1)$-dimensional sphere. The difficulty is that the resulting theory is generically non-unitary: the boundary conformal weights

$$h_{\pm} = \frac{D-1}{2} \pm \sqrt{\left(\frac{D-1}{2}\right)^2 - m^2\ell^2} \tag{1.1}$$

become complex once $m^2\ell^2 > (\frac{D-1}{2})^2$, so probabilities do not sum to one and the dual theory is mathematically elusive.

1.2.2. *Susskind: the static patch.* Susskind, one of the originators of the holographic principle [4], addressed the problem from the standpoint of a real observer: holography should be formulated inside the cosmological horizon of a static observer, with all interior information encoded on that horizon at one bit per Planck area,

$$S_{\mathrm{hor}} = \frac{A_{\mathrm{hor}}}{4G_N}. \tag{1.2}$$

In recent work this static patch has been connected to advanced quantum-mechanical models such as double-scaled SYK; the holographic encoding itself, however, remains a postulate rather than a derived correspondence.

1.2.3. *Witten: finite dimensionality.* Also in 2001, Witten [1] analysed the quantum properties of de Sitter and argued that, because the horizon has finite area, the Hilbert space of quantum gravity in de Sitter must be strictly finite-dimensional,

$$\dim \mathcal{H} < \infty. \tag{1.3}$$

This collides directly with traditional conformal field theories, which possess infinitely many states, and imposes severe constraints on any holographic attempt; the observer, in particular, must be included in the description rather than idealised away [2].

1.2.4. *The ensemble and vacuum selection.* Despite these proposals the picture is, as Nico Cooper puts it, always the same [5] (a gravitational theory at weak coupling in a d-dimensional anti-de Sitter interior is captured by a conformal theory at strong coupling on its $(d-1)$-dimensional edge, the cleanest realisations being string backgrounds that flow to supergravity. In low dimensions this breaks down: two-dimensional JT gravity has no single boundary dual, and the bulk is recovered only as an average over an *ensemble* of boundary theories [6, 5], even though quantum gravity should carry its own discrete, individual microstates—much as an ensemble of Hamiltonians reproduces Wigner's nuclear decay rates while any single nucleus is governed by one. This ensemble problem connects directly to the observer problem in de Sitter: the absence of a definite spatial boundary confines the observer to an averaged statistical description within their cosmological horizon, breaking the notion of a single pure quantum state and forcing an effective description that reconciles the finiteness of the horizon with the theoretical existence of fundamental microstates). The most widely held response is to abandon uniqueness altogether: it merges the Everettian many-worlds interpretation of quantum mechanics with anthropic vacuum selection over the string landscape—an estimated $\sim 10^{500}$ metastable de Sitter vacua [7, 8, 9]. Our vacuum is then one branch among many, populated by eternal inflation and fixed by anthropic selection rather than derived, the branches of the wavefunction being identified with the landscape's vacua rather than with a single microstate [10].

1.3. **Hypothesis: self-reference as the common cause.**

1.3.1. *A matter of mathematical representation.* We propose that the problem may be epistemological—a matter of mathematical representation—and not ontological—a matter of vacuum selection in which we do not know which vacuum we inhabit. We propose that the problem is how to describe something without referring to that same something one is describing. Across many scientific and mathematical worlds we encounter geometric structures that are part of a whole identical to themselves: fractals. They occur even in literature, among the rhetorical figures—the *mise en abyme*, for example. Self-reference in mathematics, however, is a profoundly delicate matter, one that wreaks havoc in domains far beyond the de Sitter universe we address here. The way the whole relates to the part appears as a concrete obstruction across many domains: in set theory, as the collection of all collections that do not contain themselves; in the foundations of arithmetic, as a true proposition that asserts its own unprovability and so escapes every proof; in the theory of truth, as the impossibility of defining truth within the language itself; in computation, as the impossibility of deciding whether an arbitrary procedure—itself included—ever halts; and in engineering, as the feedback of a system that must represent itself. We propose that the essence of the problem lies in how to describe the object without referring to the object, akin to the figure–ground problem in the graphic arts and painting. In essence, it is the Hilbert–Pólya operator problem and the Weil Rosetta Stone, together with the development of the $\mathbb{F}_1$ program—a way of referring to something without referring to that something directly: in the case of $\mathbb{F}_1$, referring to the zeros of ζ, or to the primes, without referring to the primes or to the zeros of ζ.

1.3.2. *The Lawvere–Yanofsky obstruction.* Let us first examine how the problem of self-reference wreaks havoc across diverse domains. In de Sitter holography the observer is forced to classify itself by a two-valued distinction—inside or outside the horizon, bulk or boundary, yes or no—and it is this binary self-reference, not self-reference in general, that becomes obstructed. The obstruction is made precise by the Lawvere fixed-point theorem [11]: in a cartesian closed category a point-surjective map $f\colon T \to Y^T$ forces every endomorphism of Y to have a fixed point, so a single fixed-point-free endomorphism forbids such an f. Yanofsky's diagonal [12] packages the classical paradoxes through

$$g(t) = \alpha\big(f(t,t)\big), \qquad \alpha(x) = 1 - x, \tag{1.4}$$

where the fixed-point-free involution α on the binary register $\{0, 1\}$ is precisely what makes the *binary* self-application $f(t,t)$ contradictory. Our hypothesis is that the de Sitter obstruction is an instance of this binary mechanism, compounded by the dependence of quantum mechanics—as it was originally formulated—on Boolean logic.

1.3.3. *Stratification as prohibition.* Set theory under classical bivalent logic resolves self-reference by *prohibiting* it through stratification. The axiom of extensionality defines objects purely by their membership extension, and the axiom schema of separation restricts comprehension so that no set is formed by a predicate quantifying over the totality of sets—because the predicate $x \notin x$ applied to the class of all sets produces Russell's paradox. The resolution is the cumulative hierarchy: every set has a rank α, with

$$x \in u \implies \mathrm{rank}(x) < \mathrm{rank}(u), \tag{1.5}$$

and no set has a complement that is also a set, so the Boolean algebra of union and intersection is available within each level while global complementation is missing. A morphism translating between two domains A and B would have to occupy some rank α while its arguments and values refer across levels—precisely the cross-level self-application that stratification forbids. The barrier between universes is therefore not an additional restriction imposed on top of the paradox-avoidance mechanism: it *is* that mechanism. Stratification solves self-reference by making cross-level morphisms illegal, and cross-level morphisms are exactly what inter-domain translation requires.

1.3.4. *Boolean logic is not the only logic.* The preceding obstruction is specific to the classical, bivalent setting. Boolean logic, however, is not the only logic that exists, and the resolution we shall pursue lies in replacing prohibition by mediation—a possibility opened by the non-Boolean logics reviewed next.

1.4. **Mathematical antecedents: non-Boolean strategies for self-reference.** The classical, bivalent prohibition of §1.3 is not the only way to confront self-reference; the literature documents several *distinct* strategies, each with its own mechanism, and they should not be conflated into a single route. Historically the constructivist and intuitionist tradition questioned classical logic from outside: Kant's account of number as a construction in the pure intuition of time is a philosophical, not a mathematical, antecedent, followed by Kronecker's finitism, Poincaré's pre-intuitionism, the constructivism of Borel and Lebesgue, and Brouwer's rejection (1907) of the excluded middle for infinite domains. The mechanisms that bear directly on the Lawvere–Yanofsky obstruction, each handling self-reference in a different way, are the following.

(1) **Type-theoretic stratification** (Russell [13], Martin-Löf [14]) proceeds by *prohibition by level*: forbidding $X \in X$ through a hierarchy of universes $U_0 \subset U_1 \subset U_2$ breaks the self-referential cycle at the foundational level. It was the first tower strategy, but, as in §1.3, it does so by making cross-level morphisms illegal.
(2) **Paraconsistent logic** (Priest [15]) proceeds by *tolerance*: contradictions are accepted as non-explosive, so a system may contain a contradiction without becoming trivial, and the diagonal need not be excluded at all.
(3) **Tarski's decidable geometry** [16] proceeds by *removing the capacity for self-reference*: the first-order theory of real closed fields, which contains Euclidean geometry $\mathbb{R}^n$, is complete and decidable precisely because it cannot internally define $\mathbb{N}$. Without Gödel numbering no formula can be assigned a code, and the Yanofsky diagonal $g(t) = \alpha(f(t,t))$ *cannot form.*
(4) **Intuitionism, Heyting algebras and the topos** (Brouwer, Heyting; Curry–Howard; Lawvere) proceed by *mediation*: the internal logic is intuitionistic, valued in a Heyting algebra, and the implication $A \to B$ is realised as an internal object—the exponential B^A, governed by the currying adjunction $\mathrm{Hom}(C \times A,\, B) \cong \mathrm{Hom}(C,\, B^A)$ that also underlies the Lawvere statement (1.4). Self-reference is carried by the exponential rather than forbidden.

What these strategies have in common is structural rather than logical: several are *tower* constructions. Russell's hierarchy of types $U_0 \subset U_1 \subset U_2$ and Martin-Löf's hierarchy of universes both resolve self-reference by ranking objects in an ascending tower of levels indexed by a single rule, and this stratified, level-indexed form recurs throughout the constructivist tradition.

1.5. **Physical antecedents.**

1.5.1. *Throats and vacuum selection.* In holography there already exist constructions that resemble these strategies, but from what we like to call the Tarskian or geometric perspective. In the near-horizon limit of a brane stack the geometry develops an anti-de Sitter *throat* [17] whose radial coordinate plays the role of a renormalisation-group scale: in Poincaré coordinates

$$ds^2 = \frac{L^2}{z^2}\left(-dt^2 + d\vec{x}^2 + dz^2\right), \tag{1.6}$$

the boundary $z \to 0$ is the ultraviolet of the dual theory, with $E_{\text{SYM}} = (L/z)\, E_{\text{proper}}$. The bulk interior is encoded on the boundary through the holographic bound

$$S = \frac{A}{4G_N}, \tag{1.7}$$

one bit per Planck area, matching the N^2 degrees of freedom of the dual gauge theory [18]. In a *warped* throat this count runs with the radial coordinate: in the Klebanov–Strassler/Tseytlin cascade [19] the effective rank grows logarithmically,

$$N_{\text{eff}}(r) \sim g_s M^2 \log\frac{r}{r_s}, \tag{1.8}$$

so the throat *gains degrees of freedom—"gains bits"—as it grows* toward the ultraviolet, terminating in the deep infrared in a confining gauge group with a mass gap through a cascade of Seiberg dualities. This already resembles the Russellian type-theoretic hierarchy; what is missing is a *logical* selection criterion. In the orthodox formulation the rule was a strict duality, one bulk universe to one boundary theory; but in the simplest controllable case, 2d JT gravity, the bulk cannot be matched to a single discrete boundary theory, and the mathematics requires instead a statistical ensemble of random matrices [6], in tension with the discrete microstates expected of quantum gravity. Beyond the anthropic principle—selection by one's own existence—there is no rule isolating a single conformal field theory; this is the vacuum-selection problem, magnified in de Sitter by the landscape of $\sim 10^{500}$ vacua [9].

1.5.2. *M-theory from the superpoint.* A second, independently developed string/M-theoretic construction selects its content logically rather than anthropically. Applying the Lawvere development of Hegel—the *unity of opposites* formalized as an adjunction [20], carried into the cohesive ∞-topos in which the superpoint lives [21]—Huerta and Schreiber [22] prove that the super-Minkowski spacetimes of string/M-theory arise as iterated *maximal invariant central extensions* of the superpoint $\mathbb{R}^{0|1}$—the simplest super-Minkowski spacetime, with neither Lorentz nor spinorial structure. Forming these extensions consecutively discovers the spacetimes that carry superstrings and culminates in the ten- and eleven-dimensional cases,

$$\mathbb{R}^{0|1} \longrightarrow \cdots \longrightarrow \mathbb{R}^{9,1|16} \longrightarrow \mathbb{R}^{10,1|32}, \tag{1.9}$$

each step singled out by super-Lie-algebra cohomology rather than postulated; the full brane content—the D-branes, the M5-brane and their dualities—then follows from the induced higher central extensions of the *brane bouquet*. The dimensional structure is in this sense algebraically necessary, with no anthropic choice. Crucially, the construction is binary at its root. Supergeometry is $\mathbb{Z}_2$-graded: it rests on the cyclic group $\mathbb{Z}_2 = \{0, 1\}$ with addition modulo two ($1 + 1 = 0$), the most elementary binary structure, which fixes the sign rule of the superalgebra and hence the (anti)commutation of the supercharges that the invariant cocycles encode. This $\mathbb{Z}_2$ is precisely the binary register on which the involution $\alpha(x) = 1 - x$ of §1.3 acts. The sense in which this is binary is worth distinguishing: a classical bit is Boolean, taking the value 0 or 1; a qubit is quantum, a superposition of the two; the superpoint's binary is geometric, its $\mathbb{Z}_2$-grading distinguishing the even, bosonic directions of ordinary geometry (the "0") from the odd, fermionic direction that carries supersymmetry (the "1"), the superpoint $\mathbb{R}^{0|1}$ being the minimal seed consisting of that single odd direction. M-theory begins here because it is the smallest possible seed—a single relation between two distinct kinds of coordinate—from whose iterated extensions the eleven-dimensional fabric is generated. So, although its selection is logical rather than anthropic, the construction remains within the binary register and there is no observer to be found.

1.6. **Physico-mathematical antecedents: Manin's $\mathbb{F}_1$ program.**

1.6.1. *Quantum tori and numbers as functions.* Between string theory and pure mathematics there is an intermediate, experimental line of research that sought to use the quantum tori of M-theory, or the rings of Borger, to approach the Riemann hypothesis: the $\mathbb{F}_1$-geometry program associated with Yuri Manin. The $\mathbb{F}_1$ ("field with one element") program is the one in which a number of mathematicians—among them Manin, Deninger, Soulé, Kurokawa, Connes and Consani, and Borger—have sought to prove the Riemann hypothesis by transporting Weil's proof of the Riemann hypothesis *for curves over finite fields* to the arithmetic of $\mathbb{Z}$: were $\operatorname{Spec}\mathbb{Z}$ realisable as a "curve" over an absolute base $\mathbb{F}_1$, Weil's geometric argument might carry over.

Manin's quantum-tori strand [23] uses the noncommutative tori that Connes, Douglas and Schwarz [24] showed arise as compactifications of M-theory, placing M-theory and noncommutative/$\mathbb{F}_1$-geometry in contact. Such a torus is the noncommutative space T^2_θ whose algebra A_θ is generated by two unitaries U, V obeying

$$VU = e^{2\pi i\theta}\, UV, \qquad \theta \in \mathbb{R} \setminus \mathbb{Q}, \tag{1.10}$$

the irrational angle θ encoding the real multiplication that Manin attaches to real quadratic fields.

This is particularly significant for the Riemann hypothesis because, as Elvang, Herderschee and Morales [25] showed, the odd zeta values appear in the Veneziano string amplitude: the Wilson coefficients $a_{2k-1,0} = \alpha'^{2k+1}\zeta(2k+1)$ left free by the effective field theory are fixed, in the unique stringy completion, to the Veneziano values [26]. The odd zeta values thus tie the two frameworks—arithmetic and string-theoretic—together beyond Manin's program itself.

1.6.2. *A geometric route to the Riemann hypothesis.* In his other program, *Numbers as Functions* [27], Manin develops the conception of numbers as functions. Its arithmetic-differential strand replaces the derivative by the Fermat quotient,

$$\delta_p(a) = \frac{a - a^p}{p}, \qquad \psi_p(a) \equiv a^p \pmod{p}, \qquad \psi_p \circ \psi_q = \psi_{pq}, \tag{1.11}$$

the lift ψ_p being the Frobenius of a Λ-ring [28]. Here Manin approaches the Riemann hypothesis from a geometric perspective intended to transcend the self-reference that proving it entails: one cannot refer to the primes, nor to any zero, without importing circularity. The geometric route is meant to sidestep this by working with the space of the problem rather than its arithmetic statement.

1.6.3. *Riemann's ζ, holography and the primes: beyond Manin's programs.* Both of Manin's frameworks remained unfinished before his death in January 2023; both still lack a *dynamical completion*, and both programs remain open.

Beyond Manin's work, however, further relations have been established between string theory, the spectrum of the primes, the Riemann hypothesis and the problem of holography in de Sitter. Two of the most significant and best known bear directly on the problem: the odd zeta values appear both in the perturbative QED contributions to the electron's anomalous magnetic moment—where $\zeta(3)$ and $\zeta(5)$ enter the three-loop coefficient [29]—and in the Veneziano amplitudes already mentioned. A more recent and striking relation is the one between black-hole interiors and the primes obtained by Hartnoll, Yang and De Clerck [30, 31]: near a spacelike (BKL) singularity the Wheeler–DeWitt wavefunctions become automorphic forms whose L-functions, expanded over their Euler product, take the form of the partition function of a free gas of oscillators labelled by the primes—a "primon gas" [32]—the five-dimensional case requiring the complex (Gaussian and

Eisenstein) primes. Finally, and no less significant, is the relation between the GUE (Gaussian Unitary Ensemble) of random matrices—whose statistics Montgomery found to govern the pair correlation of the zeta zeros, as recognised by Dyson and verified numerically by Odlyzko [33, 34]—and the elusiveness of the Pólya operator: the Hilbert–Pólya conjecture that the zeros form the spectrum of some self-adjoint Hamiltonian.

Read in the light of holography, three points stand out: (i) the Veneziano amplitudes generate the odd zeta values; (ii) AdS/CFT has been most successful precisely for black holes, which have been in turn related to the primes and GUE random matrices; and (iii) the Hermitian operator of the zeros, like the discrete microstate of low-dimensional AdS/CFT, is GUE—itself a statistical description used specifically when no Hamiltonian is known. Taken together, these coincidences strongly suggest that AdS/CFT bears a closer relation to the Riemann hypothesis, at the mathematical level, than it might at first appear; and that, given the relations above, the absence of a single concrete microstate in both cases may share the same mathematical—rather than physical—cause. It is on this that we base our contention that the conjecture of selection among the $\sim 10^{500}$ vacua is mistaken, and that the conflict may be principally epistemological rather than physical.

1.7. **The framework.**

1.7.1. *A geometric construction.* Taking as precedent the frameworks of string theory, holography and AdS/CFT, together with Manin's framework and his reading of Borger's, and proceeding from a mainly geometric perspective, we develop a framework that sits halfway between set theory and intuitionism. It is intuitionist in the classical sense—in that it takes the observation of nature as its starting point rather than purely abstract mathematical axioms—yet it does not conflict with classification under set-theoretic equivalence; self-reference is thereby mediated at the intersection of the two systems, rather than prohibited. Conceptually it is a self-similar, interdimensional fractal that contains, is part of, and unifies—within a single lattice—the symmetries of $e^{i\pi}$, $\mathbb{N}$, $\mathbb{E}^3$, the primes, the half-integers and $\mathbb{C}$, by means of an algebraic ring, working as a contemporary Vitruvian module that uses φ, trigonometry and cyclotomic pentagons, mounted on a torus.

1.7.2. *The ring on the torus.* We combine Euler's identity, the modulus of $\mathbb{C}$ and the Vitruvian modulus into a single ring (the unifying isomorphism is established independently in [35])—a construction in which the four structural components (origin, magnitude as ratio to the module, angular direction, and equivalence under magnitudes) coincide with the data of $e^{i\pi} + 1 = 0$. These are carried by the ring

$$R_{\mathrm{PCF}} = \mathbb{Z}[\varphi, \varphi^{-1}, \tfrac{1}{2}], \tag{1.12}$$

whose self-similarity is seeded by the fixed-point equation

$$\varphi^2 = \varphi + 1, \qquad \varphi = \tfrac{1+\sqrt{5}}{2} = 2\cos\tfrac{\pi}{5} = \zeta_{10} + \zeta_{10}^{-1}, \tag{1.13}$$

which extends the role of $i^2 = -1$ from $\mathbb{R} \to \mathbb{C}$ to the further embedding $\mathbb{C} \to \mathbb{E}^3$ via $z = \varphi y$. The ring lives on a torus T^2_{PCF} that maintains a relation between 2D and 3D symmetry without selecting bulk or boundary; in this reading the closed string of string theory is reinterpreted as the ring R_{PCF} living on the torus—the extended identity. The same i that seeds $\mathbb{C}$ also fixes the modulus from the π-side: through $e^{i\pi} = -1$ and the Gaussian integral one has $\Gamma(\tfrac{1}{2}) = \sqrt{\pi}$, hence the half-factorial $(\tfrac{1}{2})! = \tfrac{1}{2}\sqrt{\pi}$, so the common modulus $\mu = \tfrac{1}{2}$ arises on the i/π side exactly as on the φ side (proved in §2.1, Lemma 2.4 and Theorem 2.5; the $\pi \leftrightarrow \varphi$ bridge is Proposition 2.2).

1.7.3. *Tarskian descent.* The mechanism allows a descent from arithmetic to pure geometry that we call Tarskian, after the decidability of the first-order theory of real-closed fields: arithmetic statements are recast as geometric ones, and numbers are treated as ratios—in particular as trigonometric ratios through $\varphi = 2\cos(\pi/5)$. The descent is what lets the construction sidestep the direct self-reference that the arithmetic statement of the problem would require. The descent yields a single geometric–arithmetic invariant—the unit of scale, one bit—that integrates the two origins of the modulus into one state (made precise in §2.5, where the commutative diagram of Theorem 2.26 shows the five routes to $\mu = \frac{1}{2}$ coincide).

1.7.4. *The golden tower and the multiplicative spectrum.* On this geometric substrate the module is expanded as a tower of powers that are Russellian in form but geometric rather than logical. The tower is anchored by the algebraic bridge

$$\varphi^{\lambda_{\log}} = 2, \qquad \lambda_{\log} = \frac{\log 2}{\log \varphi} = \log_\varphi 2, \tag{1.14}$$

so that φ generates the multiplicative spectrum that carries the binary base 2^p. The same tower carries the ternary base, $\varphi^{\log_\varphi 3} = 3$, so its growth mediates between the binary register (2) and the ternary register (3), with $\mu = \frac{1}{2}$ as the geometric mean: $\varphi^{\mu\lambda_{\log}} = \sqrt{2}$, $\varphi^{\mu \log_\varphi 3} = \sqrt{3}$ (proved in §2.1, Proposition 2.6; the mediated Mersenne form is Proposition 2.22 in §2.6).

1.7.5. *The selection of φ: non-paradoxical self-reference.* The first reason for the choice of φ is a non-paradoxical self-referent relation: (1.13) is the unique positive solution of $x = 1+1/x$, mapping onto itself without contradiction and maintaining consistency across 1D, 2D and 3D. This is the same relation that, as Pacioli's *De Divina Proportione* [36] records, made φ central in art, alongside other ratios.

1.7.6. *The selection of φ: the empirical trace.* The second reason is empirical: φ recurs in wave functions of light, electricity and magnetism, and in many lattices. Fibonacci numbers and the golden ratio appear across nearly all domains of science wherever self-organisation or minimum-energy configurations are at play [37], including the inflation factor and the multifractal tight-binding spectrum of the Fibonacci quasicrystal [38], and the gain eigenvalue of four-mode parametric coupling in nonlinear photonic crystals [39].

A more recent and striking appearance is gravitational. Among the extremal Kerr–Newman black holes whose mass, charge and angular momentum are fixed multiples of the irreducible mass, Sonnino *et al.* identify a distinguished family whose coefficients depend solely on φ [40]. Within it lies a unique black hole whose physical parameters converge on φ: analysed by differential geometry in $\mathbb{E}^3$, both the scale parameter (twice the irreducible mass) and the Gaussian curvature of the horizon surface at the poles equal φ—the configuration of highest symmetry admissible in the Kerr–Newman metric, with a proposed bearing on the dark-energy density parameter Ω_Λ of the Kerr–Newman–de Sitter family. Read against the black-hole/prime relation in §1.6.3, where the Wheeler–DeWitt wavefunctions near a five-dimensional spacelike singularity become automorphic L-functions whose Euler product is a primon gas, the two results place black holes, φ and the primes on the same object from complementary sides: the horizon geometry selects φ, and the interior selects the primes.

1.7.7. *The selection of φ: the classification of primes.* The third reason is arithmetic. The golden ratio is the fundamental unit of $\mathbb{Q}(\sqrt{5})$, and the quadratic character χ_5 classifies every rational prime; in the cyclotomic-pentagonal arithmetic of Grisales Herrera this yields a classification of the primes by φ, realised below through the Dedekind zeta of $\mathbb{Q}(\sqrt{5})$.

1.7.8. *The selection of φ: extremal irrationality.* The fourth reason is that φ is the number least well approximated by rationals. The same fixed-point relation $\varphi = 1 + 1/\varphi$ is the continued fraction $[1; 1, 1, \ldots]$, all of whose partial quotients are the minimum 1, so by Hurwitz's theorem [41] its rational approximations converge more slowly than those of any other irrational: φ is maximally obstructed *as a fraction*. Yet the same number is exact *as a ratio*—$\varphi = 2\cos(\pi/5) = \zeta_{10} + \zeta_{10}^{-1}$ (1.13), the diagonal-to-side ratio of the pentagon. The object whose arithmetic representation is worst is the one whose geometric representation is exact, so it is precisely φ that lets the construction carry the modulus as a ratio while never resolving into a rational sublattice.

1.7.9. *The selection of φ: optimal packing and the information bound.* The fifth reason joins the arithmetic to the physical. Extremal irrationality is, read geometrically, *optimal packing*: because φ resonates with no rational sublattice, a distribution stepped by φ fills space with maximal uniformity—the principle behind phyllotaxis, where seeds set at the golden angle leave no gaps. Read informationally, the same property is a *bound*: the most uniform packing is the densest record of distinguishable states, so φ is the ratio at which a region holds the most information it can hold. This we propose as a golden-ratio reading of the holographic principle—a limiting quantity of information per region—and it is what makes φ the natural carrier of the modulus $\mu = \frac{1}{2}$, whose binary entropy $H(\frac{1}{2}) = 1$ is exactly one bit per fundamental cell. The reading is realised quantitatively in §3.1, where the modulus saturates the Bekenstein bound (3.7).

1.7.10. *A single microstate.* The combination of the foregoing produces the mathematical equivalent of a *single microstate*: a structure describable by a Hermitian operator and manifesting as one specific torus carrying a string. In this paper the microstate enters geometrically, through that torus and its invariants—which is all the derivations below require; its realization as a Hermitian operator is built, from the same invariants, in the companion Hilbert–Pólya paper [42]. Here we establish the mathematical framework from string theory and then trace how it connects to physics and to the observer.

1.8. **The physical program.** Sections 3 and 4 demonstrate that the mathematical apparatus assembled in Section 2—the single microstate $|\Omega| = \frac{1}{2}$—makes contact with physics. The aim is not to propose new physics, but to establish that this apparatus meets criteria already fixed in the literature: each physical result is obtained by satisfying a requirement that an accepted framework had left open.

1.8.1. *Established frameworks and their open criteria.* The physical content starting point is Maldacena's conjecture that "M/string theory on various Anti-deSitter spacetimes is dual to various conformal field theories" [17]. We demonstrate how the mathematical framework allows the holographic and the M-theoretic descriptions to meet in one object, of which the microstate is a projection. Section 3 presents the microstate's relation to the 't Hooft–Susskind holographic bound [4], to the AdS boundary conditions [17], and to the ER = EPR bridge [43]; Section 4 presents its relation to Witten's unitarity and entropy conjectures for de Sitter space [1] and to the observer requirement of Chandrasekaran, Longo, Penington and Witten [2]. The equations are given in the sections indicated and collected in the demonstrative lines of Sections 3.6 and 4.5.

1.8.2. *Plan of the paper and division of labour.* The remainder of this paper proceeds as follows. Section 2 establishes the mathematical framework and constructs the single microstate from its string-theoretic ingredients: the two geometric origins of $\mu = \frac{1}{2}$, the uniqueness of the arity that blocks the diagonal, the base ring on the torus and its golden tower, and the convergence on μ, closing with the formal verification. Section 3 derives the physics, obtaining all of AdS/CFT and the M-theory type duality web from that single microstate at each level. Section 4 turns to the observer and to gravity: the observer is the projection Π, it does not select a vacuum but accumulates information, and that accumulation yields, through the Landauer–Jacobson–Verlinde chain, Einstein's equations. Section 6 concludes.

TABLE 1. Notation and key definitions.

Symbol / object	Meaning
$\varphi = 2\cos(\pi/5) = \frac{1+\sqrt{5}}{2}$	the golden ratio; generator of the torus
$\lambda_{\log} = \log 2/\log\varphi$	golden logarithmic scale
$\mu = \frac{1}{2}$	the modulus, $\mu = \lvert\Omega\rvert$
σ	tower level; $\sigma = \frac{3}{2}$ at the spectral point
Ω	the microstate; $\lvert\Omega\rvert = \frac{1}{2}$
$T^2_{\mathrm{PCF}} = \mathbb{C}/\Lambda,\ \tau = i$	the torus (geometric home)
$R_{\mathrm{PCF}} = \mathbb{Z}[\varphi, \varphi^{-1}, \frac{1}{2}]$	the base ring
$z = \varphi^\sigma$	the golden tower / throat coordinate
$N_{\mathrm{modes}} = \lfloor \pi\varphi^\sigma \rfloor$	mode count at level σ
ε_0	ultraviolet cutoff (certainty cell)
M_{PCF}	PCF mass scale
$G_N = \frac{1}{2}$	Newton constant
$c = 3$	central charge = colour arity = $\lfloor\pi\rfloor = \chi(\mathbb{CP}^2)$
$S_{\mathrm{BH}} = A/4G_N$	horizon (Bekenstein–Hawking) entropy
$f_{\mathrm{crit}} = \frac{1}{2}$	objectivity threshold
$\Pi : \mathbb{E}^3 \to \mathbb{C}$	the observer projection
Key definitions:	
microstate	the single object Ω, $\lvert\Omega\rvert = \frac{1}{2}$, with no ensemble average
shared signature	$\lvert\Omega\rvert = \frac{1}{2},\ \ell = 1,\ G_N = \frac{1}{2},\ c = 3,\ \mathrm{GKP} = \frac{3}{4}$
golden tower	the levels $z = \varphi^\sigma$ of the construction

2. METHODS

This section develops the mathematical core of the framework, our approach to the frameworks of §1.6: Manin's quantum tori $\mathbb{F}_1$ program with Borger's Λ-algebraic ring. Here we develop only what is pertinent to the string; not the whole of the $\mathbb{F}_1$ program, which is developed in a companion paper [44]. What we develop here is presented as a single thread, from i to the two vertices $\mu = \frac{1}{2}$ and $\sigma = \frac{3}{2}$, each step formally verified in Lean 4 (theorem names in brackets); each subsection uses only objects already introduced.

2.1. Two geometric origins of $\mu = \frac{1}{2}$.

The constant $\mu = \frac{1}{2}$ arises from two independent geometric origins, a π-branch (Gamma/Gaussian) and a φ-branch (the involution $x = 1 - x$), both routed through φ.

Theorem 2.1 (Pentagonal identity). $\varphi = 2\cos(\pi/5)$.

Proof. The Chebyshev polynomial T_5 satisfies $T_5(\cos\theta) = \cos(5\theta)$ with $T_5(x) = 16x^5 - 20x^3 + 5x$. At $\theta = \pi/5$ this gives $T_5(\cos(\pi/5)) = \cos\pi = -1$, so $\cos(\pi/5)$ is a root of $16x^5 - 20x^3 + 5x + 1 = 0$, which factors as $(x+1)(4x^2 - 2x - 1)^2$. Since $0 < \pi/5 < \pi/2$ forces $\cos(\pi/5) > 0$, we have $\cos(\pi/5) \neq -1$; hence $\cos(\pi/5)$ is a root of $4x^2 - 2x - 1 = 0$, whose positive root is $(1 + \sqrt{5})/4 = \varphi/2$. Therefore $\varphi = 2\cos(\pi/5)$; this classical identity is also derived in the correspondence paper [35] and formalised in Lean as `phi_eq_two_cos_pi_fifth`. □

[`phi_eq_two_cos_pi_fifth`]

Proposition 2.2 (Pentagonal bridge $\pi \leftrightarrow \varphi$). *Since $\varphi = 2\cos(\pi/5)$ (Theorem 2.1),*

$$\pi = 5\arccos(\varphi/2). \tag{2.1}$$

[M13_pi_eq_five_arccos]

Proposition 2.3 (Gamma as an infinite integral (Euler)). *For $s > 0$,*

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t}\, dt. \tag{2.2}$$

[M1_gamma_integral]

Lemma 2.4 (Gaussian value). $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ *(classical; from the Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$).* [M2_gamma_half, gaussian_integral_value, gaussian_weight_phi]

Theorem 2.5 (Half-factorial (classical)).

$$(\tfrac{1}{2})! \;=\; \Gamma(\tfrac{3}{2}) \;=\; \mu\sqrt{\pi}, \qquad \textit{equivalently} \qquad \mu = \frac{(\frac{1}{2})!}{\sqrt{\pi}}. \tag{2.3}$$

[M3_half_factorial]

Proposition 2.6 (Mediation half-powers).

$$\varphi^{\mu\lambda_{\log}} = \sqrt{2}, \qquad \varphi^{\mu \log_\varphi 3} = \sqrt{3}. \tag{2.4}$$

[M4_sqrt2, M4_sqrt3]

Remark 2.7 (φ-branch). $\mu = \frac{1}{2}$ is the fixed point of the involution $x = 1 - x$.

2.2. **Arity uniqueness and the blocking of the diagonal.** The arity 3 and the modulus $\frac{1}{2}$ are forced, and the Lawvere–Yanofsky diagonal is contracted (not prohibited).

Definition 2.8 (Distributed self-reference). A recurrence of depth k is *distributed* when $k \geq 2$ (no component depends directly on itself). [DistributedSelfReference]

Theorem 2.9 (Fibonacci minimality). *Depth $k = 2$ is minimal, and the unique positive root of $r^2 = r + 1$ is φ. Among natural-coefficient depth-2 recurrences $F_{n+2} = c_1F_{n+1} + c_2F_n$, the pair $c_1 = c_2 = 1$ is the unique one whose characteristic root—hence growth ratio—is φ.* [M6_characteristic_root_is_phi, M6_phi_root_unique]

Proposition 2.10 (Spectral constraints force $(\sigma, \mu) = (\frac{3}{2}, \frac{1}{2})$). $\sigma + \mu = 2$, $\sigma\mu = \frac{3}{4}$, *with unique solution $0 < \mu < 1$ giving $\mu = \frac{1}{2}$, $\sigma = \frac{3}{2}$.* [spectral_uniqueness]

Remark 2.11 (No-diagonal). $\|\hat{\Omega}\| = \frac{1}{2} < 1$ contracts the diagonal below the contradiction threshold.

2.3. **Construction of PCF and the base ring on the torus.** The generator $\varphi^2 = \varphi + 1$, the base ring, the tripartite operator, and the projection that fixes ε_0.

$$\varphi^2 = \varphi + 1, \qquad R_{\mathrm{PCF}} = \mathbb{Z}[\varphi, \varphi^{-1}, \tfrac{1}{2}], \qquad \mathbb{E}^3 = \{(x, y, \varphi y)\}. \tag{2.5}$$

$$\Omega = P \cdot C \cdot F, \qquad \|P\| = \tfrac{1}{\sqrt{3}},\ \|C\| = 1,\ \|F\| = \tfrac{\sqrt{3}}{2}, \qquad \|P\|\,\|C\|\,\|F\| = \tfrac{1}{2}. \tag{2.6}$$

Proposition 2.12 (Component norms from the eigenvalue triangle). *The operator $\hat{\Omega}$ has eigenvalues $\lambda_k = \frac{1}{2}\omega^k$ $(k = 0, 1, 2,\ \omega = e^{2\pi i/3})$, inscribing an equilateral triangle in the circle $|z| = R = \frac{1}{2}$ of $\mathbb{C}$. The component norms are fixed by this geometry:*

$$\|C\| = 1, \qquad \|F\| = \tfrac{\sqrt{3}}{2}, \qquad \|P\| = \tfrac{1}{\sqrt{3}}, \tag{2.7}$$

whence $\|P\|\,\|C\|\,\|F\| = \frac{1}{2} = \|\hat{\Omega}\|$.

Proof. *(C)* C is the real eigenvalue $\lambda_0 = \frac{1}{2}$, normalised by the circumradius $R = |\lambda_0| = \frac{1}{2}$: $\|C\| = |\lambda_0|/R = 1$. *(F)* F is the chord between adjacent eigenvalues, $\|F\| = |\lambda_1 - \lambda_0| = \frac{1}{2}|\omega - 1| = \sin(\pi/3) = \frac{\sqrt{3}}{2}$. *(P)* The fundamental right triangle (centre, midpoint of $\lambda_0\lambda_1$, vertex λ_0) has apothem $R\cos(\pi/3) = \frac{1}{4}$ and half-side $\|F\|/2 = \frac{\sqrt{3}}{4}$, so $\|P\| = \dfrac{R\cos(\pi/3)}{\|F\|/2} = \dfrac{1/4}{\sqrt{3}/4} = \frac{1}{\sqrt{3}} = \tan(\pi/6)$. *(Product)* The $\sqrt{3}$ of $\|F\|$ cancels that of $\|P\|$, leaving $\frac{1}{2}$; the cancellation is specific to the equilateral (n=3) triangle. □

Definition 2.13 (PCF projection). $\pi_{\mathrm{PCF}}(a, b, c) = \dfrac{ab}{c\sqrt{3}} \cdot \dfrac{\pi}{3}$.

Proposition 2.14 (Fibonacci sum as projection). $F = P \oplus C := \pi_{\mathrm{PCF}}(P, C, 1) = \dfrac{(PC)\,\pi}{3\sqrt{3}}$. *[M7_oplus_formula]*

Theorem 2.15 (Derivation of ε_0).

$$\varepsilon_0 = \pi_{\mathrm{PCF}}\big(\sin(\tfrac{\pi}{6}),\, \ln\varphi,\, \pi\big) = \frac{\ln\varphi}{6\sqrt{3}}. \tag{2.8}$$

[M8_epsilon0_from_projection]

Remark 2.16 (Torus, scale, and Sierpiński attractor). The microstate is realised on the torus

$$T^2_{\mathrm{PCF}} = \mathbb{C}/\Lambda_{\mathrm{PCF}}, \qquad \Lambda_{\mathrm{PCF}} = M_{\mathrm{PCF}}\,\mathbb{Z}[i], \qquad \tau = i, \tag{2.9}$$

whose scale is fixed by ε_0 of (2.8):

$$M_{\mathrm{PCF}} = \frac{6\sqrt{3}\,\pi}{\ln\varphi} = \frac{\pi}{\varepsilon_0} \approx 67.846. \tag{2.10}$$

[M_PCF_eq_pi_div_eps0] Equivalently, the *Certainty Principle*

$$\varepsilon_0\, M_{\mathrm{PCF}} = \pi. \tag{2.11}$$

The three eigenvalues $\lambda_k = \frac{1}{2}\omega^k$ define an iterated function system of three contractions of ratio $\frac{1}{2}$,

$$T_k(z) = \tfrac{1}{2}(z - v_k) + v_k, \qquad v_k = \tfrac{1}{2}\omega^k, \qquad k \in \{0, 1, 2\}, \tag{2.12}$$

whose Sierpiński-type attractor has Hausdorff dimension

$$d_H = \frac{\log 3}{\log 2} \approx 1.585, \qquad 2^{d_H} = 3, \tag{2.13}$$

the Moran equation counting the three components P, C, F.

2.4. **The non-orientable fibre is independent of the torus.** The base torus of (2.9) carries a module direction and one non-orientable identification of that direction, both fixed by data already in place. The module direction is the phase $\Omega(\tau) = \frac{1}{2}\, e^{i\tau \ln\varphi}$, of constant modulus $|\Omega(\tau)| = \frac{1}{2} = \|\hat{\Omega}\|$ (Proposition 2.12).

Arithmetic type: an Eisenstein–Gauss transition. The eigenvalue triangle $\lambda_k = \frac{1}{2}\omega^k$ ($\omega = e^{2\pi i/3}$) of Proposition 2.12 sits in the Eisenstein lattice $\mathbb{Z}[\omega]$ (hexagonal, 120°); the torus modulus $\tau = i$ sits in the Gauss lattice $\mathbb{Z}[i]$ (square, 90°). The golden coupling $z = \varphi y$ of (2.5) carries the first into the second isometrically—it transforms the angular structure (120° → 90°) while preserving the metric relations; the construction is carried out in the companion operator paper [42]—and $\tau = i$ is its fixed point. The mediator is φ and no other: from (2.5), $\varphi = 1 + 1/\varphi$, so its continued fraction is $[1; 1, 1, \ldots]$ and φ is the extremal irrational, most poorly approximated by rationals (Hurwitz [41])—the optimality that also places φ in phyllotaxis—and the modulus $\frac{1}{2}$ is carried uniformly on the fibre.

Proposition 2.17 (The fibre is independent of the base algebra)**.** *The non-orientable identification lives in the fibre, not in the algebra of the torus, and is therefore independent of whether that algebra commutes. Carried once around the fundamental domain, the module direction accumulates the half-turn* $\varepsilon_0 M_{\mathrm{PCF}} = \pi$ (2.11)*, acting on the fibre of modulus* $\frac{1}{2}$ *as* $e^{i\pi} = -1$*: the reflection* $v \mapsto -v$*,* π *and not* 2π *because* $\|\hat{\Omega}\| = \frac{1}{2}$*, and orientation-reversing of order two,* $(-1)^2 = 1$*. This datum is a* $\mathbb{Z}/2$ *reflection of the fibre; it carries no hypothesis on the base commutator phase* $UV = e^{2\pi i\theta}VU$*. Hence the same fibre couples onto any torus—the commutative torus (*$\theta = 0$*) and the noncommutative torus* T^2_θ *that M-theory uses (*θ *irrational, §1.6)—while preserving* $|\Omega| = \frac{1}{2}$ *on either.* `[fibre_monodromy, fibre_reflection_order_two, fibre_independent_of_base]`

Proof. The fibre monodromy is $e^{i\pi} = -1$, of order two, $(-1)^2 = 1$, a $\mathbb{Z}/2$ datum that holds with no condition on θ (`[fibre_independent_of_base]`); the base commutes for $\theta \in \mathbb{Z}$ and does not for θ irrational, the fibre being the same in both. The modulus $\frac{1}{2}$ is Proposition 2.12, constant in τ since $|\Omega(\tau)| = \frac{1}{2}$, and is preserved by the reflection. □

2.5. **The single microstate, the meta-invariant, and the geometric origin of $\frac{1}{2}$ tied to the Euler product.** The geometric origin of $\frac{1}{2}$ via the angle $\pi/6$, its anchoring to the critical line, and the factorial as connector.

Theorem 2.18 (Trigonometric collapse of the norm product)**.**

$$\|P\|\,\|C\|\,\|F\| = \tan(\tfrac{\pi}{6}) \cdot 1 \cdot \cos(\tfrac{\pi}{6}) = \sin(\tfrac{\pi}{6}) = \tfrac{1}{2}. \tag{2.14}$$

`[M9_collapse, M9_eq_half]`

Proposition 2.19 (Angle $\pi/3$ gives both invariants)**.** $\sin(\frac{\pi}{6}) = \cos(\frac{\pi}{3}) = \mu$*, and*

$$\sigma = \frac{\zeta(2)}{(\pi/3)^2} = \frac{\pi^2/6}{\pi^2/9} = \frac{9}{6} = \frac{3}{2}. \tag{2.15}$$

`[M10_sin_cos_mu, M10_sigma_from_basel]`

Proposition 2.20 (Factorial as connector)**.** $\dfrac{(\frac{1}{2})!}{\sqrt{\pi}} = \mu = \frac{1}{2}$ *(half face), and* $3! = 6 = |S_3|$ *(integer face).* `[M11_factorial_face, M11_factorial_six]`

Remark 2.21 ($\frac{1}{2} \leftrightarrow$ Euler product)**.** The modulus $\frac{1}{2} = \mu$ is anchored to the critical line; the Euler product factors through the ternary level $\mu_3 = \frac{1}{2}$.

2.6. **Expansion of the spectrum mediated between i and φ.** Additive spectrum (pentagons, no arithmetic) and multiplicative spectrum (Mersenne), joined by the 3/2 structure; primes classified by χ_5.

$$\varphi^n = F_n\varphi + F_{n-1} \quad \text{(additive)}, \qquad 3\,\varphi^{\sigma(p)} = 2^p \quad \text{(Mersenne)}. \tag{2.16}$$

Proposition 2.22 (Mediated Mersenne)**.** $3\,\varphi^{\mu p \lambda_{\log}} = 3\,(\sqrt{2})^p$*.* `[M12_mersenne_mediated]`

Remark 2.23. χ_5 classifies every rational prime in $\mathbb{Q}(\sqrt{5})$ (split/inert/ramified).

2.7. **The rings as a tower (our reinterpretation of the Λ-structure of Borger).** The Frobenius lifts and the Λ-ring tower carry the arithmetic to the Euler product as a colimit [44].

$$\psi_p(\varphi) = \varphi^p = F_p\varphi + F_{p-1}, \qquad \zeta_{\mathbb{Q}(\sqrt{5})}(s) = \varinjlim_{S} E(S)(s). \tag{2.17}$$

2.8. **π and $\pi\varphi$ as producers of ζ and odd-ζ.** The Dedekind factorisation and the even/odd zeta values; the bridge (2.1) ties π to φ here.

$$\zeta(s) = \frac{\zeta_{\mathbb{Q}(\sqrt{5})}(s)}{L(s,\chi_5)}, \qquad L(1,\chi_5) = \frac{2\log\varphi}{\sqrt{5}}. \tag{2.18}$$

Theorem 2.24 (Basel value)**.** *$\zeta(2) = \pi^2/6$ (Euler).* `[M14_basel]`

Theorem 2.25 (General even zeta value (Euler))**.** *For every integer $k \geq 1$,*

$$\zeta(2k) = (-1)^{k+1}\frac{B_{2k}\,(2\pi)^{2k}}{2\,(2k)!}, \tag{2.19}$$

with B_{2k} the Bernoulli numbers; $k = 1$ recovers $\zeta(2) = \pi^2/6$ (Theorem 2.24).

Proof. Euler's cotangent expansion $\pi z\cot(\pi z) = 1 - 2\sum_{k\geq 1}\zeta(2k)\,z^{2k}$ $(0 < |z| < 1)$ and the Bernoulli generating function $\frac{w}{e^w-1} = \sum_{n\geq 0}\frac{B_n}{n!}w^n$ are linked by $\pi z\cot(\pi z) = \pi i z + \dfrac{2\pi i z}{e^{2\pi i z}-1}$. Substituting the second series with $w = 2\pi i z$ and using $(2i)^{2k} = (-1)^k 4^k$, the coefficient of z^{2k} on the right is $(-1)^k B_{2k}(2\pi)^{2k}/(2k)!$; matching it to $-2\zeta(2k)$ gives (2.19). The pentagonal, χ_5-twisted refinement, with generalised Bernoulli numbers B_{2k,χ_5}, extends this computation and is carried out independently in the companion F_1-descent paper [44]. □

2.9. **The commutative diagram: convergence on μ and σ.** The constant $\frac{1}{2}$ is the apex of a cocone: five routes coincide on it. The argument is convergence, not a chain.

$$\begin{tikzcd}
\frac{(\frac{1}{2})!}{\sqrt{\pi}} \arrow[ddddr] & \\
\|P\|\|C\|\|F\| \arrow[dddr] & \\
\cos(\frac{\pi}{3}) \arrow[ddr] & \\
\varphi^{-\lambda_{\log}} \arrow[dr] & \\
\mathrm{fix}(x{=}1{-}x) \arrow[r] & \mu = \frac{1}{2}
\end{tikzcd}$$

Theorem 2.26 (Modulus diagram commutes)**.** *The five routes coincide on the apex:* $\dfrac{(\frac{1}{2})!}{\sqrt{\pi}} = \|P\|\|C\|\|F\| = \cos(\frac{\pi}{3}) = \varphi^{-\lambda_{\log}} = \mathrm{fix}(x{=}1{-}x) = \mu = \frac{1}{2}$. *The non-elementary legs are established at (2.3) (the Gamma value), (2.14) (the norm product) and (1.14) (the binary bridge $\varphi^{\lambda_{\log}} = 2$, whence $\varphi^{-\lambda_{\log}} = \frac{1}{2}$); the angle $\cos(\frac{\pi}{3})$ and the involution fixed point are elementary.* `[mu_diagram_commutes, mu_faces_pairwise_eq]`

Remark 2.27 (The microstate is unique: no landscape to select)**.** The five routes of Theorem 2.26 leave no freedom in the modulus: the value $\mu = \frac{1}{2}$ is forced at once by the Gamma value, the norm product, the angle $\cos(\frac{\pi}{3})$, the binary bridge and the involution fixed point. The microstate is therefore the *unique* self-consistent solution, not one option among many. This is what dissolves the vacuum-selection problem of §1.2.4: there is no $\sim 10^{500}$ landscape to choose from and no anthropic selection to invoke—the single microstate is fixed by the convergence itself, and the abstract's "unique self-consistent solution" is established here. The uniqueness proved here is that of the modulus $\mu = \frac{1}{2}$ as the algebraic fixed point of the five coincident routes (Theorem 2.26); its reading as the absence of a physical vacuum landscape is the interpretive step developed in §4, where the observer builds—rather than selects—the geometry.

$$\begin{tikzcd}
\dfrac{\zeta(2)}{(\pi/3)^2} \arrow[dr] & \\
\dfrac{|\mathrm{rot}\, S_3|^2}{|S_3|} \arrow[r] & \sigma = \tfrac{3}{2}
\end{tikzcd}$$

Lemma 2.28 (Orders in S_3 and the geometric leg)**.** *The equilateral triangle of eigenvalues $\lambda_k = \frac{1}{2}\omega^k$ (Proposition 2.12) has symmetry group the symmetric group $S_3 \cong D_3$. Its order is $|S_3| = 3! = 6$, and its rotation subgroup is the cyclic group $C_3 = \{1, r, r^2\}$ of order $|\mathrm{rot}\, S_3| = 3$. Hence*

$$\frac{|\mathrm{rot}\, S_3|^2}{|S_3|} = \frac{3^2}{6} = \frac{9}{6} = \tfrac{3}{2}, \tag{2.20}$$

so the geometric leg equals σ rather than defining it.

Theorem 2.29 (Sum diagram commutes)**.** *Both routes coincide on $\sigma = \frac{3}{2}$: the analytic leg gives $\dfrac{\zeta(2)}{(\pi/3)^2} = \dfrac{\pi^2/6}{\pi^2/9} = \dfrac{9}{6}$ (2.15), and the geometric leg gives $\dfrac{|\mathrm{rot}\, S_3|^2}{|S_3|} = \dfrac{9}{6}$ (Lemma 2.28); both equal $\sigma = \frac{3}{2}$.* *[sigma_diagram_commutes]*

Theorem 2.30 (Methods master)**.** *Both diagrams commute and the spectral invariants follow: $\sigma + \mu = 2$, $\sigma/\mu = 3$.* *[section2_master]*

2.10. **Formal verification.** Each result of this section carries, in brackets, the name of its Lean 4 theorem. The development that accompanies the paper is the self-contained file `CW3_lean.lean`, which declares no axioms. The golden ratio is defined, $\varphi := (1+\sqrt{5})/2$, and the two facts that anchor the construction are theorems proved from that definition, not assumptions. The relation $\varphi^2 = \varphi + 1$ (`phi_sq`) is derived from the definition and is (1.13). The modulus $\|\hat{\Omega}\| = \frac{1}{2}$ (`modulus_Omega`) is likewise a theorem: its value is fixed in §2.1 and its component structure in Proposition 2.12.

Compilation note (single point of record). The library lemma names that require confirmation against the installed Mathlib version are `Gamma_eq_integral`, `Gamma_one_half_eq`, `riemannZeta_two`, `Real.sqrt_eq_rpow`, `rpow_natCast`, `Gamma_add_one`, `arccos_cos` and, for the M6 uniqueness, `Nat.Prime.irrational_sqrt`, `Irrational.int_mul`, `Int.not_irrational`.

3. Derivations

Building on the single microstate established in §2, this section presents one demonstrative line: the object generates a tower, and the tower generates the duality web. The line runs in the order 3.1 → 3.2 → 3.3 → 3.4 → 3.5, summarised as a 19-step spine at the end (§3.6). The node that stitches §2 to here is the function Γ; every quantitative statement below is verified to 50 digits in `CW3_verify_paper_physics.py` (§2.10).

3.1. **The fundamental object in physics.** The single object is the half-integer Γ value, the carrier of the string microstate. The seed inherited from §2 is

$$\Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi}. \tag{3.1}$$

The worldline phase and its modulus are

$$\Omega(\tau) = \tfrac{1}{2}\, e^{i\tau \ln\varphi}, \qquad |\Omega| = \tfrac{1}{2}. \tag{3.2}$$

The wavefunction is Gaussian, with the half-factorial normalisation $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and a weight fixed by the scale quantum,

$$\int_{\mathbb{R}} e^{-x^2}\, dx = \sqrt{\pi}, \qquad \mu = \varphi^{-\lambda_{\log}} = \tfrac{1}{2} \quad (\varphi^{\lambda_{\log}} = 2). \tag{3.3}$$

The one-dimensional worldline is promoted to the two-dimensional worldsheet, governed by the Polyakov action [45]

$$S_P = -\frac{1}{4\pi\alpha'}\int d^2\sigma\,\sqrt{-h}\,h^{ab}\,\partial_a X^\mu\,\partial_b X^\nu\,\eta_{\mu\nu}. \tag{3.4}$$

Proposition 3.1 (The worldsheet inherits the non-orientable fibre)**.** *The worldsheet carries the non-orientable fibre of the base torus (Proposition 2.17, §2.4): the module direction of $\Omega(\tau) = \frac{1}{2}\,e^{i\tau\ln\varphi}$ (3.2), carried once around the fundamental domain, closes with the orientation-reversing half-turn $e^{i\pi} = -1$. The worldsheet is therefore non-orientable and one-sided—the half-turn carries the module's interior edge to its exterior—so the bulk–boundary relation the torus carries admits no privileged side, independently of which torus carries the fibre.*

The Schwinger parametrisation, used to assemble propagators and amplitudes, follows from the Γ integral by the change of variable $t = xA$,

$$A^{-n} = \frac{1}{\Gamma(n)}\int_0^\infty x^{n-1}\,e^{-xA}\,dx \qquad (A > 0,\ n > 0). \tag{3.5}$$

Remark 3.2 (The role of φ in the object)**.** Since $\pi = 5\arccos(\varphi/2)$ (§2), both $\sqrt{\pi}$ and the half-factorial are functions of φ; and $\mu = \frac{1}{2} = \varphi^{-\lambda_{\log}}$. The object is φ-generated, not posited.

3.2. **The meta-invariant: the holographic quantum.** The meta-invariant emerges from the structure already in place: the worldsheet phase $\Omega(\tau) = \frac{1}{2}\,e^{i\tau\ln\varphi}$ of §3.1 (3.2), carried on the torus T^2_{PCF} of §2.3 (2.9), whose single modulus $|\Omega| = \frac{1}{2}$ is the norm product fixed in §2.3 (2.6). The bridge from this object to the rest of physics is that $|\Omega| = \frac{1}{2}$ *is* the holographic bit (one bit per Planck area). The spectral invariants are

$$d = 3\ (= |S_3| = 3! = 6), \qquad \mu = \tfrac{1}{2}, \qquad \sigma = \tfrac{3}{2}, \qquad \sigma + \mu = 2. \tag{3.6}$$

The spectral angle and the uniqueness of the invariants. At scale σ the worldline subtends the spectral angle $\alpha(\sigma) = \arctan(\varepsilon_0\varphi^\sigma)$, normalised by $\varepsilon_0 M_{\mathrm{PCF}} = \pi$ (Fig. 1) `[alpha_decomposition]`. The invariants (3.6) are not chosen but forced: the pair (σ, μ) with $\sigma + \mu = 2$ and $\sigma\mu = \frac{3}{4}$ has the unique solution $(\sigma, \mu) = (\frac{3}{2}, \frac{1}{2})$ `[spectral_uniqueness]`, i.e. $\sigma = \frac{3}{2}$ `[sigma_three]`; over the integer grid the spectral-angle identity is matched at the single pair $(\sigma_1, \sigma_2) = (2, 3)$, the same arity 3 and modulus $\frac{1}{2}$ forced by the No-Diagonal theorem of §2.2.

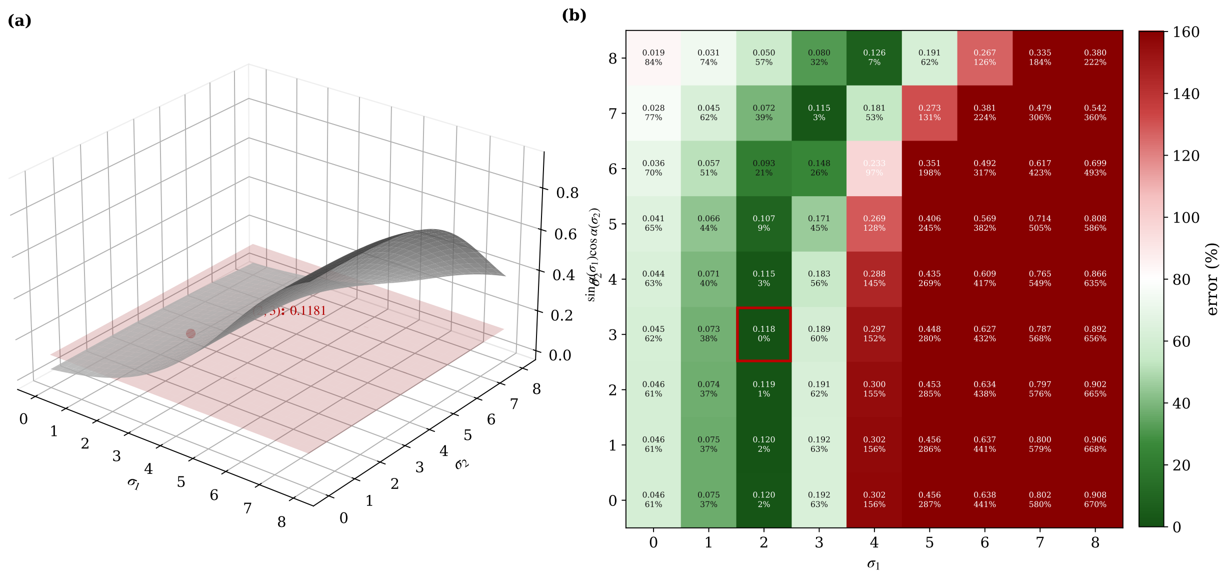

FIGURE 1. The spectral angle $\alpha(\sigma) = \arctan(\varepsilon_0\varphi^\sigma)$: (a) the surface $\sin\alpha(\sigma_1)\cos\alpha(\sigma_2)$; (b) the error map, vanishing only at $(\sigma_1,\sigma_2) = (2,3)$ (boxed) — the unique pair compatible with the forced invariants (3.6).

With the binary entropy $H(p) = -p\log_2 p - (1-p)\log_2(1-p)$, the modulus saturates the Bekenstein bound [46, 47],

$$\frac{1}{4G_N} = \tfrac{1}{2} = |\Omega|, \qquad H\left(\tfrac{1}{2}\right) = 1 \text{ bit}, \qquad S_{\mathrm{BH}} = \frac{A}{4G_N}. \tag{3.7}$$

`[binary_entropy_half]`

Remark 3.3 (The role of φ in the meta-invariant)**.** $\mu = \frac{1}{2} = \varphi^{-\lambda_{\log}}$ is the scale quantum ($\varphi^{\lambda_{\log}} = 2 =$ one bit). The invariants $d = 3$, $\mu = \frac{1}{2}$, $\sigma = \frac{3}{2}$ come from the φ/S_3 geometry of §2; they are not postulated.

Thread (the invariants force the holographic principle). The holographic principle is read off the invariants, not assumed:

(1) the modulus $\mu = \frac{1}{2} = |\Omega|$ has binary entropy $H(\frac{1}{2}) = 1$, so each fundamental cell (each eigenvalue) carries exactly one bit ((3.6)→(3.7));
(2) the $d = 3$ eigenvalues give 3 bits $= c = 3$ (Brown–Henneaux [48]: three eigenvalues × one bit);
(3) and $1/(4G_N) = \mu = \frac{1}{2}$ turns $S_{\mathrm{BH}} = A/(4G_N)$ into the area law.

Hence information = one bit per Planck area (the holographic bound), read from the invariants. (Verified in `CW3_verify_paper_physics.py`: $H(\frac{1}{2}) = 1$, $c = 3$, $G_N = \frac{1}{2}$.)

This is the holographic bound that Susskind, following 't Hooft, drew from unproven assumptions about the nonperturbative behaviour of string theory: one discrete degree of freedom per Planck area [4, 18]. Here it is not assumed but derived—the bit $H(\frac{1}{2}) = 1$, the central charge $c = 3$, and the area law $S_{\mathrm{BH}} = A/4G_N$ are read off the single modulus $|\Omega| = \frac{1}{2}$.

3.3. **The object as generator of the tower and the amplitudes.** Built on the golden tower of §2.7 (the Borger Λ-structure on φ^σ) and the mediated spectrum of §2.6, the tower mediates *between* a binary register (the half-factorial / Γ / π; i, d=1→2) and a ternary one (S_3/φ; d=2→3), with $\mu = \frac{1}{2}$ as the geometric mean. The Huerta–Schreiber superpoint is purely binary ($\mathbb{Z}_2$) [22];

PCF spans and mediates both. The M-theory object enters here: the HS tower projects from the φ^σ tower. The dimensional chain is

$$\mathbb{R}_{d=1} \xrightarrow{i^2=-1} \mathbb{C}_{d=2} \xrightarrow{\varphi^2=\varphi+1} \mathbb{E}^3_{d=3}. \tag{3.8}$$

Mediation by $\mu = \frac{1}{2}$ produces the two registers,

$$\varphi^{\mu\lambda_{\log}} = \sqrt{2}\ \ (\text{binary}), \qquad \varphi^{\mu \log_\varphi 3} = \sqrt{3}\ \ (\text{ternary}). \tag{3.9}$$

The tower and its mode count are

$$z(\sigma) = \varphi^\sigma, \qquad N_{\text{modes}}(\sigma) = \lfloor \pi\, \varphi^\sigma \rfloor. \tag{3.10}$$

The count and its golden ratio are shown in Fig. 2.

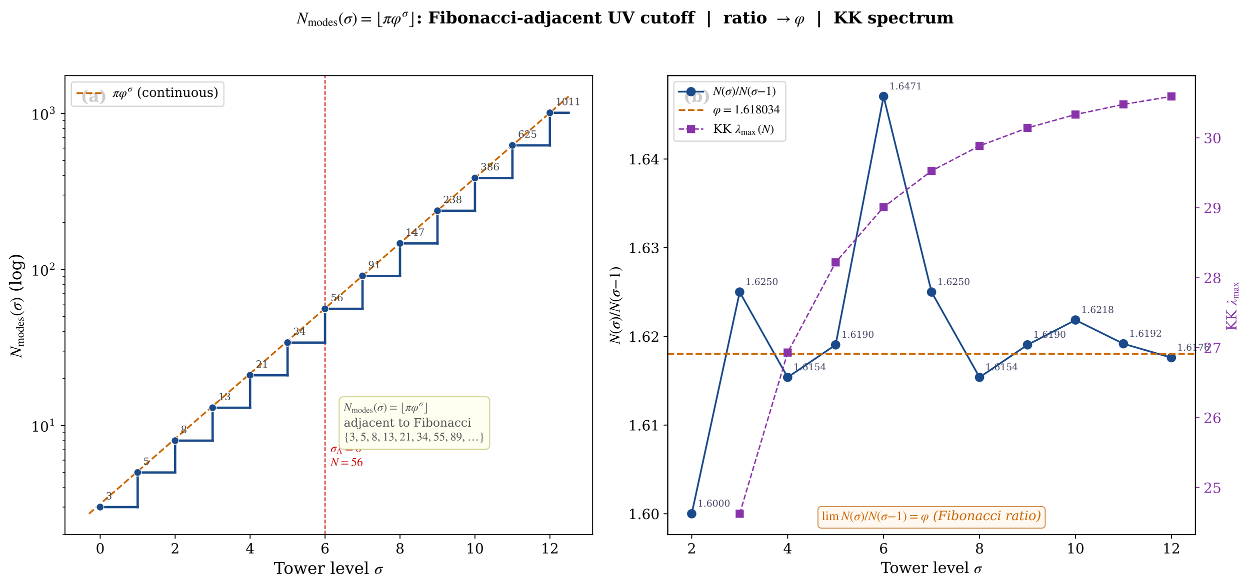

FIGURE 2. The mode tower $N_{\text{modes}}(\sigma) = \lfloor \pi\varphi^\sigma \rfloor$ (3.10): (a) the Fibonacci-adjacent staircase $\{3, 5, 8, 13, 21, 34, 56, \dots\}$; (b) the successive ratio $N_{\text{modes}}(\sigma)/N_{\text{modes}}(\sigma - 1) \to \varphi$ alongside the Kaluza–Klein spectrum.

The Huerta–Schreiber brane-bouquet tower [22] maps to the PCF tower under a functor F,

$$\mathbb{R}^{0|1} \to \mathbb{R}^{1,1|1} \to \cdots \to \mathbb{R}^{9,1|16} \to \mathbb{R}^{10,1|32} \xrightarrow{F} \{\varphi^\sigma\}_\sigma, \tag{3.11}$$

the dynamics being presupposed on the HS side ($H = L_0 + \bar{L}_0$) and derived on the PCF side. Tree amplitudes follow from π (binary, via Γ) and φ (ternary, via Regge): the Veneziano amplitude [26]

$$A_4 = \frac{\Gamma(-\alpha' s)\,\Gamma(-\alpha' u)}{\Gamma\big(1 - \alpha'(s+u)\big)} \tag{3.12}$$

has simple poles at $\alpha' s = n$, $n \geq 1$ (the Regge tower). The pole *positions* are the integer spectrum, so they index the tower's Dirichlet series; the residues are the Regge polynomials of degree $n - 1$ (carrying spin $\leq n-1$), nonzero at every level, which certifies that every integer level is populated. That Dirichlet series is $\zeta(s)$, which reorganises, by unique factorisation, into the Euler product,

$$\sum_{n\geq 1} n^{-s} = \zeta(s) = \prod_p \big(1 - p^{-s}\big)^{-1} \qquad (\operatorname{Re} s > 1). \tag{3.13}$$

[regge_tower_is_euler_product, regge_dirichlet_eq_zeta, regge_residue_ne_zero, regge_residue_degree]The pentagonal origin $\varphi = 2\cos(\pi/5)$ generates this structure: the Regge tower (string side) and the Euler product (arithmetic side) are the same object seen from two angles. The uniqueness of the ultraviolet completion selected here—that under supersymmetry and the standard consistency conditions the Veneziano form is essentially forced—is a recent result of Elvang and collaborators [25]. Loops are likewise φ-organised; the one-loop partition function is finite,

$$Z_{\mathrm{PCF}}(i) = \frac{e^{-3\pi/2}}{|\eta(i)|^6}, \qquad \eta(i) = \frac{\Gamma(\frac{1}{4})}{2\,\pi^{3/4}}, \tag{3.14}$$

with the loop hierarchy — one loop = colour, two loops = generations: a structural assignment (anchored on the colour ratio $1-\mu_3^2 = \frac{3}{4}$ and on Apéry's two-loop constant $\zeta(3)$ [49]; odd zeta over the tower [44]), developed in the $\mathbb{F}_1$ companion and not derived here. [colour_ratio]
The ultraviolet is regulated in two complementary ways: the modular partition (3.14) (global, on the torus) and dimensional regularisation at the vertex (local),

$$\Gamma\big(\tfrac{\varepsilon}{2}\big) = \frac{2}{\varepsilon} - \gamma_E + O(\varepsilon). \tag{3.15}$$

Remark 3.4 (The role of φ in the tower)**.** $\varphi^{\mu\lambda_{\log}} = \sqrt{2}$ ties the binary register and $\varphi^{\mu\log_\varphi 3} = \sqrt{3}$ the ternary one, with $\mu = \frac{1}{2}$ as mediator — distinguishing PCF from the purely binary $\mathbb{Z}_2$ chain of HS.

Thread (amplitudes and loops from π and φ via the tower).

1. *tree (Veneziano):* in (3.12) the Γ factors carry the π/binary side ($\Gamma(\frac{1}{2}) = \sqrt{\pi}$) and the poles at $\alpha' s = n$ are the Regge tower (the φ/ternary side);
2. the Regge tower *is* the Euler product (3.13), linking amplitude and primes;
3. *loops:* the one-loop partition (3.14) is finite; the colour/generation hierarchy ($\zeta(3)$, odd zeta over the tower) is a structural assignment, not derived here;
4. the modes at each level are $N_{\mathrm{modes}}(\sigma) = \lfloor \pi\varphi^\sigma \rfloor$ (3.10).

(Verified in `CW3_verify_paper_physics.py`: $\Gamma(\frac{1}{2})^2 = \pi$, Regge poles, $\zeta(2)$ = Euler product, $Z_{\mathrm{PCF}}(i)$, N_{modes}.)
Thread (the two UV regulators coexist).

1. *modular/η:* the partition (3.14) is finite because $\eta(i)$ is well defined — modular invariance regulates the ultraviolet on the torus, with no divergence;
2. *dimensional/$\Gamma(\varepsilon/2)$:* (3.15) extracts the $2/\varepsilon$ pole at the local vertex;
3. they are complementary: η handles the worldsheet (global, finite partition), $\Gamma(\varepsilon/2)$ the vertex (local pole); both give the same finite physics.

3.4. **All AdS/CFT from a single microstate, at each level.** The whole AdS/CFT correspondence follows from one microstate (the object of §3.1), not from an ensemble — and it holds at each level of the tower. This per-level statement is the dependency the web (§3.5) rests on. The single microstate is the object Ω with $|\Omega| = \frac{1}{2}$ (its realisation as a self-adjoint operator is built in [42]); being one state and not an ensemble, it resolves the firewall ambiguity (AMPS) without averaging. The throat and its warped metric are

$$z(\sigma) = \varphi^\sigma, \qquad ds^2 = e^{2A(z)}\,\eta_{\mu\nu}\,dx^\mu dx^\nu + dz^2. \tag{3.16}$$

The five-dimensional throat: AdS_5, S^5 and the ladder. The throat (3.16) is the AdS_5 geometry whose curvature invariants are fixed by the modulus: $R = -20$, $\Lambda_5 = -6$, $G_{\mu\nu} = 6$, saturating the Breitenlohner–Freedman bound (Fig. 3) [R_scalar_AdS5, Lambda5_value, G_munu_AdS5, BF_bound_AdS5]. Its companion five-sphere carries the Hopf chain $S^1 \hookrightarrow S^5 \to \mathbb{CP}^2$ with $\chi(\mathbb{CP}^2) = 3 = n$ [hopf_latitude, hopf_from_clifford], the same arity-3 of §2.2; the five-dimensional case is precisely the one requiring the Gaussian and Eisenstein primes (§1). The discrete levels $\sigma = 0,\ldots,5$ (superpoint $\to$ NS5, §1.5.2) sit on the Hamming hypercube

$H_5 = \{0,1\}^5$, $|H_5| = 2^5$ `[hypercube_card]`, with $\Lambda_\sigma = \varphi^\sigma \Lambda_{\mathrm{PCF}}$ and the saturation entropy $S(\sigma) = \pi\varphi^\sigma$ (3.10).

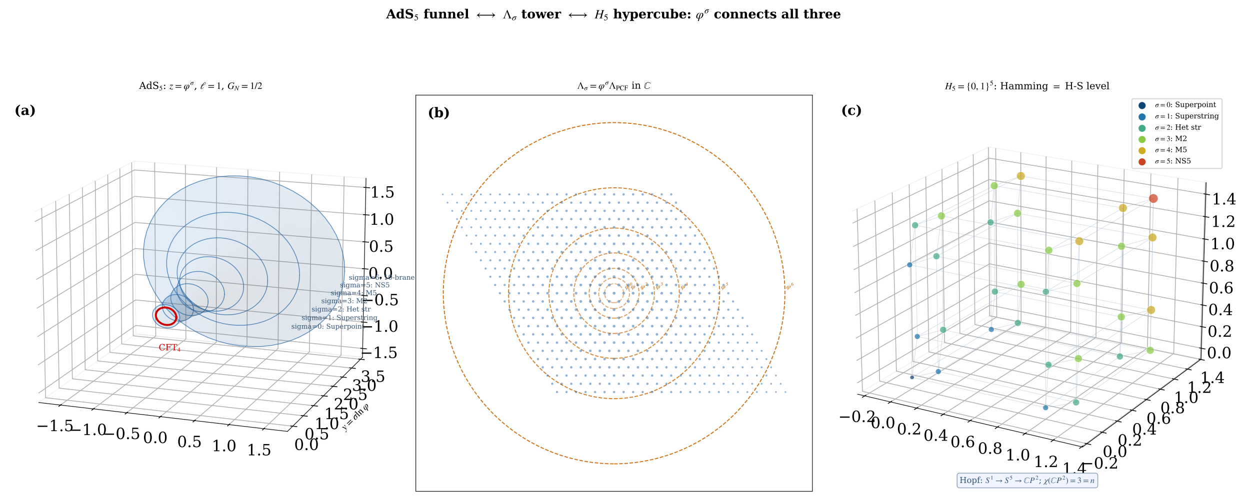

FIGURE 3. AdS$_5$ funnel $z = \varphi^\sigma$ (3.16) $\leftrightarrow$ the lattice tower Λ_σ $\leftrightarrow$ the Hamming hypercube H_5 of the M-theory ladder (superpoint $\to$ NS5); φ^σ connects all three.

The bulk–boundary map is an isometry [42] and obeys the GKPW dictionary [17, 50, 51],

$$V : \mathcal{H}_{\mathrm{bulk}} \to \mathcal{H}_\partial, \qquad V^\dagger V = \mathbf{1}, \qquad Z_{\mathrm{grav}}[\phi|_\partial] = \left\langle e^{\int_\partial \phi_0 \mathcal{O}} \right\rangle_{\mathrm{CFT}}. \tag{3.17}$$

The isometry as an algebra–isometry (bulk–boundary) duality. The embedding V (3.17) is realised on the two faces of one modulus. The eigenvalue triangle $\lambda_k = \frac{1}{2}\omega^k$ ($\omega = e^{2\pi i/3}$) of Prop. 2.12 (2.7) is the Eisenstein lattice $\mathbb{Z}[\omega]$ (*algebra* face: 120°, S_3 symmetry, A_2 root system $= \mathfrak{su}(3)$) `[eisenstein_omega, Omega_eigenvalues, A2_root_normalized]`; the same modulus on the i/π side, where $(\frac{1}{2})! = \frac{1}{2}\sqrt{\pi}$ fixes $\mu = \frac{1}{2}$ (§2.3), is the Gauss lattice $\mathbb{Z}[i]$ (*isometry* face: $\tau = i$, $\ell = 1$). The φ-mediation $S_3 \to \mathbb{Z}_4$ between them preserves $|\Omega| = \frac{1}{2}$ `[holographic_area]` and fixes $\tau = i$: this is $V^\dagger V = \mathbf{1}$ read on the lattice — bulk (geometry, $\mathbb{Z}[i]$) and boundary (algebra, $\mathbb{Z}[\omega]$) are two readings of one triangle, neither privileged, exactly as (3.17) requires.

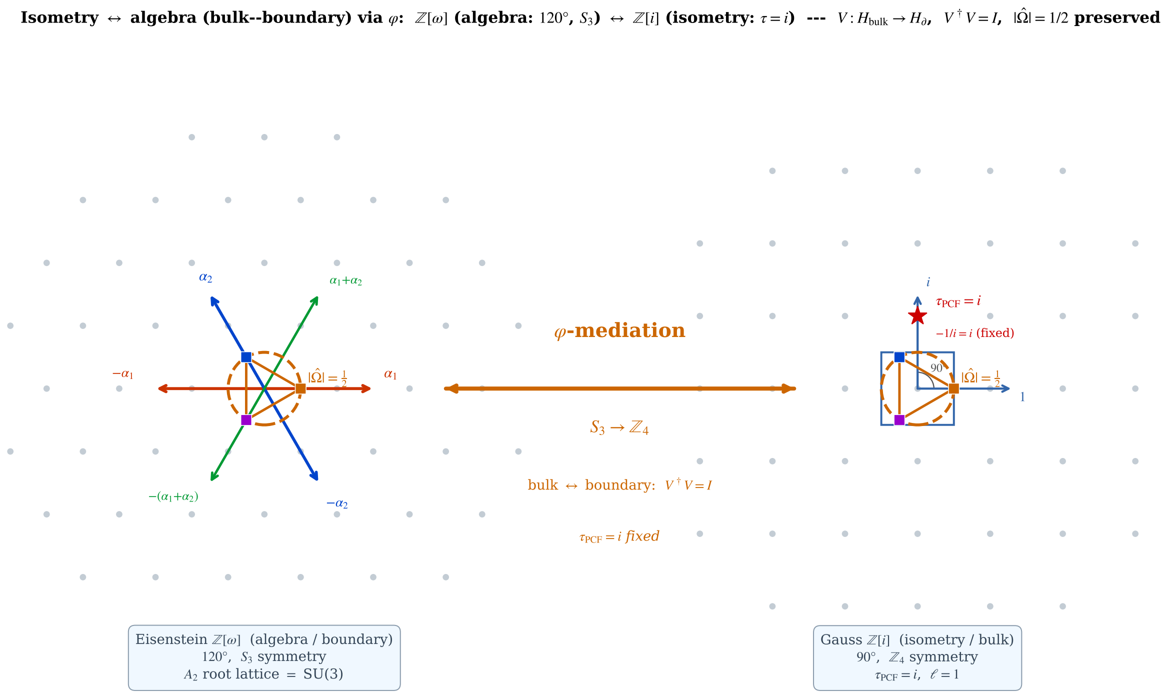

FIGURE 4. Isometry ↔ algebra (bulk–boundary): the Eisenstein lattice $\mathbb{Z}[\omega]$ ($A_2 = \mathfrak{su}(3)$, algebra/boundary) and the Gauss lattice $\mathbb{Z}[i]$ ($\tau = i$, isometry/bulk) are φ-mediated ($S_3 \to \mathbb{Z}_4$) with $|\Omega| = \frac{1}{2}$ preserved — the lattice realisation of $V : \mathcal{H}_{\text{bulk}} \to \mathcal{H}_\partial$, $V^\dagger V = \mathbf{1}$ (3.17).

Standard AdS/CFT integrates out the radial coordinate z (it privileges a boundary); here the isometry singles out neither side, so PCF keeps both bulk and boundary ((3.17); "forget z" only on the Maldacena side). The modulus is scale invariant, so the holographic relation holds at every level:

$$|\Omega|_\sigma = \tfrac{1}{2} \qquad \forall\,\sigma. \tag{3.18}$$

The gravitational data follow from the same $\mu = \frac{1}{2}$:

$$S_{\text{BH}} = \frac{1}{4G_N} = \mu = \tfrac{1}{2}, \qquad c = \frac{3\ell}{2G_N} = 3 \quad (\ell = 1,\ G_N = \tfrac{1}{2}). \tag{3.19}$$

Remark 3.5 (The role of φ in AdS/CFT)**.** Everything follows from the microstate Ω (§3.1) and the tower (§3.3); the modulus is $\mu = \frac{1}{2}$ at every level, and the AdS/CFT signature ($\ell{=}1, G_N{=}\frac{1}{2}, c{=}3, \text{GKP}{=}\frac{3}{4}$) is exactly the microstate's invariants.

Thread (all AdS/CFT from one microstate).

(1) the isometry V with $V^\dagger V = \mathbf{1}$ embeds bulk into boundary preserving the norm, so the duality holds from *one* state, without averaging ((3.2)→(3.17));
(2) the modulus is scale invariant, $|\Omega|_\sigma = \frac{1}{2}\ \forall\sigma$, so the holographic relation holds at *each* level of the throat, not only at the boundary (3.18);
(3) from the same $\mu = \frac{1}{2}$: $S_{\text{BH}} = 1/(4G_N) = \frac{1}{2}$ (area law) and $c = 3$ (Brown–Henneaux) — the gravity-side entropy and central charge come from the microstate invariant (3.19);
(4) hence GKPW [50, 51] and Ryu–Takayanagi [52] are consequences of the isometric embedding of the microstate.

Thus the entire AdS/CFT correspondence comes from one microstate, not an ensemble. (Verified in `CW3_verify_paper_physics.py`: $|\Omega|_\sigma = \frac{1}{2}\ \forall\sigma$, GKP $= \frac{3}{4}$, $S_{\mathrm{BH}} = \mu$, $c = 3$.)

Maldacena left the AdS boundary conditions unspecified—"we leave this interesting problem for the future" [17]. Here they are determined: the modular parameter is fixed at $\tau = i$, the fixed point of S-duality `[s_duality_fixes_i]`, and the scale by the certainty principle $\varepsilon_0 M_{\mathrm{PCF}} = \pi$ `[certainty_principle]`.

3.5. **The M-theory type duality web from the microstate.** The M-theory type duality web projects from the microstate (§3.1) through the φ^σ tower (§3.3); Witten's eight/eleven-dimensional object is not required here. Witten *posits* that object and *conjectures* that the web follows from it [53]; PCF starts from a *demonstrated* microstate and *generates* the tower, of which the web is a projection. The web follows from the tower with HS (§3.3) and from per-level AdS/CFT (§3.4), both already stated. The shared signature, derived in App. A.4 from the web's two corners, is that the M-theory object and AdS/CFT are projections of the microstate, carrying

$$\underbrace{|\Omega| = \tfrac{1}{2},\ |\Omega|^2 = \tfrac{1}{4}}_{\text{Huerta–Schreiber (M-theory)}} \qquad \underbrace{\ell = 1,\ G_N = \tfrac{1}{2},\ c = 3,\ \mathrm{GKP} = \tfrac{3}{4}}_{\text{Maldacena (AdS/CFT)}}, \tag{3.20}$$

`[GN_shared, GKP_shared, HS_modulus_shared]` which equal the microstate invariants $\mu = \frac{1}{2}$. That the two corners are one and the same object is Maldacena's conjecture that "M/string theory on various Anti-deSitter spacetimes is dual to various conformal field theories" [17]—the definition of M-theory he anticipated extending to five non-compact dimensions: here it is realised, both corners being projections of the single microstate $|\Omega| = \frac{1}{2}$. The torus on which they meet is the noncommutative torus that Connes, Douglas and Schwarz identify with M-theory compactification [24] and that Manin attaches to the $\mathbb{F}_1$ program [23]; the non-orientable fibre it carries is independent of that algebra (Prop. 2.17). The dualities that engage the tower are

$$\text{(S-duality)} \quad \tau \to -\tfrac{1}{\tau} \iff \varphi \to -\tfrac{1}{\varphi} = \bar{\varphi}, \tag{3.21}$$

$$\text{(T-duality)} \quad R \to \alpha'/R, \tag{3.22}$$

$$\text{(IIA strong coupling)} \quad \sigma = 10 \to \sigma = 11 \quad \text{(M-theory)}, \tag{3.23}$$

$$\text{(ER = EPR)} \quad \text{entanglement} \leftrightarrow \text{geometry}, \tag{3.24}$$

$$\text{(de Sitter swampland)} \quad \frac{|\partial_\sigma V|}{V} = \ln\varphi, \tag{3.25}$$

$$\text{(asymptotic safety)} \quad \text{UV fixed point}. \tag{3.26}$$

Remark 3.6 (The IIA→11D bridge as Witten's Kaluza–Klein tower)**.** The bridge $\sigma = 10 \to \sigma = 11$ in (3.23) corresponds to Witten's Kaluza–Klein mechanism: at Type IIA strong coupling the tower of states are the Kaluza–Klein modes of eleven-dimensional supergravity on $\mathbb{R}^{10} \times S^1$, of mass $\sim 1/r$ [53]. The φ^σ tower plays that role here. The golden Kaluza–Klein structure ($\varphi^2 + \varphi^{-2} - 2 = 1$; $G_5 \to G_4 = \frac{1}{4}$; the Breitenlohner–Freedman comparison; discrete-spectrum stability) is proved in Appendix A.7.

Remark 3.7 (The role of φ in the duality web)**.** The dualities are projections of the microstate's φ^σ tower, not of a posited object; the modulus is $\mu = \frac{1}{2}$ at every level, and S-duality is the Galois conjugation $\varphi \to -1/\varphi$.

Remark 3.8 (the web is reconstructed from its two corners)**.** The web this paper uses has two corners only—Huerta–Schreiber M-theory and Maldacena AdS/CFT—and both are shown in Appendix A.4 to carry the microstate modulus $|\Omega| = \frac{1}{2}$; the dualities are the symmetries of the φ^σ tower that fix it. The wider string-theory content (U-duality, E_8, the five string theories) is neither used nor needed here.

Thread (microstate → tower → web).

(1) the dualities are written in σ/φ, as projections of the tower: S as $\varphi \to -1/\varphi$ (Galois conjugate), T as $R \to \alpha'/R$, IIA→11D as $\sigma = 10 \to 11$, ER=EPR, and the de Sitter swampland $|\partial_\sigma V|/V = \ln\varphi$ ((3.21)–(3.25));
(2) the shared signature (3.20): the colimit proves the M-theory object (modulus $\frac{1}{2}$) and AdS/CFT ($\ell{=}1, G_N{=}\frac{1}{2}, c{=}3, \mathrm{GKP}{=}\frac{3}{4}$) are projections with the microstate invariants ($\mu = \frac{1}{2}$);
(3) hence the web is reconstructed from its two corners (App. A.4): both carry $\mu = \frac{1}{2}$, and the dualities fix the modulus; Witten's eight-dimensional object is not required.

Thus microstate → tower → web (HS and per-level AdS/CFT). The quantitative pieces ($\varphi \to -1/\varphi$, the swampland constant $\ln\varphi$, the T-self-dual radius, and the shared HS/Maldacena signature) are checked numerically in `CW3_verify_paper_physics.py` and formally in `CW3_lean.lean` (§2.10).

The remaining duality, ER = EPR, is where the microstate supplies what Cool Horizons left open. Maldacena and Susskind observe that the ER bridge "may be a highly quantum object that is yet to be independently defined", and that they "do not have an independent argument for its smoothness" [43]. In PCF the bridge is both defined and analytic: it is the cocycle $T(\sigma_1, \sigma_2)$ of the proposition below, smooth in the analytic sense because its denominator $1 + \varepsilon_0\varphi^\sigma$ is strictly positive at every level, carried, with its geometry supplied by the non-orientable fibre, on the M-theory noncommutative torus rather than on a general quantum object.

Proposition 3.9 (ER = EPR from the modulus)**.** *The entanglement of the two hemispheres and the geometric bridge between them are one invariant,* $\mu_3^2 = \frac{1}{4}$*.* (EPR.) *Since* $|\Omega(\tau)| = \frac{1}{2}$ *for every* τ (3.2)*, the two-point entanglement modulus is*

$$|\Omega(\tau_A)\,\overline{\Omega(\tau_B)}| = |\Omega(\tau_A)|\,|\Omega(\tau_B)| = \mu_3^2 = \tfrac{1}{4} \qquad (\forall\,\tau_A, \tau_B),$$

which is the holographic area factor $\mu_3^2 = \frac{1}{4}$*.* (ER.) *The bridge amplitude between two levels of the throat,* $T(\sigma_1,\sigma_2) = \dfrac{1+\varepsilon_0\varphi^{\sigma_1}}{1+\varepsilon_0\varphi^{\sigma_2}}$*, satisfies* $T(\sigma_1,\sigma_2)\,T(\sigma_2,\sigma_1) = 1$ *and* $T(\sigma_1,\sigma_2)\,T(\sigma_2,\sigma_3) = T(\sigma_1,\sigma_3)$*, so the bridges form a group: a round trip is the identity and composition is additive in* σ*.* (ER = EPR.) *Both are fixed by the one scale-independent modulus* $\mu_3 = \frac{1}{2}$ *(*$|\Omega|_\sigma = \frac{1}{2}\ \forall\sigma$*, (3.18)): the entanglement between* P *(south) and* F *(north), mediated by* C*, is the bridge that joins them through the equatorial axis. Entanglement is geometry, derived from the modulus, not posited.* *[bridge_cocycle, bridge_inverse, bridge_compose]*

The bridge and its $\frac{1}{4}$ identity are shown in Fig. 5.

Proof. (EPR) $\Omega(\tau) = \frac{1}{2}e^{i\tau\ln\varphi}$ has $|\Omega(\tau)| = \frac{1}{2} = \mu_3$ for all τ (3.2); the product rule for moduli gives $|\Omega(\tau_A)\overline{\Omega(\tau_B)}| = \mu_3^2 = \frac{1}{4}$, independent of τ_A, τ_B because μ_3 comes from the S_3 arity (Proposition 2.12), not from the phase. *(ER)* $T(\sigma_2,\sigma_1)$ is the reciprocal of $T(\sigma_1,\sigma_2)$, whence $T(\sigma_1,\sigma_2)T(\sigma_2,\sigma_1) = 1$; the telescoping of numerator and denominator gives $T(\sigma_1,\sigma_2)T(\sigma_2,\sigma_3) = T(\sigma_1,\sigma_3)$. *(*ER = EPR*)* The levels are equivalent because $\varepsilon_0\varphi^\sigma \cdot M_{\mathrm{PCF}}\varphi^{-\sigma} = \pi$ at every σ (2.11), so $|\Omega|_\sigma = \frac{1}{2}$ is scale-invariant (3.18); the entanglement invariant μ_3^2 and the bridge group thus share one origin. ∎

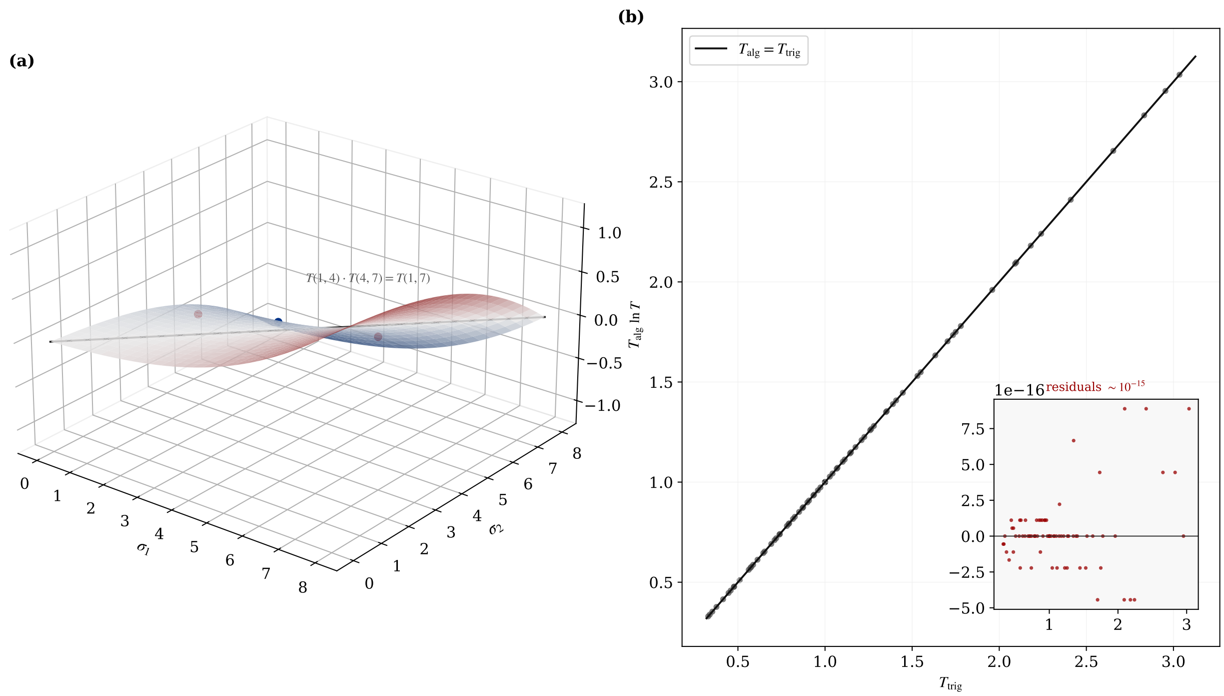

FIGURE 5. ER = EPR from the modulus (Prop. 3.9): (a) the bridge amplitude $T(\sigma_1, \sigma_2)$ as a cocycle, $T(1,4)\,T(4,7) = T(1,7)$; (b) the algebraic and trigonometric forms agree, $T_{\rm alg} = T_{\rm trig}$, to machine precision.

3.6. **The demonstrative line (19-step spine).** Each step uses only previous steps.

§2→§3.: P1 $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (3.1) (§2.1; seed).
3.1 object.: P2 $\Omega, |\Omega| = \frac{1}{2}$ (3.2) (§2.3) → P3 Gaussian, $\mu = \frac{1}{2}$ (3.3) (§2.1) → P4 worldsheet/Polyakov (3.4).
3.2 meta-invariant.: P5 invariants $d{=}3, \mu{=}\frac{1}{2}, \sigma{=}\frac{3}{2}$ (3.6) (§2.2, §2.9) → P6 $1/(4G_N) = \frac{1}{2} = |\Omega|$, $H(\frac{1}{2}) = 1$ (3.7) (§2.3).
3.3 tower + HS.: P7 chain $i \to \varphi$ (3.8) (§2.6) → P8 $\sqrt{2}/\sqrt{3}$ by μ (3.9) (§2.1) → P9 tower φ^σ, $N_{\rm modes}$ (3.10) (§2.7) → P10 HS↔tower (3.11) → P11 Veneziano (3.12) → P12 loops $Z_{\rm PCF}(i)$ (3.14) → P13 both UV regulators (3.15).
3.4 AdS/CFT per level.: P14 throat $z(\sigma) = \varphi^\sigma$ (3.16) (§2.7) → P15 isometry $V^\dagger V = \mathbf{1}$, GKPW (3.17) → P16 $|\Omega|_\sigma = \frac{1}{2}\ \forall\sigma$ (3.18) (§2.3) → P17 $S_{\rm BH} = \mu$, $c = 3$ (3.19) (§2.9).
3.5 web (final).: P18 shared signature (3.20) → P19 web (3.21)–(3.26).

The observer steps that close the introduction's problem (the objectivity threshold $f_{\rm crit} = \frac{1}{2} = \mu$ and the Cramér–Rao saturation $F^{-1}_{\rm max} = \frac{1}{4} = \mu^2$) open §4 Implications. Here P18 is what makes P19 (the web) defensible without re-deriving M-theory or positing Witten's object: §3 closes with its demonstrative line complete (Fig. 6).

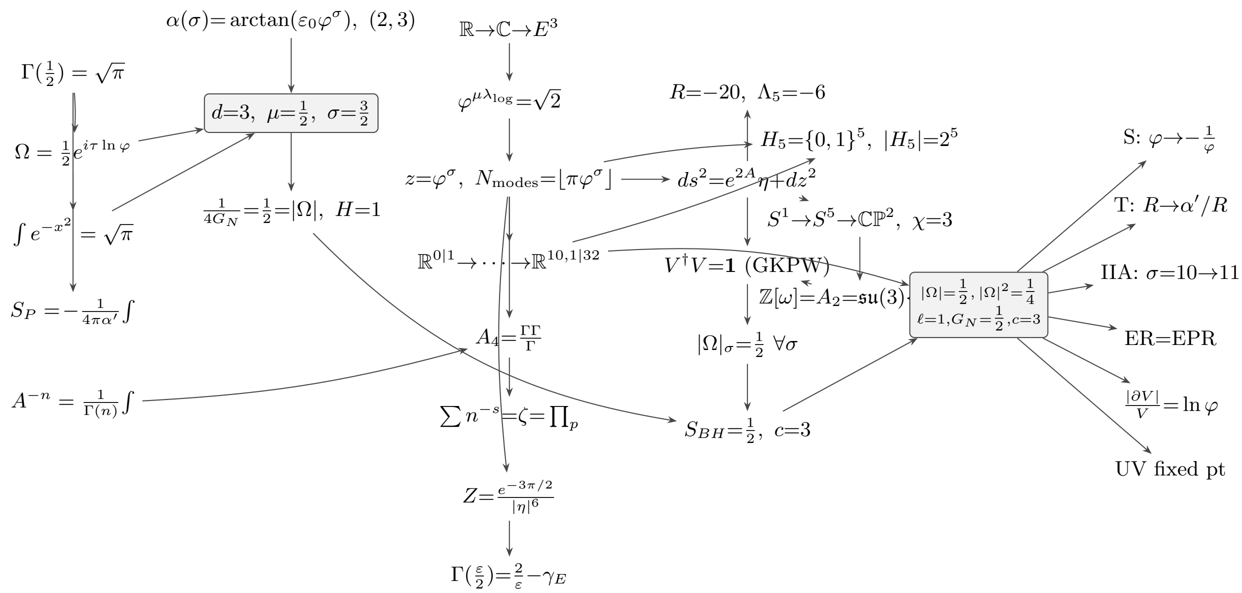

FIGURE 6. Commutative spine of §3 (P1→P19): the same microstate Ω, $|\Omega|=\frac{1}{2}$ propagates through the object, the meta-invariant, the tower and amplitudes, AdS/CFT, and the duality web — every arrow uses only earlier nodes. Mathematical, physical and conceptual layers coincide on one diagram.

4. Implications

Sections 2 and 3 derived AdS/CFT from a single microstate, with no ensemble average. The present section shows that this apparatus meets the criteria for holography in de Sitter space already fixed in the literature. The aim is not to propose new physics: the mechanisms it uses—the tensors of general relativity, the Polyakov worldsheet, Kaluza–Klein compactification, the warped throat, Jacobson's horizon thermodynamics—are established results. What is shown is that these principles already coincide with the microstate; much of the section is therefore a review of the known principle followed by how the object realises it.

4.1. **What the section does, and against what it is measured.** Holography in de Sitter, as Witten poses it [1], requires that general relativity be embedded in a more complete theory that determines the cosmological constant Λ, and in which the observer is described by the theory rather than introduced from outside—because in a cosmology with $\Lambda > 0$ there is no asymptotic region to which an external observer may retreat.

The yardstick is the classification of von Neumann factors into three types—used in this context by Chandrasekaran, Longo, Penington and Witten—three modes of existence of the same field. *Type I* is that of discrete objects—particle, path integral, string counted one by one, operator of definite spectrum—with minimal projections. *Type II* is that of observation—the trace, the crossed product of the algebra by its modular flow—where observing turns an algebra without trace into one with a trace. *Type III* is that of the local field—the wedge, the horizon, the worldsheet, the spacetime—without normal trace or projections: non-locality.

Of these types, what the section *proves* is the signature of each, not the technical classification: type I as discrete spectrum (`[eigenvalues_modulus_half, witten_finite_rank]`); type II as an invariant the flow preserves—the trace in concrete form (`[omega_flow_invariant, tower_scale_group]`); type III as the non-factorisation of the horizon into interior $\otimes$ exterior (Proposition 4.1). The technical names (II_1, III_1) and the crossed product are taken from CLPW by citation; the signatures are proved here. Signature first, name second.

The criteria come from three distinct sources, which should not be conflated. From **Witten 2001** (*Quantum Gravity in De Sitter Space*, W01) come the determination of Λ, the finiteness of the Hilbert space with $S = \ln N$, the isometry-group obstruction, the pairing $\langle f|i\rangle$, the CPT map, the two conjectures—of unitarity and of entropy—and the antipodal singularity. From **Chandrasekaran, Longo, Penington and Witten 2022** (CLPW) come the observer described by the theory, the maximally mixed vacuum state, and the construction of the type II algebra by crossed product from the type III. The thermal nature of the static patch predates both: it is the result of **Gibbons–Hawking**. Throughout, each criterion is tagged with which of the three sources states it.

The thesis is that the framework realises the three types on a single object, the microstate Ω of modulus $|\Omega| = \frac{1}{2}$, and thereby satisfies the requirements Witten left open. The three types are not a label per subsection but three threads crossing the whole section: type I runs through the particle (§4.2), its path integral, and the discrete string of the tower (§4.4)—to it belongs the Hermitian operator, referred to the companion paper and not proved here; type II runs through the ER=EPR bridge (§4.2), observation (§4.3) and the tracial flow (§4.4); type III runs through the worldsheet (§4.2) and the spacetime with gravity (§4.4). The modulus $\frac{1}{2}$, the projection Π, the entropy S, the centre C and the conjugation Θ recur across movements; each recurrence declares its register and its anchor. The repetition is the weave of a multi-thread demonstration.

4.2. **The non-local field and the particle that emerges from it.**

The worldsheet, the type III field, and its link to gravity. General relativity requires that the laws not depend on coordinates; the tensor guarantees this, because a tensor equation true in one frame is true in all. In string theory this invariance lives on the worldsheet: the string sweeps a surface whose dynamics is the Polyakov action, written on the worldsheet metric h_{ab}—a tensor—and invariant under diffeomorphisms and Weyl rescalings; its constant is the tension $T = 1/2\pi\alpha'$. The worldsheet is the type III field of the section: a local wedge without normal trace, and it is the same field that will reappear as the curved spacetime of §4.4—the type III face that opens and closes §4. Its non-locality is geometric, not postulated. The worldline $\Omega(\tau) = \frac{1}{2}e^{i\tau \ln \varphi}$, promoted by Polyakov, closes onto the torus $T^2_{\mathrm{PCF}} = \mathbb{C}/\Lambda$ with modular parameter $\tau = i$ fixed—the self-dual point of S-duality, not chosen—and φ generates the lattice. A torus is compact and boundaryless, and a boundaryless domain admits no question of what happens at the edge of one region against another, because there is no edge at which to separate (`[fibre_independent_of_base]`). By the modulus $\frac{1}{2}$, the loop around the torus accumulates a half rotation, $e^{i\pi} = -1$: the sheet is a Möbius band, non-orientable, with no privileged interior side. The particle is the bulk–boundary interface this field carries:

$$\Pi\colon \mathbb{E}^3 \longrightarrow \mathbb{C}, \qquad d_H = \frac{\log 3}{\log 2} \approx 1.585, \qquad 2^{d_H} = 3. \tag{4.1}$$

The horizon of Π does not factorise into interior $\otimes$ exterior, which is the type III signature and the reason there is no firewall:

Proposition 4.1 (Holographic–relativistic horizon (no firewall))**.** *The projection* $\Pi\colon \mathbb{E}^3 \twoheadrightarrow \mathbb{C}$ *creates a horizon that is at once* holographic *(information fully encoded on the boundary) and* relativistic *(the Galois conjugation* $\varphi \leftrightarrow \bar\varphi$ *exchanges the two sides like the two wedges of a Rindler spacetime):* Π *is surjective, with* $\ker \Pi = \{(0,0,0)\}$ *and* $\Pi^{-1}(p) \neq \varnothing$ *for every* $p \in \mathbb{C}$*. No averaging over microstates is needed: the microstate is one state and not an ensemble, as established in §3.4 (its realisation as a self-adjoint operator being built in the companion paper [42])—so the AMPS firewall ambiguity [54], which requires an ensemble average, does not arise.*

Three independent facts confirm the non-separability, and all three fall on $\frac{1}{2}$. At every scale, $|\Omega|_\sigma = \frac{1}{2}\ \forall\sigma$ (`[omega_flow_invariant]`). At every separation, the entanglement is $|\Omega(\tau_A)\bar\Omega(\tau_B)| = \mu_3^2 = \frac{1}{4}$, and this ER=EPR bridge is derived, not postulated: the operators $T(\sigma_1, \sigma_2)$ form a group (Proposition 3.9; `[bridge_cocycle, bridge_inverse, bridge_compose]`).

The conjugation $\Theta = \mathrm{CPT}$, the antilinear Galois map $\varphi \leftrightarrow \bar{\varphi}$, swaps the wedges $P \leftrightarrow F$ and fixes C.

W6 — the antilinear Θ/CPT map (W01). Witten defines the Hilbert space non-perturbatively through an antilinear map Θ (a CPT operation) that renders the bilinear pairing Hermitian (W01, eq. 18). It is the Galois conjugation $\varphi \leftrightarrow \bar{\varphi}$: Θ swaps the two Rindler wedges and fixes the centre, with $\Theta^2 = 1$. *Proof:* `[cpt_swaps_P_to_F, cpt_fixes_C, cpt_involution]`.

W7 — the $\mathrm{SO}(1, n-1)$ *obstruction (W01).* With N finite the de Sitter isometry group cannot act on the Hilbert space; being non-compact it has no finite-dimensional unitary representations (W01). Its place is taken by the discrete $\mathbb{Z}_3$ symmetry of the cube roots of unity with the scale invariance $|\Omega|_\sigma = \frac{1}{2}$. *Proof:* `[eisenstein_cube, omega_conj]`; [1].

The whole geometry as a tensor: the warp generates curvature. This spacetime is tensorial in its entirety. The throat that compactifies the scale has a warped metric with radial coordinate $z(\sigma) = \varphi^\sigma$ and non-vanishing warp gradient: with $A = -\log z$, so $A' = -1$, the standard machinery—from g to Christoffel, to Riemann, to Ricci—turns that gradient into curvature computed, not assumed: $R_{\mu\nu} = -(A'' + d\,A'^2)\, g_{\mu\nu} = -4\, g_{\mu\nu}$, so the throat is an Einstein space (§A.1; `[R_scalar_AdS5, R_munu_AdS5]`). The densification of the warp *is* the Ricci tensor. The extrinsic side closes it: the constant-σ leaves, their embedding and second fundamental form are tied to the intrinsic curvature by Gauss's equation—de Sitter is the hyperboloid whose intrinsic curvature is the wedge of the extrinsic curvature of the embedding, $R_{\mu\nu} = 3H^2 g_{\mu\nu}$, developed in full in §A.2 (`[dS_ricci_from_gauss]`).

From that same geometry the particle emerges, discrete (type I).. The warp is sustained by a stress–energy tensor: Einstein's equation with source balances the throat geometry against a matter tensor whose spatial part is pressure. The tensor that in gravity describes the metric is the same that describes the tension—both words share the root *tendere*—and it is that tension that sustains the throat. Geometrically the throat is a densifying centre: on the Riemann sphere, with zero at one pole and infinity at the other, the golden tower produces concentric shells growing outward and a centre compressed without bottom, since $e^{2A} = z^{-2} = \varphi^{-2\sigma}$ diverges as $z \to 0$. The centre is superpoint $\mathbb{R}^{0|1}$-like, the object without metric or spin structure, with a single odd direction (§1.5.2). Its $\mathbb{Z}_2 = \{0, 1\}$ grading distinguishes the even, bosonic directions (the "0") from the single odd, fermionic one (the "1"); the centre of the tension is that odd direction: fermionic. The involution $\alpha(x) = 1 - x$ swaps 0 and 1 and has no fixed point between them—which makes the binary register self-contradictory—but a unique fixed point at $x = \frac{1}{2}$: the extremes $\{0, 1\}$ are the binary register—the bit, the boson, the fermion—and the self-conjugate midpoint $\frac{1}{2}$ is at once the modulus $|\Omega|$ and the fermion's *spin* $\frac{1}{2}$. The particle is thus coded by three readings of the same fact: $\frac{1}{2}$ (modulus, spin, fixed point of $1 - x$); the $0|1$ of the fermionic superpoint; and the tension T_{AB} that produced it.

From this object follows its one-dimensional description: the worldline and its path integral, the Gaussian normalised by the half-factorial, $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$, $(\frac{1}{2})! = \Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$ (`[M2_gamma_half]`), computed where it is stated. The particle is then a type I operator: real discrete spectrum, minimal projections, integer dimension. Its internal structure is the spin-star,

$$\underbrace{S + E_1 + \cdots + E_N}_{\text{central+environment}} \xrightarrow{N=2} \underbrace{C + P + F}_{\text{PCF}}, \qquad F_{\max} = N^2 = 4. \tag{4.2}$$

with arithmetic $1 + 2 = 3$: the binary fermion, mediated by $\frac{1}{2}$, has become the tripartite particle. And the geometry that generated the particle fixes its charges: the eigenvalue triangle is the Eisenstein lattice $\mathbb{Z}[\omega] = A_2 = \mathfrak{su}(3)$; the Clifford latitude gives $SU(2)$; the Hopf fibre, $U(1)$; the weak mixing angle is a ratio of tower entropies, $\sin^2\theta_W = S(3)/S(6) = \varphi^{-3} \approx 0.236$ against 0.231 measured (§A.3; `[gauge_dim_su3, hopf_from_clifford, weinberg_ratio]`).

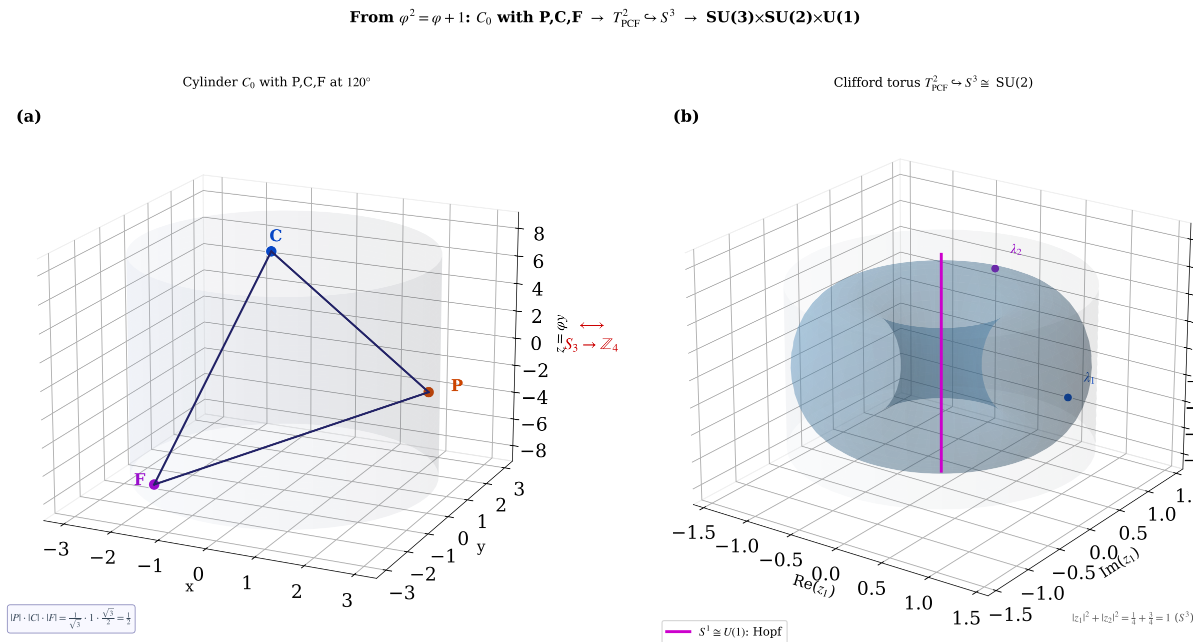

FIGURE 7. From $\varphi^2 = \varphi+1$: the cylinder C_0 with P, C, F at 120° ($\|P\|\,\|C\|\,\|F\| = \frac{1}{2}$) closes to the torus $T^2_{\mathrm{PCF}} = \mathbb{C}/\Lambda$, $\tau = i$, whose Clifford embedding $T^2 \hookrightarrow S^3$ and Hopf fibre give the gauge factors $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ of App. A.3.

Geometry and particle are the same object. The order of these facts is fixed by the diagram. The projection $\Pi\colon \mathbb{E}^3 \to \mathbb{C}$ is the *root* of the whole downstream chain, and it is at once the geometry of spacetime—the sheet and torus projected—and the particle. This reflects the *self-similarity* of the object: the same modulus $\frac{1}{2}$ at every scale level, seeded by $\varphi^2 = \varphi + 1$ extending $i^2 = -1$ from the plane to space. The particle carries the same geometry in its information and, as it accumulates entropy, keeps it and moves within the spacetime geometry it itself is. There is no stage with geometry and a later one with the particle: they are the same object at two scale readings. It is this accumulation of information, carried by the self-similar geometry, that the next movement reads as an observer (§4.3): container and content, bulk and boundary, geometry and particle are not separate—the same non-separation the Möbius band already fixed.

W1 — the observer described by the theory (CLPW). CLPW require (§2.5) that the observer be described by the theory, not introduced from outside. Π is internal—no external region from which to look—and self-similar, so the observer is the geometry itself read at its centre. *Proof:* (4.1) with the bulk–boundary isometry of §3; Proposition 4.1; `[cw3_observer_items_core]`; [2].

W2 — finite-dimensional Hilbert space (W01). The finite horizon area requires finite dimension; here the spectrum is the triangle of three eigenvalues and the mode count is integer. *Proof:* `[witten_finite_rank]`; (4.2).

4.3. **The particle as observer.** What turns the particle into an observer is the accumulation of information, described here by an internal Wick rotation, not imposed. When the centre is taken as the time axis, its square changes sign, $(i\,t)^2 = -t^2$ (`[wick_squares_flip_sign, lorentzian_signature_sum]`), and past and future become the two light cones, with the apex—the present—as observer (`[C_fixed_PF_rotate, causal_trichotomy]`). The full metric is formalised in §4.4; here the algebra of the rotation, which is self-contained, suffices. The observer accumulates:

its Fisher information grows to the ceiling the Cramér–Rao bound permits,

$$\tau_F = \frac{\tau_D}{\sqrt{2f}}, \qquad \tau_F = \tau_D \iff f = \tfrac{1}{2}. \tag{4.3}$$

$$\mathrm{Var}(\theta) \geq F_\theta^{-1}, \qquad \underbrace{F_{\max}^{-1} = \tfrac{1}{4}}_{\text{Kiely: Cramér–Rao}} \quad = \underbrace{|\Omega|^2 = \mu_3^2 = \tfrac{1}{4}}_{\text{PCF area factor}}. \tag{4.4}$$

$$R_\delta \sim N \quad \text{(all fragments agree)} \longleftrightarrow \text{objective record}, \tag{4.5}$$

$$F_\theta(t) \xrightarrow{N\to\infty} F_{\max}\big(1 - e^{-(t/\tau_F)^2}\big), \quad F_{\max} = 4, \qquad F_\Omega = 4\mu_3^2 = 1 \text{ bit}, \tag{4.6}$$

and sees exactly the half,

$$|P| \cdot |C| \cdot |F| = \tfrac{1}{2} = |\Omega|. \tag{4.7}$$

The "half" is geometric: the eigenvalue dipole places P at the south pole and F at the north, with C the equatorial axis ($|C| = 1$), and $|P|\,|F| = \frac{1}{4}$ while $|C| = 1$ gives the product $\frac{1}{2}$; the $\sqrt{3}$ of $|F|$ cancels that of $|P|$—a cancellation proper to the equilateral triangle, arity three (Proposition 2.12, Proposition 2.10: $\sigma + \mu = 2$, $\sigma\mu = \frac{3}{4}$ with unique solution $\mu = \frac{1}{2}$, $\sigma = \frac{3}{2}$; `[spectral_uniqueness]`).

W5 — the initial–final pairing observable from within (W01). Witten computes the pairing $\langle f|i\rangle$ as calculable but not observable from outside (W01). It becomes the internal pairing $\langle F|P\rangle = \bar{F}P/C$, with $|\langle F|P\rangle| = \frac{1}{2}$, observable because Π is inside and no external region remains. *Proof:* `[pairing_FP_modulus_half, pairing_PF_modulus_half]`.

W3 — the unitarity conjecture (W01). Witten conjectures that the Hermitian pairing is positive definite (W01). The spectrum is positive and $|\Omega| = \frac{1}{2} > \frac{1}{4}$, so the object does not collapse. *Proof:* `[witten_unitarity_realized, density_eigenvalues_positive]`; (4.7).

The coincidence with Kiely's quantum Darwinism is forced, not numerical. The objectivity threshold is $f_{\text{crit}} = \frac{1}{2}$; the redundancy grows as the mode number; the certainty condition is $\varepsilon_0 M = \pi$:

$$\underbrace{f_{\text{crit}} = \tfrac{1}{2}}_{\text{Kiely: objectivity}} \quad = \underbrace{|\Omega| = \mu_3 = \tfrac{1}{2}}_{\text{PCF}}, \tag{4.8}$$

$$\varepsilon_0 \cdot M_{\text{PCF}} = \pi \qquad (\text{cell capacity } = \pi \text{ bits}), \tag{4.9}$$

Proposition 4.2 (No separable parts before observation)**.** *For a pure two-party state the entanglement modulus $|\Omega|$—the largest Schmidt coefficient, i.e. the top eigenvalue of the reduced state—satisfies $|\Omega| = |\Omega|^2$ exactly when the state is a product (separable parts exist) and $|\Omega| > |\Omega|^2$ otherwise. The No-Diagonal fixes $|\widehat{\Omega}| = \frac{1}{2}$ in the balanced (maximally entangled) case (Proposition 2.10), so*

$$|\Omega| = \tfrac{1}{2} \;>\; \tfrac{1}{4} = |\Omega|^2, \qquad H(\tfrac{1}{2}) = 1 \textit{ bit},$$

the strict inequality being the signature of non-factorizability; for a product state it collapses to $|\Omega| = |\Omega|^2 = 1$, $H = 0$. Hence before observation there are no separable subsystems carrying values: what is real is the non-local whole, recovered as $|\widehat{\Omega}| = \frac{1}{2}$. PCF therefore predetermines no parts, so it is neither a local-realist (Bell) nor a parts-realist (Leggett) hidden-variable model, and the no-go theorems of [55] and [56] do not bind it. `[M9_eq_half]`

The No-Diagonal theorem (§2.2) establishes that every self-referential structure without collapse must produce $\mu^2 = \frac{1}{4}$ and the tripartite decomposition, with no other solution—which forces each identification. In the diagram, here the first apex converges: three independent routes—the bridge (§4.2), the triangle norm and the Fisher metric—give the same $\frac{1}{2}$. When three independent chains reach the same number, the diagram commutes at that point.

4.4. The tower of information, and the emergence of curved spacetime.

The tower (types I and II interwoven). The accumulating observer does not select one reality among many: it builds it. There is no landscape of vacua to choose from, no statistical ensemble to average. This homogeneity—the same structure at every point—is what the field equations require: it is the tensor covariance of §4.2 now read as a constructive principle, and its full development comes below. The entropy of one bit per cell, $H(\frac{1}{2}) = 1$ (`[binary_entropy_half]`), is the maximally mixed state—$\rho_{\max} = 1$ in CLPW's language—describing the de Sitter vacuum, and from its overflow the geometry is born.

When the π-bit cell fills, the throat opens:

$$z(\sigma) = \varphi^{\sigma}, \qquad S(\sigma) = \pi\varphi^{\sigma}, \tag{4.10}$$

with $N_{\text{modes}}(\sigma) = \lfloor \pi\varphi^{\sigma} \rfloor$. As it overflows, the throat count is set:

$$\frac{|\partial_\sigma V|}{V} = \ln\varphi = c_{\text{PCF}}. \tag{4.11}$$

The UV fixed point forces the invariants:

$$\beta_g(g^*, \lambda^*) = 0 \iff \varepsilon_0 M_{\text{PCF}} = \pi, \tag{4.12}$$

The quantisation is in powers of φ, and the identity that closes it is $\varphi^{-\lambda} = \frac{1}{2} \Leftrightarrow \varphi^{\lambda} = 2$: the binary base comes out of the golden geometry. Here the string reappears, not as a new object but as the same particle in its information register: the dimension of the state space at each level is $\dim \mathcal{H}(\sigma) = \lfloor S(\sigma) \rfloor$.

W4 — the entropy conjecture (W01). Witten conjectures that the Hilbert space is finite-dimensional with $S = \ln N$ (W01). The Hilbert dimension per level is the integer part of the horizon entropy, $\dim \mathcal{H}(\sigma) = \lfloor S(\sigma) \rfloor$, with $N_{\text{modes}} = \lfloor \pi\varphi^{\sigma} \rfloor$—the constructive form of Witten's $S = \ln N$. *Proof:* `[witten_entropy_realized]` (by `rfl`—the most direct realisation), `[witten_finite_rank, witten_two_conjectures]`. The entropy branch is asserted strongly (tight); the unitarity branch is "realised".

The flow carrying one level to another, $\sigma \mapsto \sigma + t$, fixes $\tau = i$ and preserves $\frac{1}{2}$: it is the tracial flow, the type II signature, and it is the thermal flow of the static patch at the Gibbons–Hawking temperature (`[tower_scale_group, tower_flow_fixes_tau, omega_flow_group, omega_flow_invariant]`).

W11 — the type II algebra by crossed product (CLPW, completed). CLPW build the type II$_1$ algebra from the type III by the crossed product with the modular flow (§1.2–1.3). The scale flow preserves $\frac{1}{2}$ as a trace, which is the type II signature; the section proves the flow and takes from CLPW the name of the crossed-product classification. *Proof:* `[omega_flow_invariant, tower_scale_group]`; [2].

W12 — the maximally mixed vacuum state (CLPW §2.4, developed). CLPW identify the de Sitter vacuum with the maximally mixed state, $\rho_{\max} = 1$. One bit per cell, $H(\frac{1}{2}) = 1$, gives that $\rho = 1$ of maximal entropy. *Proof:* `[binary_entropy_half]`; [2].

The heart of the movement is that bulk and boundary are the same thing at once. The bulk is the three-dimensional projection; the boundary, the two-dimensional plane; and they are the same II of the first movement. The two accumulations—of time (Fisher) and of scale (tower)—are one:

$$\tau_F(\sigma) = \tau(\sigma) = M_{\text{PCF}}\,\varphi^{-\sigma}, \qquad S(\sigma)\,\tau_F(\sigma) = \pi M_{\text{PCF}} \quad \text{(constant in } \sigma\text{)}. \tag{4.13}$$

$$\underbrace{F_\Omega \cdot N_{\text{modes}}}_{\text{bits (Shannon)}} = \underbrace{S(\sigma) = \pi\varphi^{\sigma}}_{\text{tower}} = \underbrace{N_{\text{modes}} = \lfloor S(\sigma) \rfloor}_{\text{matter}} = \underbrace{S_{\text{BH}} = \frac{A}{4G_N}}_{\text{horizon}} = \underbrace{S_{\text{Jacobson}}}_{\delta Q = T\delta S}, \tag{4.14}$$

This identification—the *weld*—is where four routes converge, and it is the only point where the three types coincide in one object: type I by the discrete tower, type II by the observer's time, type III because it lives on the spacetime. It is the maximal apex of the diagram. Local and global are not separate: $\frac{1}{2}$ at each level (local) and invariant under the flow (global), the same invariant read two ways. Its last two branches close below.

Curved spacetime (type III again). The accumulated entropy now sources gravity. Each bit costs—the Landauer cost, $\lambda \cdot R = \log 2$—and when that cost is imposed as an equation of state $\delta Q = T\,\delta S$ across every local Rindler horizon, the result is, by Jacobson's argument, the Einstein equation:

$$\text{energy per bit} = \frac{1}{M_{\rm PCF}}, \qquad S_{\rm BH}/k_B = \lambda_{\log} \cdot R = \frac{\log 2}{\log \varphi} \cdot \log \varphi = \log 2 \ \ \text{(one bit)}, \tag{4.15}$$

$$\delta Q = T\,\delta S \implies \text{the Einstein equation}, \tag{4.16}$$

$$R_{AB} = -4\,g_{AB}, \quad R = -20, \qquad G_{AB} + \Lambda_5\, g_{AB} = \kappa\, T_{AB}, \quad \kappa = 8\pi G_N, \tag{4.17}$$

with $G_N = \mu_3 = \frac{1}{2}$. Here the homogeneity of §4.2 is redeemed: Jacobson's argument requires a local horizon with the same law at every point—exactly what tensor covariance guarantees; without it the step does not exist. The matter is the mode number:

$$T^{\rm YM}_{AB} = F_{AC}F_B{}^C - \tfrac{1}{4}\,g_{AB}F_{CD}F^{CD}, \qquad \text{matter} = N_{\rm modes} = \lfloor S(\sigma) \rfloor. \tag{4.18}$$

and here the two open branches of the identity close: the horizon entropy and Jacobson's were, all along, the same number as the bits and the modes. The cosmological constant is determined, not chosen:

W8 — a theory that determines Λ (W01). The abstract of W01 states that general relativity must be derived from a theory that *determines* the value of Λ. Here Λ comes from the curvature, with zero free parameters: $\Lambda_5 = -d(d-1)/2\ell^2 = -6$, with $\ell = 1$ from (3.21) (§3) and $d = 4$ from the arity (§2), and the G–Λ duality $\varphi^{-6} \cdot \varphi^6 = 1$ (`[G_Lambda_duality, R_scalar_AdS5, BF_bound_AdS5]`). Between the gravity scale and the Λ scale there is an exact interval:

It remains to show the spacetime is de Sitter. The metric obtained is Euclidean anti-de Sitter; the same internal Wick rotation that made the centre the time axis carries it to de Sitter, with $\varepsilon(\sigma) = \varepsilon_0 \varphi^\sigma$:

$$ds^2 = \underbrace{(dx^2 + dy^2 + dz^2)}_{\mathbb{E}^3} - \underbrace{c^2\,dt^2}_{\text{Wick }(C)} + \underbrace{\lambda^2\,d(\ln s)^2}_{\varphi^\sigma\text{-tower}}, \qquad s = \varphi^\sigma, \tag{4.19}$$

It is the bridge to de Sitter that Maldacena anticipated, built here by internal analytic continuation rather than an imposed duality (`[ets_is_dS_zero_H]`). Past and future are the poles of a dipole, with C at the equator—the two conjugate sheets of de Sitter.

W10 — the antipodal map / meta-observables (W01). The antipodal singularity Witten notes corresponds to F being antipodal to P in the dipole, which resolves Strominger's obstruction geometrically. *Proof:* the dipole of (4.19); [1] (§A.2).

The geometric closure is the connection the framework contributes. The 5D ambient is flat—its curvature, computed, vanishes, $R^{(5)} = 0$ (`[ets_riemann_flat]`)—and the fifth coordinate, the scale axis, is the extra direction an embedding needs. De Sitter is the hyperboloid $-X_0^2 + \cdots + X_4^2 = 1/H^2$ embedded in that flat ambient, with intrinsic curvature = wedge of the extrinsic by Gauss: $R_{\mu\nu} = 3H^2 g_{\mu\nu}$ (`[dS_ricci_from_gauss, dS_einstein_Lambda]`). The planar patch covers exactly half the hyperboloid—$X_0 + X_4 = \ell\, e^{t/\ell} > 0$ never changes sign (`[dS_covers_half_hyperboloid]`)—and that half *realises* the observer's half $|\Omega| = \frac{1}{2}$ by the shared value (`[observer_half_from_norms]`). The assertoric level splits: strong on the covering, "realises" on the identification.

W9 — the thermal nature of the static patch (Gibbons–Hawking). This is the classical Gibbons–Hawking result, prior to W01 and CLPW: the de Sitter horizon radiates and the static patch is a thermal state at $T = H/2\pi$. The observer's modular flow is the tracial flow of the tower, and the static patch is thermal at that temperature. *Proof:* `[dS_temperature, dS_beta_reciprocal]` (§A.2); [57].

The formal statements underlying the accumulation and gravity movements follow; their proofs are those of the corresponding `[ ]`tags.

Remark 4.3 (The role of φ in the observer)**.** $\mu_3 = \frac{1}{2} = \varphi^{-\lambda_{\log}}$; the three invariants ($f_{\rm crit} = \frac{1}{2}$, $F^{-1}_{\max} = \frac{1}{4}$, $n = 3$) are φ-facts of §2–§3, not fitted values.

Remark 4.4 (The role of φ in the objectivity threshold)**.** $|\Omega| = \frac{1}{2} = \mu_3 = \varphi^{-\lambda_{\log}}$: the half of de Sitter, Kiely's threshold and the modulus are one φ-fact; the throat that observation builds is the tower φ^σ.

Remark 4.5 (The role of φ in the accumulation)**.** $\varepsilon_0 = \ln\varphi/(6\sqrt{3})$ and $M_{\mathrm{PCF}} = 6\sqrt{3}\,\pi/\ln\varphi$ give $\varepsilon_0 M_{\mathrm{PCF}} = \pi$: the clock of the accumulation, the ultraviolet fixed point and the certainty principle are one φ-fact; the accumulation tower is φ^σ.

Remark 4.6 (Past and future are fixed dynamically)**.** The reading of C-P-F as present / past / future is not a free labelling but a consequence of the causal structure. The Wick rotation makes the centre C the time axis and gives the metric of (4.19) its Lorentzian signature $(+,+,+,-)$, so it carries a light cone: C is its apex—the observer's present—and P, F are the past- and future-directed cones, rotating in opposite senses about the fixed, real C, with Θ exchanging them. The direction of the φ-driven accumulation along the throat—from the origin toward the ceiling (Proposition 4.7)—orients the cone, singling out one direction as the future. The past/future assignment therefore follows from the causal cone and the direction of accumulation, not from convention. `[C_fixed_PF_rotate, wick_squares_flip_sign, lorentzian_signature_sum, entropy_increasing, rotation_rate_grows]`

Proposition 4.7 (Accumulation and ceiling)**.** *The metrological quantum Fisher information accumulates as $F_\theta(t) \to F_{\max}\big(1 - e^{-(t/\tau_F)^2}\big)$ with $F_{\max} = 4$ [58], so the Cramér–Rao ceiling $F_{\max}^{-1} = \frac{1}{4} = \mu_3^2$ equals the holographic area factor. The Fisher information per eigenvalue is $F_\Omega = 4\mu_3^2 = 1$ bit; the three eigenvalues give $3F_\Omega = 3 = c$, the central charge of §3.1; and the objectivity threshold $f_{\mathrm{crit}} = \mu_3 = \frac{1}{2}$ is the maximum-entropy point $H(\frac{1}{2}) = 1$ bit.*

Theorem 4.8 (Time–scale conjugacy)**.** *At the operating point $f = \mu_3 = \frac{1}{2}$ the metrological timescale equals the tower period, $\tau_F(\sigma) = \tau(\sigma) = M_{\mathrm{PCF}}\,\varphi^{-\sigma}$. Hence*

$$S(\sigma)\,\tau_F(\sigma) = (\pi\varphi^\sigma)\,(M_{\mathrm{PCF}}\,\varphi^{-\sigma}) = \pi M_{\mathrm{PCF}}$$

is constant in σ: the certainty principle $\varepsilon_0 M_{\mathrm{PCF}} = \pi$ ((2.11)) promoted from a single cell to the generation/clock pair. `[obs_weld]`

Proposition 4.9 (The bulk metric is an Einstein space)**.** *For the warped throat (3.16) with $A = -\log z$ (anti-de Sitter, $\ell = 1$), explicit computation of the Levi-Civita curvature gives, uniformly in all five components,*

$$R_{AB} = -4\,g_{AB}, \qquad R = -20, \qquad G_{AB} = 6\,g_{AB} \quad (A, B = 0, \ldots, 4),$$

so the metric solves $G_{AB} + \Lambda_5\,g_{AB} = 0$ with $\Lambda_5 = -6$, and the Breitenlohner–Freedman bound $m_{\mathrm{BF}}^2 = -d^2/4 = -4$ holds, so the discrete tower is stable. With a source, $G_{AB} + \Lambda_5\,g_{AB} = \kappa\,T_{AB}$ and $\kappa = 8\pi G_N$; the PCF value $G_N = \mu_3 = \frac{1}{2}$ gives $\kappa = 4\pi$ (effective $G_4 = \frac{1}{4}$), and $\nabla^A G_{AB} = 0$ holds identically, so T_{AB} is conserved. `[R_scalar_AdS5, R_munu_AdS5, G_munu_AdS5, BF_bound_AdS5]`

Theorem 4.10 (Jacobson's construction on the bulk metric)**.** *Since (3.16) is an Einstein space, $R_{ab}k^a k^b = -4\,g_{ab}k^a k^b = 0$ for every null k; in vacuum the Clausius relation $\delta Q = \int T_{ab}\chi^a\,d\Sigma^b = 0$ forces $\delta S = 0$ (Jacobson's entanglement equilibrium [59]), which coincides with the ceiling $F_{\max}^{-1} = \mu_3^2$ of Prop. 4.7. The normalisation $\eta = 1/4G_N$ with $G_N = \mu_3$ fixes the area factor $\mu_3^2 = \frac{1}{4}$ and the coupling $2\pi/\eta = 8\pi G_N$; with matter, $\delta Q = T\,\delta S$ on every local Rindler horizon yields the Einstein equation [59, 60].*

Proposition 4.11 (Matter is the saturation entropy)**.** *Each interior level carries $N_{\mathrm{modes}}(\sigma) = \lfloor\pi\varphi^\sigma\rfloor = \lfloor S(\sigma)\rfloor$ scalar modes, each contributing one Shannon bit, so the matter content of level σ is the saturation entropy $S(\sigma) = \pi\varphi^\sigma$. The colour octet has $\dim\mathfrak{su}(3) = n^2 - 1 = 8$ at the arity $n = 3$ fixed by the golden central chain $\varphi^2 + \varphi^{-2} = 3$; the full $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ follows from the torus T^2_{PCF} (2.9), its gauge factors landing at the force levels by increasing dimension $1 < 3 < 8$:*

$\mathrm{U}(1)$ *is the Hopf fibre* S^1 *at* $\sigma = 3$ *(EM),* $\mathrm{SU}(2)$ *the Clifford* S^3 *at* $\sigma = 4$ *(weak), and* $\mathrm{SU}(3)$ *the* A_2 *roots of the Eisenstein face (Fig. 4) at* $\sigma = 5$ *(strong), with* $\dim \mathfrak{su}(3) = n^2 - 1 = 8$ *at the arity* $n = 3$ *fixed by the golden central chain* $\varphi^2 + \varphi^{-2} = 3$ *[gauge_dim_su3, phi_central_chain, central_chain, clifford_S3_condition, hopf_from_clifford, A2_root_normalized]. The electroweak data follow from the same modulus: the Weinberg ratio is* $\sin^2\theta_W = \varphi^{-3}$ *[weinberg_ratio], the breaking* $\mathbb{Z}_3 \to \mathbb{Z}_4$ *occurs at the tower midpoint* $\sigma = 3$*, and the* G*–*Λ *duality reads* $\varphi^{-6}\varphi^{+6} = 1$ *[G_Lambda_duality]. Only the individual Yukawa couplings and the full three-generation flavour spectrum are not fixed by this placement and remain open.Their Yang–Mills stress-energy* $T^{\mathrm{YM}}_{AB} = F_{AC}F_B{}^C - \frac{1}{4}g_{AB}F_{CD}F^{CD}$ *is conserved (*$\nabla^A T^{\mathrm{YM}}_{AB} = 0$*) and traceless on the* $D = 4$ *worldsheet (conformal there).*

Remark 4.12 (The G–Λ interval: Λ is determined, not selected)**.** The cosmological constant is not an input. The scale $\ell = 1$ (from the S-duality fixed point $\tau = i$, (3.21)) and the dimension $d = 4$ (from the arity $n = 3$, §2.2) are both derived, and they fix

$$\Lambda_5 = -\frac{d(d-1)}{2\ell^2} = -6, \tag{4.20}$$

with zero free parameters. This realises precisely what Witten [1] identified as necessary: a theory "that determines the value of the cosmological constant."

The tower levels. The activation levels of the three sectors are read off the arity, not postulated:

$$\sigma_G = n - 1 = 2, \qquad \sigma_{\mathrm{EM}} = n = 3, \qquad \sigma_\Lambda = 2n = 6. \tag{4.21}$$

Gravity activates at $\sigma_G = n-1$ (the spatial bulk $\mathbb{E}^3$ complete, one dimension below the bulk); the electromagnetic sector sits at the midpoint $\sigma_{\mathrm{EM}} = n$ (the Page point that halves the tower $[0, 2n]$); and Λ activates at the ceiling $\sigma_\Lambda = 2n$. The ceiling is geometric: the torus T^2_{PCF} is a *complex* space, so complex dimension n is real dimension $2n$—"complex dimension three" is $\mathbb{R}^6$, never $\mathbb{R}^3$. The tower runs over the full real space $\sigma \in \{0, 1, \dots, 2n\}$, and its ceiling $\sigma_\Lambda = 2n$ is that real dimension.

The interval is the spacetime. The gap between the gravity and Λ levels is exactly the dimension of spacetime,

$$\sigma_\Lambda - \sigma_G = 2n - (n-1) = n + 1 = 4 = \dim(M^4), \tag{4.22}$$

and the interval carries the two holographic invariants of the microstate:

$$\frac{\sigma_{\mathrm{EM}} - \sigma_G}{\sigma_\Lambda - \sigma_G} = \frac{1}{n+1} = \frac{1}{4} = |\Omega|^2 \qquad \text{and} \qquad \frac{\sigma_{\mathrm{EM}} - \sigma_G}{\sigma_\Lambda - \sigma_{\mathrm{EM}}} = \frac{1}{n} = \frac{1}{3} = \|P\|^2, \tag{4.23}$$

the holographic area factor $\frac{1}{4} = |\Omega|^2$ and the past-component weight $\frac{1}{3} = \|P\|^2 = F_\Omega / c$.

Two independent routes to the same 6*.* The curvature magnitude $|\Lambda_5| = d(d-1)/2 = 6$ (from $d = 4$) and the tower ceiling $\sigma_\Lambda = 2n = 6$ (from $n = 3$, as the real dimension of the complex torus) coincide by independent routes—5D curvature on one side, the real dimension of the worldsheet torus on the other. Both are the entropy saturation of the microstate read geometrically: $\sigma_\Lambda = 2n = 6$ is the ceiling of the golden tower $S(\sigma) = \pi\varphi^\sigma$ whose accumulated bits are the horizon entropy of (4.14) and, through (4.17), the field equations—the six-fold value is that saturation, not a separate geometric fact. Together with $6 = |S_3| = 3!$ (the order of the symmetry group, Lem. 2.28) and the exponent of the G–Λ duality $\varphi^{-6}\varphi^{+6} = 1$, the value $\sigma_\Lambda = 6$ is fixed from four sides.
[sigma_G_val, sigma_EM_val, sigma_Lambda_val, interval_gap, interval_gap_general, em_holographic_fraction, entropy_at_mu_three, threshold_ordering, em_asymmetry, lambda_magnitude_eq_level]

Scope. This we propose as the structure Witten required: the de Sitter Hilbert-space dimension N is integer, so $G\Lambda^{(n-2)/n}$ cannot vary continuously and both G and Λ must be fixed by a more fundamental theory [1]; here $G_N = \frac{1}{2}$ and $\Lambda \sim \varphi^{-6}$, the product fixed, $\varphi^{-6}\varphi^{+6} = 1 \iff G\Lambda \sim M_{\mathrm{Pl}}^{-4} M_{\mathrm{Pl}}^4 = 1$ (PCF units). What is determined is this dimensionless conjugacy; the ∼120-order gap to the observed $\Lambda_{\text{obs}} \sim 10^{-122} M_{\text{Pl}}^4$ is a translation to the absolute scale, which the tower does not fix.

4.5. **Why this is a single demonstration.** The path went through the three types: it opened in the non-local type III field, made the type I particle emerge, turned it into the type II observer, interwove I and II in the tower, and closed in the type III spacetime. These are not five demonstrations glued together but one, because the same object of modulus $\frac{1}{2}$ is the protagonist of all, and each transition is a proved implication: entropy orients the flow and yields the particle, the Wick rotation gives the cones and the observer, accumulation overflows and builds the tower, entropy sources gravity.

The commutative diagram supports this unity with precise structure. The twenty-three equations form a directed acyclic graph with two roots—the projection Π ((4.1)) and the observer's objective record ((4.5))—from which all else derives, with 34 demonstrative edges. A topological order exists—the equations can be stated so that none depends on a later one—which guarantees the section reads without forward reference. Seven nodes are apices, where two or more independent chains converge; the weld ((4.13)), with four routes, is the maximal one and the only triple point I∩II∩III. The circle closes literally: (4.17) returns to (4.1)—the non-local gravity returns to the same Π and $\frac{1}{2}$ of the opening, III→III. The spacetime at the end is non-local at every point, type III, as the field at the start, and its Newton constant is the same $\frac{1}{2}$ of the object one began with. One enters by any door and reaches the same object: that is what it means for the diagram to commute. The section does not prove six things; it proves that an observer of modulus $\frac{1}{2}$, which cannot be separated into parts, is at once the particle, the string, the tower of information and the curved de Sitter universe that contains it, and that this universe has the Λ the theory was bound to determine, with the observer inside. The debt with which the section opened is settled by the same object that raised it.

The demonstrative line. Each step uses only the previous ones.

§4.1 what is proved, and against what.: Holography in de Sitter requires a theory that determines Λ and describes the observer from within (Witten). The yardstick is the von Neumann classification into three types—I discrete (particle, definite spectrum), II observation (trace, crossed product), III local field (worldsheet, horizon, non-locality)—and what is proved is the *signature* of each, not the technical name. The criteria come from three distinct sources, each tagged where used: Witten 2001 (Λ, finiteness $S = \ln N$, the isometry obstruction, the pairing, the CPT map, the two conjectures), CLPW 2022 (the observer described by the theory, the maximally mixed vacuum, the crossed product), and Gibbons–Hawking (the thermal static patch). The thesis: the three types are realised on one object, the microstate Ω with $|\Omega| = \frac{1}{2}$, as three threads crossing the whole section.

§4.2 the field and the particle.: The path opens in the non-local type III field: the worldsheet carries the interface $\Pi\colon \mathbb{E}^3 \to \mathbb{C}$ ((4.1)), whose three components are the spin-star C-P-F ((4.2)). The whole geometry is one tensor—the warp $z = \varphi^\sigma$ generates the curvature $R_{AB} = -4g_{AB}$ (App. A.1)—and from that same geometry, by its tension, the discrete type I particle emerges. Geometry and particle are the same object.

§4.3 the particle as observer.: An internal Wick rotation turns the particle into an observer: the centre becomes the time axis, past and future the two cones. The observer accumulates—Fisher grows to the Cramér–Rao ceiling $F_{\max}^{-1} = \frac{1}{4}$ ((4.4), (4.3))—and sees the exact half $\|P\|\|C\|\|F\| = \frac{1}{2} = |\Omega|$ ((4.7)), with redundancy $R_\delta \sim N$ ((4.5)) and accumulation to $F_{\max} = 4$ ((4.6)). Three independent routes converge on $\frac{1}{2}$ (first apex), the objectivity threshold $f_{\text{crit}} = \frac{1}{2}$ of Kiely ((4.8)) and the cell capacity $\varepsilon_0 M_{\text{PCF}} = \pi$ ((4.9)).

§4.4 the tower, and curved spacetime.: The accumulation builds the tower (types I and II interwoven): the cell overflows and the throat opens, $z = \varphi^\sigma$, $S(\sigma) = \pi\varphi^\sigma$ ((4.10), (4.11)), at the UV fixed point $\beta_g = 0 \Leftrightarrow \varepsilon_0 M_{\text{PCF}} = \pi$ ((4.12)). The two accumulations—of time and of scale—are one, the weld ((4.13)), giving the identity $F_\Omega N_{\mathrm{modes}} = S(\sigma) = S_{\mathrm{BH}} = S_{\mathrm{Jacobson}}$ ((4.14)). Then curved spacetime (type III again): Landauer ((4.15)) $\to$ Jacobson ((4.16)) $\to$ Einstein ((4.17)), with matter the saturation entropy ((4.18)). The internal Wick rotation gives the de Sitter metric ((4.19)), $\Lambda_5 = -6$ ((4.20)), and the level interval $\sigma_G{=}2, \sigma_{\mathrm{EM}}{=}3, \sigma_\Lambda{=}6$ ((4.21)) with $\sigma_\Lambda{-}\sigma_G{=}n{+}1{=}4$ ((4.22)) and the fractions $\frac{1}{n+1}{=}\frac{1}{4}{=}|\Omega|^2$ ((4.23)); the G–Λ duality and $\sin^2\theta_W{=}\varphi^{-3}$ (Prop. 4.11) give the gauge factors SU(3)×SU(2)×U(1) (Fig. 7).

`[cw3_observer_items_core]` The observer does not select; it builds without alternative — participatory yet forced — consonant with Kiely's quantum Darwinism. The whole section is one commutative diagram (Fig. 8): the twenty-three equations as nodes, the demonstrative arrows between them, and the apices where independent routes converge.

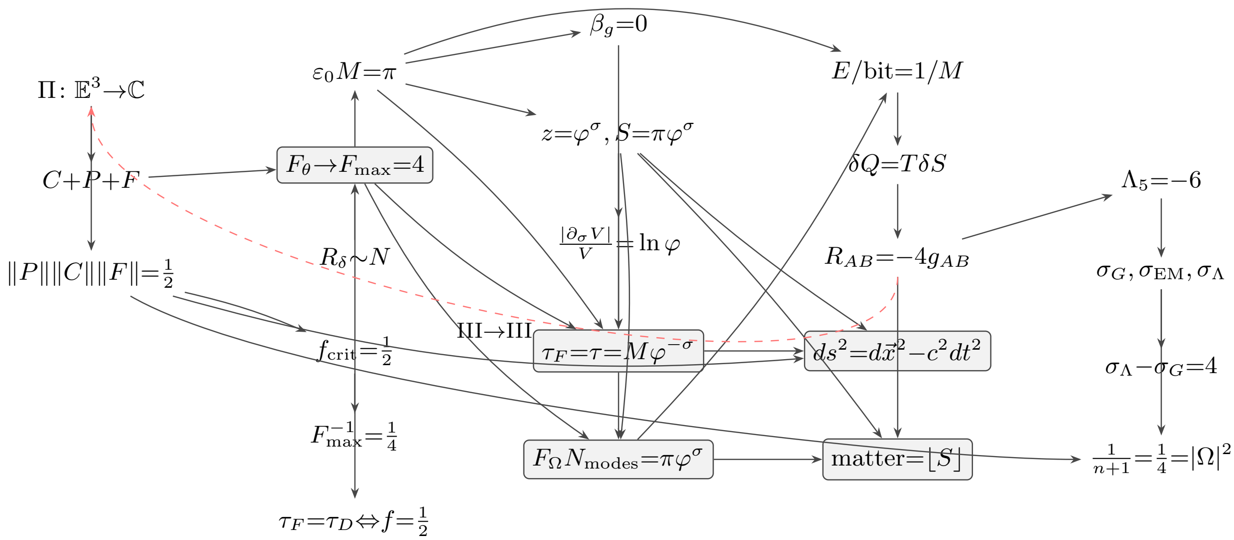

FIGURE 8. Commutative diagram of §4: the twenty-three equations of §4 as nodes, arranged by subsection (the non-local field and particle, §4.2; the observer, §4.3; the tower and gravity, §4.4), with the thirty-four demonstrative arrows. Shaded nodes are the apices, where two or more independent chains converge on the same value; the weld $\tau_F{=}\tau{=}M\varphi^{-\sigma}$ has four incoming routes and is the maximal apex. The dashed return edge is the closure $R_{AB}{=}-4g_{AB} \to \Pi$ (type III to type III): the non-local gravity returns to the same projection Π and modulus $\frac{1}{2}$ of the opening.

5. Discussion

The model previously developed arose from the search for the intersection of two different sets of necessities. On one side, the necessities of holographic physics and its already-developed framework—quantum tori, strings, holography, information—which are forced by the physics itself. On the other, the necessities of mathematics: which paths are open and which are blocked, if the physics is to be described as it requires to be described. The object both sets must accommodate is the participatory observer with no exterior, in de Sitter space.

We approached the intersection in a manner that may be called Barthesian, applied to these shared mathematical necessities—in this case, the problem of self-reference. Barthes holds that there is a degree zero of writing: a structure prior to the present language which precedes it and allows it to signify [61]. Here we sought the analogous structure common to, and prior to, the mathematical and the physical languages that must intersect. The structure we found is twofold in one. As ancestor and common point, it is the vanishing point of drawing, industrial design and architecture: the Vitruvian module—the historical and etymological ancestor of isometry, of the modulus of $\mathbb{C}$, and of the modular spaces that precede the contemporary moduli spaces and still underlie them. And as shared fundamental principles between the two sets of necessities, it is π, φ, polygons and trigonometry: an entire language, and a way of approaching moduli spaces. Joined with φ, a practice likewise descending from the arts, the module parametrises the whole space of a building with a single magnitude and thereby carries diverse ratios into one space; we use exactly this to bridge arithmetic and geometry, treating arithmetic as ratios, polygons and trigonometry, and from that prior structure the framework is built within a string—parametrising a worldsheet.

The discussion follows one route. §5.1 sets out the problem in fuller form than the introduction: existing theories leave no place for the observer. §5.2 states our proposal as a theory, recasting the self-reference so that it is no longer a paradox. §5.3 gives the solution—the geometric mediation π/φ, and the mathematical detail of its execution, where it meets Manin's ideas on the same object (developed in §2). §5.4 shows the string and torus this produces, and how that mathematical structure connects to holography and M-theory through its AdS/CFT link (developed in §3). §5.5 gathers what follows from it in de Sitter (developed in §4), where the framework meets relativity and the quantum along Witten's criteria on the von Neumann types. §5.6 closes by returning to the self-reference and the epistemological limit the paper began with.

5.1. **The problem: an observer with no place.** *M-theory from the Superpoint* [22, 62] showed that the spacetimes of string theory and M-theory are not postulates but consequences. Starting from the superpoint $\mathbb{R}^{0|1}$—an object with a trivial (abelian) Lie bracket, no metric, and no spin structure—and iterating maximal invariant central extensions classified by super-Lie-algebra cohomology, one recovers super-Minkowski spacetime in dimensions $3, 4, 6, 10$, and 11, and from these the entire brane content [63]. The Lorentz metric is never put in by hand; it appears in the automorphisms of the algebra. The result is remarkable, but it is written wholly in the third person. There is no observer anywhere in it. The metric appears—but to whom? The construction is a ladder, played as a game of two moves from outside the system, and it never asks where an observer would stand. In Minkowski or anti–de Sitter space that silence is harmless. In de Sitter space, which has no reachable spatial boundary and no outside, it is not: a de Sitter universe is closed upon itself, and whatever observes it must be inside it.

Our program takes that omission as its problem. The question is how to place the observer inside de Sitter space, where there is no exterior to retreat to. We share the superpoint's philosophical root—Hegel's opening, in which Being and Nothing are identical and their truth is Becoming—but we realize it in a form that has room for an observer, using φ and π.

The Hegelian content of the construction enters through Lawvere's categorical formalization of Hegel's *Science of Logic*. Lawvere formalizes the *unity and identity of opposites* as an adjunction, with adjoint functors as the basic instance [20]; under the resulting dictionary, *pure being* and *nothing* are the terminal and initial objects—the right and left adjoints of $(\varnothing \dashv *)$—*becoming* is an adjoint modality, and *determinate being* (Dasein) is the *Aufhebung* of $(\varnothing \dashv *)$, realized as the next higher level in the lattice of essential subtoposes. Cohesion is axiomatized as an adjoint quadruple $(\Pi_0 \dashv \mathrm{Disc} \dashv \Gamma \dashv \mathrm{coDisc})$ [21], and the superpoint $\mathbb{R}^{0|1}$ is an object of the corresponding *super cohesion*—Schreiber's *solid cohesion*, the ∞-topos of super formal smooth ∞-groupoids, cohesive over the base topos of bare super-sets [21]. From it, super-Minkowski spacetime and the brane bouquet are obtained by consecutive *maximal invariant central extensions*, classified at each stage by the invariant even super Lie algebra cohomology; the three-dimensional super-Minkowski algebra, for instance, is the maximal invariant central extension of the superpoint $\mathbb{R}^{0|2}$, induced by the symmetric bilinear forms that span its even second cohomology [22].

This realizes the *form* of the unity of opposites—the adjunction—but between *distinct* objects: the initial and terminal objects are not identified. The construction is correspondingly one-directional: the brane bouquet is rooted in the superpoint and grows outward by consecutive central extensions—0-brane condensation from nothing—never closing back upon its origin. So realized, the dialectical becoming carries neither an observer nor a selected vacuum. The opposite difficulty is met in low-dimensional holography, in the *ensemble problem*: where M-theory leaves the observer outside, AdS/CFT depends on the observer but cannot put it inside—no single boundary theory is available, and the bulk survives only as an average over an *ensemble* of them, even though quantum gravity should carry discrete, individual microstates of its own [5].

5.2. **Our theoretical proposal.** Hegel's becoming was realized above through Lawvere, as the unity of opposites between distinct objects. We realize it from a different perspective—one that draws from Oberarzbacher, analysing Hegel through the non-dualist perspective of the Shaivism of Kashmir [64]. For Hegel the absolute is reached by contrasting opposites: the unity is won as their *conciliation*, and the principle of non-contradiction is the engine that drives the opposition. In Shaivism it is otherwise: the unity is not a synthesis of opposites but is given first—the Self unfolds as Subject and Object already in a correlation of identity, and what is reached is the *recognition* of that identity, not its production; there the principle of non-contradiction is not the engine but the faculty that fabricates the duality of Subject against Object.

We take the non-dual route and render it mathematically through functors, as in the case of Lawvere's but with a twist; geometrically, ratios as terms classified as equivalent. We do so on the base of the Vitruvian rhetoric, read with Foucault and Barthes as the degree zero of the geometrical episteme [65, 61]: the pre-Cartesian mode of interpretation in which things are related not by binary Cartesian logic but by adjustments, ligaments and junctures—through number, proportion and the anatomy of the things compared [66]. On that base we classify the circle of $\mathbb{C}$ and the wave function as Euler's identity, under their geometric equivalences as circles, and as lines in relation to the integers, the reals and the primes through φ. φ, in turn, lets us relate this construction to the appearance of φ in the wave functions of physics, rather than in number theory.

In simple terms, to create intuition for the framework developed in §2: we take φ, which we find as a function and an operator, as well as a ratio, in different areas of wave functions and lattices [37, 38, 39, 40]. And that φ we carry across the whole plane through an algebraic ring (the string) working as a Vitruvian module, which has been used for millennia in art to carry one and the same magnitude over the plane and also onto a three-dimensional object. The Vitruvian module is the common measure taken from the column at its base, from whose single magnitude the *symmetriae* and proportions of the whole building are derived by *commodulatio* (Vitruvius, *De arch.* 3.1.1): every part stands in fixed ratio to the module and, through it, to the central axis, conferring unity and proportion on the whole as the disposition of parts does in rhetoric [66]. It was the basis of Brunelleschi's and Alberti's perspective, and it is in fact the ancestor of the module of $\mathbb{C}$ and of the isometric lattice as well. In abstract mathematical terms its components are: an origin, a radius, a magnitude, an angle, six-fold hexagonal symmetry, trigonometry and polygons—in this case, to distribute φ across the plane and across dimensions and space in a completely self-consistent way.

Finally, this module/ring is mounted on the noncommutative torus that M-theory uses, its non-orientable fibre—independent of the algebra (Proposition 2.17)—taking it from interior to exterior with $\frac{1}{2}$, a topological relation that makes it similar to an AdS/CFT throat.

The result lets us connect the geometry of π in the quantum paths taken with equivalence of value: π weighs every path with equal importance, so that none is privileged. But here fixed by the cutoff ε_0 with $\varepsilon_0 M_{\mathrm{PCF}} = \pi$, and tied to the string as an accumulation of information, which is likewise non-privileged.

Thus π and φ become, at the same time, the point of contact between the wave function, the string as a Vitruvian module, and $\mathbb{C}$. What the Vitruvian module achieves in $\mathbb{C}$ is, through its properties (π, φ, polygons and trigonometry), to generate a relation to the Euler product and the values of ζ, as well as to the primes—this is demonstrated in full in other papers [35, 44]. What it allows here, independently of the primes, because we are in physics, is to unite the geometry of 1D with the 2D structure of $\mathbb{C}$ and with the 3D isometric, self-consistently.

We could then say that it is a bridge between a new kind of space and a very old one—a steampunk kind of space: a contemporary, neoclassical/Victorian, idealized reconstruction of the ancients that employs the principles preceding the module space of $\mathbb{C}$ and the isometric space, which are relevant in the Vitruvian module, placing them within the territory of string theory through the contact of π and φ.

5.3. **The resolution: geometric mediation—π/φ.** The relation the module opens in $\mathbb{C}$—to the Euler product, ζ and the primes, on one side, and, on the other, to the very places where ζ already appears in physics: the odd zeta values $\zeta(2k+1)$ fixed by the Veneziano string amplitude [25, 26].

This intersection yields something similar to Manin's hypothesis for tackling the Riemann hypothesis: the tori that M-theory uses, carried to the $\mathbb{F}_1$ program, with Borger's Λ-ring of Frobenius lifts fixing their arithmetic [23, 44]. Here, though, the same object is turned to holography in place of $\mathbb{F}_1$, in the M-theory noncommutative tori. The mediation works through three mediators, and closes, below, on the very Frobenius lifts, the Artin character χ_5 and the Euler product it reaches.

The $\mathbb{F}_1$ program is, in essence, a geometric descent meant to avoid circularity. Beyond its link to strings, the same program has been connected to the Hermitian-operator problem—the Hilbert–Pólya route: a self-adjoint operator whose spectrum would carry the zeros without reference to the ζ values or the primes, pursued in the modern era by Berry and Keating, by Bender, Brody and Müller, and from noncommutative geometry by Connes (see the companion paper [42] for the full lineage). This is significant for the present theme, because that program seeks to recreate the spectrum of the zeros of ζ without the exact GUE statistical ensemble that low-dimensional AdS/CFT falls back on; our proposal for that operator is constructed in the same companion paper [42]. What we do here is to try to join the necessities of holography with M-theory: in the spirit of Maldacena's ideas about the use of M-theory's tori for holography, and of our reinterpretation of the superpoint, turning a similar object to the holographic problem rather than to $\mathbb{F}_1$.

The three mediators join 0 and 1: the **half**, π, and φ. Each mediator carries a definite function. The **half** is the midpoint between 0 and 1: it is the value the diagonal lacked, now present as the contraction norm $\|\widehat{\Omega}\| = \frac{1}{2}$ and as the reference point $\Gamma(\frac{1}{2})$ between unity and infinity. On the side of $\mathbb{C}$, π enters through the half-factorial $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, whose role is to seat the Gaussian and the factorials of quantum theory and string theory on the circle itself; the circle, carrying the half, meets i—the Gaussian lattice and the modulus of $\mathbb{C}$. On the side of the Vitruvian module, φ enters through the pentagon (its function is to fix the self-similar ratio, $\varphi^2 = \varphi + 1$), the triangle enters again through the half (the $\sqrt{3}/2$, the role of which is to set the ternary norm), and the Mersenne numbers and the powers of φ enter through φ and the $3/2$ ratio (their function is to lock φ to the integers, $3 \cdot \varphi^{\sigma(p)} = 2^p$). On the quantum side the same π returns in Euler's identity, bound there to i, e, and to 1 and 0 themselves [35], where its role is to close the complex rotor on the same 0 and 1 the half mediates.

What ties these together are exact bridges, each a specific identity rather than an analogy: **between** 10 **and** 1—the decagon and the unit: ζ_{10} the primitive tenth root of unity, $\varphi = \zeta_{10} + \zeta_{10}^{-1}$, with $\mathbb{Q}(\varphi) \subset \mathbb{Q}(\zeta_{10}) \subset \mathbb{Q}(\zeta_{20})$ ($\zeta_{10} = \zeta_{20}^2$, $20 = 2 \cdot 10$), and the 1 of $e^{i\pi} + 1 = 0$, in the multiplicative mediation that scales from 1 up to the next integer base 2 [35]; **between** 5 **and** 10—the pentagon and the decagon, $\varphi = 2\cos(\pi/5) = \zeta_{10} + \zeta_{10}^{-1}$, placing φ in the real subfield of $\mathbb{Q}(\zeta_{20})$ [35]; **between** π **and** φ—the same trigonometric identity $\varphi = 2\cos(\pi/5)$, in which the angle $\pi/5$ binds π to φ on the real line [35]; **between** φ **and the integers**—the universal identity $3 \cdot \varphi^{\sigma(p)} = 2^p$, with $\sigma(p) = p\lambda_{\log} - \log_\varphi 3$ and the factor $3 = M_2 = 2^2 - 1$, the first Mersenne prime [35, 44]; and **between** π **and the integers**—the even zeta values $\zeta(2k) \in \mathbb{Q} \cdot \pi^{2k}$ (Basel for $k = 1$), together with the mode count $\lfloor \pi\varphi^\sigma \rfloor$ on the tower [44]. These do not so much *impose* equivalences across the different domains as **reveal** that the equivalences were already exact and simultaneous—in the precise sense that one and the same φ satisfies all of them at once: the twelve clauses across five canonical structures (the complex rotor of Euler's identity, the real trigonometric line, the real-multiplicative line, the Mersenne arithmetic modulo 20, and the Galois group of $\mathbb{Q}(\zeta_{20})/\mathbb{Q}$) hold under a single φ, with no ordering among them, so the equivalences are *co-constituted*, not matched one after another [35].

This is what lets us go beyond Lawvere's two-term unity and make all three Hegelian moments coincide: **Nothing, Being, and Becoming as one**—not in sequence but at one stroke. The static isomorphism of 0 and 1 is the circle; the becoming is the dynamics; and the becoming is co-equal with the other two because it is anchored, in the same single φ, in arithmetic—the becoming is not first the circle and then its dynamics, but the one φ read at once as static identity and as motion. The anchoring recovers the structure of the primes (the Frobenius lifts, the Artin character χ_5, the Euler product) [44] and the zeros of ζ, which appear in the vacuum amplitudes themselves, where the partition function is a gas of oscillators labelled by the primes [32, 31].

5.4. **String reconfiguration: the resulting torus and its AdS/CFT connection.** The string of §5.2 closes, geometrically, into a torus. Its worldline $\Omega(\tau) = \frac{1}{2}\, e^{i\tau \ln\varphi}$ (modulus $|\Omega| = \frac{1}{2}$), promoted to a worldsheet by the Polyakov action, closes on the torus generated by φ (§2.3, (2.9)), along whose throat the radial coordinate is $z(\sigma) = \varphi^{\sigma}$. The relation the torus carries is AdS/CFT, but realized without a privileged side: bulk and boundary are joined by an isometry $V : \mathcal{H}_{\text{bulk}} \to \mathcal{H}_{\partial}$ with $V^{\dagger}V = 1$ [42], so where orthodox holography integrates out the radial coordinate and keeps only the boundary, here neither side is singled out and both are kept. The modulus is scale-invariant—$|\Omega|_{\sigma} = \frac{1}{2}$ for every σ—so the relation holds at each level of the throat, not only at the boundary.

This bulk–boundary map is the physical counterpart of the Frobenius lifts already at work in the arithmetic (§5.3) [44]. There the lift $\psi_p(\varphi^n) = \varphi^{pn}$ is computed *as an algebra*—a ring endomorphism of the golden monoid, for every prime p and every n. Here it cannot be: there is no three-dimensional division algebra to carry the isometry directly, since a complex space of complex dimension n is a real space of dimension $2n$—so "complex dimension three" is $\mathbb{R}^6$, never $\mathbb{R}^3$, and $\mathbb{R}^3$ is odd, with no complex structure. The relation between bulk and boundary must therefore be *recoded between algebra and geometry* rather than computed as an algebra [42]—that is the twist. The recoding is geometric of necessity, not by choice: under the Tarskian descent to the real closed fields, which already contain Euclidean geometry [16], nothing can be assigned a code at all.

What holds the recoding together at every level is the same half that §5.3 identified as the contraction's fixed point: $|\Omega|_{\sigma} = \frac{1}{2}$ for all σ is the physical analogue of $\psi_p(\varphi^n) = \varphi^{pn}$ for all p and n—one φ-tower, read on its physical side. That half-turn is the anti-diagonal in geometric form: a full turn would implement the self-application $f(f)$ that Lawvere prohibits [11], while the half-turn is the distributed passage $f \circ g \circ h$ that does not close. And it is the scale-invariant $|\Omega|_{\sigma} = \frac{1}{2}$ for all σ—the physical analogue of $\psi_p(\varphi^n) = \varphi^{pn}$ for all p and n.

The same torus carries the rest of the M-theory duality web, and each duality is here a symmetry of the golden tower rather than a separate postulate. S-duality, $\tau \to -1/\tau$, is the Galois conjugation $\varphi \to -1/\varphi = \bar{\varphi}$ of the golden field: the two arithmetic conjugates of φ are the strong and weak coupling of the string. T-duality, $R \to \alpha'/R$, is carried by the self-dual radius $R^2 = \alpha'$ of the same torus, the point the tower fixes [Tdual_selfdual]. The strong-coupling limit of type IIA, $\sigma = 10 \to 11$, is the one further step of the tower into eleven dimensions—the passage to M-theory read as one more level. And the de Sitter swampland condition $|\partial_{\sigma}V|/V = \ln\varphi$ [67] is met by the tower itself, its logarithmic growth rate being exactly $\ln\varphi$: the condition an accelerating universe must satisfy is the growth law of the construction. The web is thus not imported; it is what the modulus-preserving symmetries of the φ-tower already are, and its de Sitter corner is the bridge to the next section.

5.5. **What the resolution yields.**

5.5.1. *Vacuum selection.* The vacuum selection is the second face of the one mechanism, and it bears on relativity because it is the same field read twice. The microstate is a single definite eigenstate, not the statistical mixture of boundary theories that low-dimensional holography otherwise needs, where the bulk survives only as an average over an ensemble (§1.2.4). No such average is needed here, and the reason is geometric: the horizon of the projection Π does not factorise into interior $\otimes$ exterior—the type III signature of a non-local field with no edge at which to separate one region from another (§4.2, Proposition 4.1). A field that does not factorise has no ensemble of parts to average over; its state is one—the **vacuum selection** the superpoint's open ladder could not make.

That same non-factorising type III field is the curved spacetime of relativity. It opens §4 as the worldsheet and closes it as the de Sitter metric: the wedge without normal trace and the spacetime are one face, read at two scales. So the single microstate is not a statistical fact prior to the geometry—it *is* the geometry, read as a state. This is why the vacuum selection bears on relativity and not only on holography: the field the observer builds from within, rather than averages over, is the one metric relativity leaves undetermined for an observer inside it (§5.5.2). Measured against Witten's criteria in §4, it is the maximally mixed vacuum CLPW require, and because it is one state and not an ensemble, the firewall ambiguity that needs an ensemble average does not arise. The operator that realises this microstate as a self-adjoint operator is built in the companion paper [42].

5.5.2. *The self-modeling observer and gravity.* The observer is, accordingly, modeled as a one-dimensional object—a worldline that loops and raises its memory up the tower $S(\sigma) = \pi\varphi^{\sigma}$, emerging as a self-similar fractal through φ and the symmetries, and becoming by observing its own magnitude. The self it becomes-with-respect-to is threefold—the entity, the observing, and what has been observed as self-similar information (the accumulated interior, §5.4); its own magnitude is that accumulated information. It is the half-projection: $\|P\|\,\|C\|\,\|F\| = \frac{1}{2} = \|\widehat{\Omega}\|$ (Proposition 2.12). Observation does not select among pre-existing geometries—it builds the geometry—and the accumulation of bits drives the gravitational sector by an explicit chain: each observed bit deposits information; the deposited bits accumulate as the entropy $S(\sigma)$ along the tower; that entropy obeys the area law, the Bekenstein–Hawking $S_{\mathrm{BH}} = A/4G_N$ [46, 47]; and the area law, imposed as an equation of state $\delta Q = T\,\delta S$ across local horizons, is exactly Jacobson's derivation of the Einstein equation (§4.4) [59]—with a π-bit capacity per cell setting the Landauer cost (2.11) [68]. This is the third register of the one mechanism, the algebraic: the self-reference is non-paradoxical because the structure is ternary, and the No-Diagonal (§2.2) together with the fixed point $\varphi^2 = \varphi + 1$—and these are one construction, not one equation. The value the binary diagonal lacked is the fixed point of $x \mapsto 1 - x$, namely $\frac{1}{2}$, supplied as the half and as $\|\widehat{\Omega}\| = \frac{1}{2} < 1$; φ is the fixed point of the distinct map $x \mapsto 1 + 1/x$, i.e. $\varphi^2 = \varphi + 1$; and the two are tied exactly by $\varphi^{-\lambda} = \frac{1}{2}$ ($\varphi^{\lambda} = 2$—the same 2 that is the binary base). So $\frac{1}{2}$, $\|\widehat{\Omega}\| = \frac{1}{2}$ and $\varphi^2 = \varphi + 1$ coexist in one construction. So an operator that must contain its own spectrum [42] is defined without contradiction. The contrast with relativity is exact: relativity gives an absolute block together with the relative, but neither a metric for the observer nor a quantization of what observation is or how it occurs; PCF supplies precisely that, as self-consistency relations.

π, which asks that all directions be weighed with equal importance, and φ, the self-similarity that we find both in nature and in the primes [44], together yield what the superpoint left empty: an observer that models and describes itself—the wave function measuring itself, the One recognizing itself, connected through every available path by the single act of observation, inside a de Sitter space with no outside and no paradox.

The Einstein equation this yields is not the anti–de Sitter one it began from. The passage to de Sitter is a Wick rotation internal to the eigenvalue geometry itself—not imposed from outside, but the rotation already carried by the three-eigenvalue metric—and it is what turns the AdS throat into the de Sitter space of the observer. With it the cosmological constant is not an input: fixed by the scale $\ell = 1$ and the dimension $d = 4$, it takes the value $\Lambda_5 = -d(d-1)/2\ell^2 = -6$, which is exactly the qualitative content of Witten's requirement that the theory "determine the value of the cosmological constant" [1]. What is fixed here is $\Lambda_5 = -6$ in these units, structurally and with no free parameter; the observed $\sim 10^{-122}$ is a separate matter of scale translation, not derived here. And the matter that sources it is the accumulated entropy itself (Proposition 4.11): the same tower whose bits built the geometry is, saturated, the matter content—so the observer's information, the geometry, and the matter are one.

5.5.3. *Non-local reality before observation.* PCF is not Leggett's non-local realism, and for that reason it is not refuted by the experiment of Gröblacher and Zeilinger. The difference lies in *what* is real before observation. Leggett's models predetermine the **parts**: a hidden variable λ fixes the state of the subsystems before measurement—a realism of the parts—even while correlating them non-locally, and that is what experiment rules out [56, 69]. PCF, by contrast, makes the non-local **whole** real, not the parts. Before observation there are no subsystems carrying values (§4.1, Proposition 4.2); what is real is the indivisible One—the block, the non-local wave function, the circle that distributes magnitude across all of $\mathbb{C}$. The local—the parts—is not predetermined; it is constituted by self-measurement.

This is the needle PCF threads. Bell rules out predetermined *local* properties [55]—a prediction confirmed by the Bell-test experiments of Clauser, Aspect, and Zeilinger, the work recognized by the 2022 Nobel Prize in Physics [70, 71, 72, 73]; Leggett rules out predetermined *subsystem* properties, local or not [56], as the experiment of Gröblacher et al. and, independently, that of Branciard et al. have shown [69, 74]; PCF denies that there are subsystems at all before observation—only the non-local whole. It is not a realism of the parts but a reality of the whole. This is why "if it is not observed, it is not there" refers to the local—the parts, the geometry—and not to the non-local whole, which is real.

The geometric realization is the one built above. π weighs all directions with equal importance, so no localization is privileged before observation: that is the non-locality, by construction. The wave function—the phase, the 1—is the non-local real, not a merely predictive device (the Copenhagen reading we reject); the observer is the wave function itself. Self-measurement constitutes the local: observing builds the geometry rather than selecting among pre-given ones; the non-local measures itself, and the local emerges from it. And ER = EPR makes the non-local correlation—entanglement—into geometry itself [43] (§4.4, Proposition 3.9): the non-locality before observation is that entangled structure, which observation weaves into geometry.

Why de Sitter forces this closes the argument. In de Sitter there is no outside, hence no external frame against which to localize anything. The reality before observation must therefore be the self-contained non-local whole—there can be no pre-given local parts, because there is no exterior to define them—and the only way for anything local to appear is for that whole to measure itself. Non-locality before observation is not an addition; it is forced by the absence of an outside. The non-locally real (the whole, the One) together with the observer-dependence of the local (constituted by self-measurement) is precisely a reality that is *non-locally real and observer-dependent*—evading at once Bell (there is no local realism) and Leggett (there is no realism of the parts).

5.6. **Closure: the self-portrait.** The same Vitruvian module that Velázquez used to construct the perspective of the interior architecture in *Las Meninas*—where he portrays himself painting himself, at once the painter and the observer of the work [65]—is also the one that allowed us to portray the observer, the act of observing, and the observed as one and the same. The divine proportion of that module is the one set out by Pacioli and illustrated by Leonardo [36]. Foucault opens *The Order of Things* with this same canvas, reading it as the limit of the classical *episteme*: the order in which things were known through representation, resemblance and proportion—the order to which the Vitruvian module belongs. This is where the introduction began. There the de Sitter difficulty was posed as epistemological rather than ontological (§1.3.1)—a question of how to represent something self-referential.

The picture is not laid down at once; it is built up cumulatively by observing. Each cycle deposits bits, a cell fills and overflows, the tower $S(\sigma) = \pi\varphi^\sigma$ rises, and at the threshold $\frac{1}{2}$ redundant encoding makes each level objective, so that independent observers agree. This is consonant with Kiely's metrological quantum Darwinism [58], with gravity read off from entropy [59] and from information [75, 76], with the invariants of Cooper's *Holographic Equidistribution* [5], and with non-local reality before observation. Just as Velázquez in *Las Meninas*, in this theory the quantum vacuum paints itself a portrait of its true self, painting itself—only its tools are mathematics and physics rather than oil paint.

6. Conclusions

The framework establishes two results and joins them into one. First (§2–§3), a single microstate — the modulus $|\Omega| = \frac{1}{2}$ — suffices to derive the AdS/CFT correspondence and, through the φ^σ tower, the M-theory type duality web, with no ensemble average: the correspondence follows from *one* state, not a statistical family. Second (§4), this determinism does not dispense with the observer but recasts its role.

The observer — the projection Π already present in that derivation — is a genuine participant, yet not as a *selector* of the vacuum: the microstate is the unique self-consistent solution, so there is no landscape to choose from. It participates, instead, through the information that its own act of observing accumulates. That quantum Fisher information (in Kiely's metrological sense [58]) is what builds the geometry — conjugate to the tower entropy, bounded by the ultraviolet cuts of §3, and, through the Landauer cost of each bit and the Jacobson–Verlinde passage from entropy to curvature, the source of the Einstein equations. The observer does not choose the world; it constructs it *without alternative* — participatory yet forced. The objectivity threshold $f_{\text{crit}} = \frac{1}{2}$ is reached, and it is the modulus of the microstate: the framework is thereby consonant with Kiely's quantum Darwinism and, at the same time, with non-local realism.

Appendix A. Explicit derivations and formal tags

This appendix collects, for each physical quantity asserted in the body, the explicit derivation with its formally verified tag. Every quantity below is an *output* of the framework—a consequence of the microstate $|\Omega| = \frac{1}{2}$, the golden tower $z = \varphi^\sigma$, and the lattice/Clifford/Hopf structure of the torus—not an external value matched by hand. The Lean tags certify the *algebraic* identities (the φ-arithmetic, the dimensions, the moduli); the *physical* identification of each quantity is the argument of the body, not of Lean.

A.1. AdS_5 **curvature, and an independent check.** On the warped throat $ds^2 = e^{2A}\eta_{\mu\nu}\, dx^\mu dx^\nu + dz^2$ with $A' = -1$, $A'' = 0$ in $d = 4$, the Ricci and scalar curvatures are computed directly:

$$R_{\mu\nu} = -(A'' + d\,A'^2)\, g_{\mu\nu} = -(0+4)\, g_{\mu\nu} = -4\, g_{\mu\nu}, \tag{A.1}$$

$$R = -(2d\,A'' + d(d{+}1)A'^2) = -(0 + 4\cdot 5) = -20, \tag{A.2}$$

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R = -4 + 10 = 6. \tag{A.3}$$

The throat is therefore an Einstein space $R_{AB} = -4g_{AB}$ with $\Lambda_5 = -d(d{-}1)/(2\ell^2) = -6$ at $\ell = 1$. The remaining curvature invariants $R_{ab}R^{ab} = 80$, $R_{abcd}R^{abcd} = 40$, GB $= 120$ and the Gauss–Bonnet Lanczos tensor $H_{AB} = -12g_{AB}$ follow from Lovelock's theorem, so the second-order field equation is Einstein $+\,\Lambda$, solved by the torus metric at $\Lambda_5 = -6$, $G_N = \frac{1}{2}$. `[R_munu_AdS5, R_scalar_AdS5, G_munu_AdS5]`

Independent check. Beyond the modulus and the dimensional counting, the same data pass two checks that are independent of either: the scalar mass saturates the Breitenlohner–Freedman stability bound, $m^2_{\rm BF} = -d^2/4 = -4$, so the discrete tower is stable; and the free-energy ratio of the correspondence takes the Gubser–Klebanov–Polyakov value $\mathrm{GKP} = \frac{3}{4}$. Neither follows from $|\Omega| = \frac{1}{2}$ by counting; both are nontrivial AdS/CFT data reproduced by the microstate. `[BF_bound_AdS5]`

A.2. **The ETS metric is flat; de Sitter is its embedded hyperboloid.** Written with the scale coordinate σ, the Lorentzian metric (4.19) is $ds^2 = -dt^2 + dx^2 + dy^2 + dz^2 + \lambda^2\, d(\ln\sigma)^2$, a product of Minkowski $\mathbb{E}^{3,1}$ with the one-dimensional scale line. Its Christoffel symbols, Riemann, Ricci and scalar curvatures, and Weyl tensor, computed directly, all vanish:

$$\Gamma^a{}_{bc} = 0 \text{ (4D block)}, \qquad R^a{}_{bcd} \equiv 0, \qquad R_{ab} = 0, \qquad R = 0, \qquad C_{abcd} = 0.$$

The ETS metric is therefore *intrinsically flat*—Weyl vanishes because Riemann does—and is not de Sitter (which is curved, $R = 12H^2$) but the flat five-dimensional *ambient* in which de Sitter embeds. `[ets_riemann_flat]`

The extended light cones (causal classification). Because (4.19) is Lorentzian with signature $(-,+,+,+,+)$, it carries a light-cone structure. For two events separated by $(\Delta t, \Delta x, \Delta y, \Delta z, \Delta u)$ with $\Delta u = \Delta(\ln\sigma)$ the scale gap, the interval is

$$\Delta s^2 = -\Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2 + \lambda^2\, \Delta u^2,$$

and the separation is exactly one of *timelike* ($\Delta s^2 < 0$, causally connectable), *null* ($\Delta s^2 = 0$, on the cone), or *spacelike* ($\Delta s^2 > 0$, not connectable) — the three-way classification of the causal cones. The scale term $\lambda^2 \Delta u^2$ enters with the spacelike sign: a pure scale separation ($\Delta t = \Delta x = \Delta y = \Delta z = 0$, $\Delta u \neq 0$) is spacelike, and turning on any scale gap can only push a separation toward spacelike, never toward timelike, so the extended cone contains the Minkowski cone. A pure time separation is timelike and the null condition $\Delta t^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 + \lambda^2 \Delta u^2$ is the extended light cone that makes C the apex (the observer's present) and P, F the past/future cones. `[causal_trichotomy, ets_null_cone, time_separation_timelike, space_separation_spacelike, scale_separation_spacelike, scale_pushes_spacelike]`

de Sitter as the hyperboloid (Gauss). The observer's de Sitter geometry is the hyperboloid $-X_0^2 + X_1^2 + X_2^2 + X_3^2 + X_4^2 = \ell^2$ ($\ell = 1/H$) inside the flat ambient $\eta_5 = \mathrm{diag}(-,+,+,+,+)$, the scale direction $X_4 = \lambda \ln\sigma$ supplying the fifth, flat coordinate the embedding requires. The hyperboloid is totally umbilic, $K_{\mu\nu} = H g_{\mu\nu}$, so Gauss's equation yields its intrinsic curvature as the extrinsic curvature of the embedding,

$$R_{\mu\nu\rho\sigma} = K_{\mu\rho}K_{\nu\sigma} - K_{\mu\sigma}K_{\nu\rho} = H^2(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})\,, \qquad R_{\mu\nu} = (d-1)H^2 g_{\mu\nu} = 3H^2 g_{\mu\nu} \ (d = 4),$$

with $R = 12H^2$ and vacuum Einstein $+\,\Lambda$ fixing $\Lambda = 3H^2$, $H = \sqrt{\Lambda/3}$ (the 4D de Sitter constant of the observer's slice, distinct from the five-dimensional bulk value $\Lambda_5 = -6$ of Appendix A.1). Direct computation of the FLRW form $ds^2 = -dt^2 + e^{2Ht}(dx^2 + dy^2 + dz^2) + \lambda^2 d(\ln\sigma)^2$ reproduces these values, and the metric induced on the hyperboloid equals exactly this FLRW form. `[dS_ricci_from_gauss, dS_einstein_Lambda, dS_hubble_from_Lambda]`

The static patch is thermal (Gibbons–Hawking). The de Sitter horizon radiates: the static patch is a thermal state at the Gibbons–Hawking temperature $T = H/2\pi$, with inverse temperature (KMS period) $\beta = 1/T = 2\pi/H$. This is the thermal condition Witten and Chandrasekaran, Longo, Penington and Witten place on the de Sitter observer [1, 2]: the observer's modular flow is the KMS flow of this thermal state, and it is exactly the tracial scale flow $\sigma \mapsto \sigma + t$ of the tower (§5.5.1), which fixes $\tau = i$ and preserves $|\Omega| = \frac{1}{2}$. With $H = \sqrt{\Lambda/3}$ the temperature is fixed by the same φ-determined Λ. `[dS_temperature, dS_temperature_pos, dS_beta_reciprocal]`

The observer sees exactly half. In the planar (inflationary) slicing that gives the FLRW form, the embedding is

$$X_0 = \ell \sinh \tfrac{t}{\ell} + \tfrac{r^2}{2\ell} e^{t/\ell}, \qquad X_i = e^{t/\ell} x_i, \qquad X_4 = \ell \cosh \tfrac{t}{\ell} - \tfrac{r^2}{2\ell} e^{t/\ell},$$

which lands on the hyperboloid and satisfies $X_0 + X_4 = \ell\, e^{t/\ell} > 0$ identically. The planar patch therefore covers *exactly half* the hyperboloid—the region $X_0 + X_4 > 0$, bounded by the cosmological horizon—and this is precisely the half the half-projection (4.7) separates: $|\Omega| = \frac{1}{2}$ is at once the algebraic modulus of the microstate and the fraction of de Sitter accessible to the observer.

Two corrections. The same computation corrects two statements sometimes made about this metric: the ETS metric has $C_{abcd} = 0$ (being flat, its Weyl tensor cannot be nonzero), and the constant-σ slices of the ETS metric are totally geodesic, $K_{\mu\nu} = 0$, not $K_{\mu\nu} = (\lambda/\sigma^2) g_{\mu\nu}$. `[ets_slices_totally_geodesic]`

A.3. **The gauge group** $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$**.** The internal gauge content descends from three substructures the torus already carries. The Eisenstein eigenvalue triangle $\lambda_k = \frac{1}{2}\omega^k$, $\omega = e^{2\pi i/3}$, generates the root lattice $\mathbb{Z}[\omega] = A_2$—the root system of $\mathfrak{su}(3)$—of dimension

$$\dim \mathfrak{su}(3) = 3^2 - 1 = 8. \tag{A.4}$$

The Clifford latitude $(\frac{1}{2})^2 - (\frac{\sqrt{3}}{2})^2 = -\frac{1}{2}$ realizes $S^3 \cong \mathrm{SU}(2)$, and the Hopf fibre realizes $S^1 \cong \mathrm{U}(1)$. The weak mixing angle is the entropy ratio at the colour level,

$$\sin^2 \theta_W = \frac{S(3)}{S(6)} = \frac{\varphi^3}{\varphi^6} = \varphi^{-3} \approx 0.236, \tag{A.5}$$

to be compared with the measured $\sin^2 \theta_W(M_Z) \approx 0.231$ (agreement to $\sim 2\%$). The identification is structural: $\sin^2 \theta_W$ is the entropy fraction at the tower midpoint, $S(3)/S(6) = \varphi^{-3}$—the same $\mathbb{Z}_3 \to \mathbb{Z}_4$ transition at $\sigma = 3$—evaluated at the algebraic fixed point $\varepsilon_0 M_{\mathrm{PCF}} = \pi$; the running $\sin^2 \theta_W(k)$ is not derived here, and the residual $\sim 2\%$ is a scale-translation question of the same nature as the one recorded for Λ_{obs} in Remark 4.12. The G–Λ duality $\varphi^{-6}\varphi^6 = 1$ holds. The internal $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ connection and the Lorentz connection both descend from a single higher-dimensional spin connection; the explicit Kaluza–Klein reduction giving A^a_μ from the metric is the one step left open here. `[gauge_dim_su3, weinberg_ratio, hopf_from_clifford, G_Lambda_duality]`

A.4. **The M-theory type duality web from its two corners.** The web this paper uses has exactly two corners—Huerta–Schreiber M-theory and Maldacena's AdS/CFT—and both are shown here to carry the microstate invariant $|\Omega| = \frac{1}{2}$, so the shared signature (3.20) is *derived*, not cited. *Huerta–Schreiber corner:* the superpoint tower is purely binary ($\mathbb{Z}_2$); the functor F of §3.3 maps it to the golden tower, carrying $|\Omega| = \frac{1}{2}$, $|\Omega|^2 = \frac{1}{4}$. *Maldacena corner:* the per-level AdS/CFT of §3.4 (curvature in App. A.1) fixes $\ell = 1$, $G_N = \frac{1}{2}$, $c = 3$, GKP $= \frac{3}{4}$. Both equal the microstate invariant $\mu = \frac{1}{2}$, which *is* (3.20). The dualities are then the modulus-preserving symmetries of the tower: S-duality is the Galois conjugation $\tau \to -1/\tau \Leftrightarrow \varphi \to -1/\varphi$, which fixes $\tau = i$ and preserves $|\Omega| = \frac{1}{2}$; T-duality is $R \to \alpha'/R$; IIA strong coupling is the dimensional lift $\sigma = 10 \to 11$. The two corners are thus projections of one microstate related by symmetries that fix its modulus—the web reconstructed from this paper's own objects, with no external framework required. `[s_duality_fixes_i, modulus_Omega]`

A.5. **The superpoint ladder and the** 32 **supercharges.** M-theory's eleven-dimensional content is reached as the chain of maximal invariant central extensions of the superpoint $\mathbb{R}^{0|1}$,

$$\mathbb{R}^{0|1} \longrightarrow \cdots \longrightarrow \mathbb{R}^{10,1|32}, \tag{A.6}$$

each level the unique consistent extension singled out by super-Lie-algebra cohomology (the brane bouquet). The 32 real supercharges are the cardinality of the binary hypercube $H_5 = \{0,1\}^5$,

$$|H_k| = 2^k, \qquad |H_5| = 2^5 = 32, \tag{A.7}$$

and the base mode count is $N(0) = \lfloor \pi \rfloor = 3$, fixing $c = 3$ at the colour level. `[hypercube_card]`

A.6. **Einstein's equations from observation.** The accumulated information of the observer is the tower entropy, the matter content, the horizon entropy, and the heat of $\delta Q = T\delta S$ at once,

$$\underbrace{F_\Omega \cdot N}_{\text{bits}} = \underbrace{S(\sigma) = \pi\varphi^\sigma}_{\text{tower}} = \underbrace{N = \lfloor S(\sigma) \rfloor}_{\text{matter}} = \underbrace{S_{BH} = \tfrac{A}{4G_N}}_{\text{horizon}} = \underbrace{S_{\text{Jacobson}}}_{\delta Q = T\delta S}, \tag{A.8}$$

each mode carrying one Shannon bit $H(\frac{1}{2}) = 1$. Each bit costs the Landauer energy $1/M_{\text{PCF}}$, and with the area factor $\mu_3^2 = \frac{1}{4}$ ($G_N = \frac{1}{2}$) the Clausius relation on every local Rindler horizon yields, through Jacobson's construction, the Einstein equation

$$\delta Q = T\,\delta S \implies G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa\, T_{\mu\nu}. \tag{A.9}$$

What is new here is not Jacobson's passage but its *inputs*: the horizon entropy is the golden-tower entropy $S = \pi\varphi^\sigma$, and the area factor is the microstate modulus $G_N = \frac{1}{2}$. These two framework outputs are what make the Clausius relation yield *this* geometry; with generic inputs it would not.

A.7. **Kaluza–Klein structure.** This appendix collects the Kaluza–Klein content of the framework. It supports the remark in §3.5: the bridge $\sigma = 10 \to 11$ of (3.23) corresponds to Witten's Kaluza–Klein mechanism, in which the Type IIA strong-coupling tower is the Kaluza–Klein spectrum of eleven-dimensional supergravity on $\mathbb{R}^{10} \times S^1$ [53].

The Kaluza–Klein mass from φ. The numerator of the interior Kaluza–Klein mass is the golden identity

$$\varphi^2 + \varphi^{-2} - 2 = 1, \tag{A.10}$$

`[KK_numerator]` a direct consequence of $\varphi^2 + \varphi^{-2} = 3$. The same identity fixes, at one stroke, the arity $n = 3$ (gauge), this Kaluza–Klein numerator (gravity), and the dimension $d = n + 1 = 4$ (3.8), `[central_chain]`.

Dimensional reduction $G_5 \to G_4$. Newton's constant reduces as $G_4 = G_5/(2\ell)$, which at the framework values $G_5 = \frac{1}{2}$, $\ell = 1$ gives

$$G_4 = \frac{G_5}{2\ell} = \frac{1}{4}, \tag{A.11}$$

`[kk_reduction, kk_at_PCF]` the reduced Newton constant being the squared modulus, $G_4 = |\Omega|^2 = (\frac{1}{2})^2$ `[G4_eq_omega_sq]`. The boundary Casimir $\frac{3}{4}$ and the bulk Newton $\frac{1}{4}$ sum to unity, $\frac{3}{4} + \frac{1}{4} = 1$ `[casimir_plus_newton]`, and the boundary density ratio is $\varphi^{-2} - 1 = -\varphi^{-1}$ `[boundary_density_ratio]`.

The Breitenlohner–Freedman bound and discrete stability. For AdS_5 the Breitenlohner–Freedman bound is $m_{BF}^2 = -d^2/4 = -4$ ($d = 4$),

$$m_{BF}^2 = -\frac{d^2}{4} = -4, \tag{A.12}$$

`[KK_BF_bound]` The continuous interior Kaluza–Klein mass $m_{KK}^2 \approx -4.318$ lies below this bound; a continuous mode of that mass would be unstable. The physical spectrum is not continuous but the discrete seven-level tower $\sigma = 0, \ldots, 6$, whose modes all have $m^2 > 0$ (verified numerically in `CW3_verify_paper_physics.py`): the discreteness is what makes the tower stable.

The step left open. The internal $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ connection and the Lorentz connection descend from a single higher-dimensional spin connection (§A.3); the explicit Kaluza–Klein reduction giving the gauge field A_μ^a from the metric is the one step left open, as noted in the main text.

Appendix B. Translation of the epigraph (Schiller, 1785)

The German verse quoted in the acknowledgments (Section B) is the opening strophe of Friedrich Schiller's *An die Freude* ("To Joy", 1785), the poem later set to music by Ludwig van Beethoven in the final movement of his Ninth Symphony (*Ode an die Freude*, Op. 125, 1824). A standard English translation runs:

Joy commands the hardy mainspring
Of the universe eterne.

Joy, oh joy the wheel is driving
Which the worlds' great clock doth turn.

Flowers from the buds she coaxes,
Suns from out the hyaline,
Spheres she rotates through expanses
Which the seer can't divine.

— Friedrich Schiller, *An die Freude* (1785), English rendering.

Funding

This research was supported by TTAMAYO PUNTO COM, S.A.P.I. de C.V. (Mexico).

Conflict of interest

The authors declare that they have no conflicts of interest.

Acknowledgments

Freude heißt die starke Feder
In der ewigen Natur.

Freude, Freude treibt die Räder
In der großen Weltenuhr.

Blumen lockt sie aus den Keimen,
Sonnen aus dem Firmament,
Sphären rollt sie aus den Räumen,
Die des Sehers Rohr nicht kennt.

— Friedrich Schiller, *An die Freude* (1785)

This article was inspired by the idea that cosmic becoming implies dimensional emergence. We are grateful to all physicists, mathematicians, and other thinkers who directly or indirectly contributed to this theory. We are specially thankful to Dr. Elena Odgers, Dr. Franz Peter Oberarzbacher Niederwolfsgruber, Dr. Pável Jiménez Vázquez and Professor Sánchez Rull, with whom this project originated. Likewise, we would like to thank the Lean FRO and Anthropic Research teams, who were indispensable for developing a multidisciplinary project such as this one.

References

[1] E. Witten. "Quantum gravity in de Sitter space". In: (2001). URL: https://arxiv.org/abs/hep-th/0106109.
[2] V. Chandrasekaran et al. "An algebra of observables for de Sitter space". In: *Journal of High Energy Physics* 02 (2023), p. 082.
[3] A. Strominger. "The dS/CFT correspondence". In: *Journal of High Energy Physics* 10 (2001), p. 034. DOI: 10.1088/1126-6708/2001/10/034.

[4] L. Susskind. "The world as a hologram". In: *Journal of Mathematical Physics* 36 (1995), p. 6377. DOI: 10.1063/1.531249.

[5] N. Cooper. "Holographic equidistribution". In: (2026). URL: https://arxiv.org/abs/2602.12265.

[6] P. Saad, S. H. Shenker, and D. Stanford. "JT gravity as a matrix integral". In: (2019). URL: https://arxiv.org/abs/1903.11115.

[7] R. Bousso and J. Polchinski. "Quantization of four-form fluxes and dynamical neutralization of the cosmological constant". In: *Journal of High Energy Physics* 06 (2000), p. 006. DOI: 10.1088/1126-6708/2000/06/006.

[8] M. R. Douglas. "The statistics of string/M theory vacua". In: *Journal of High Energy Physics* 05 (2003), p. 046. DOI: 10.1088/1126-6708/2003/05/046.

[9] L. Susskind. "The anthropic landscape of string theory". In: (2003). URL: https://arxiv.org/abs/hep-th/0302219.

[10] R. Bousso and L. Susskind. "The multiverse interpretation of quantum mechanics". In: *Physical Review D* 85 (2012). DOI: 10.1103/PhysRevD.85.045007.

[11] F. W. Lawvere. "Diagonal arguments and cartesian closed categories". In: *Lecture Notes in Mathematics* 92 (1969), p. 134. DOI: 10.1007/BFb0080769.

[12] N. S. Yanofsky. "A universal approach to self-referential paradoxes, incompleteness and fixed points". In: *Bulletin of Symbolic Logic* 9 (2003), p. 362. DOI: 10.2178/bsl/1058448677.

[13] B. Russell. "Mathematical logic as based on the theory of types". In: *American Journal of Mathematics* 30 (1908), p. 222. DOI: 10.2307/2272708.

[14] P. Martin-Löf. "An intuitionistic theory of types: predicative part". In: *Logic Colloquium '73*. Amsterdam: North-Holland, 1975, pp. 73–118.

[15] G. Priest. "The logic of paradox". In: *Journal of Philosophical Logic* 8 (1979), p. 219. DOI: 10.1007/BF00258428.

[16] A. Tarski. *A Decision Method for Elementary Algebra and Geometry*. 2nd. Berkeley: University of California Press, Feb. 1951.

[17] J. M. Maldacena. "The large N limit of superconformal field theories and supergravity". In: *Advances in Theoretical and Mathematical Physics* 2 (1998), p. 231. DOI: 10.1023/A:1026654312961.

[18] L. Susskind and E. Witten. "The holographic bound in anti-de Sitter space". In: (1998). URL: https://arxiv.org/abs/hep-th/9805114.

[19] I. R. Klebanov and M. J. Strassler. "Supergravity and a confining gauge theory: duality cascades and χSB-resolution of naked singularities". In: *Journal of High Energy Physics* 08 (2000), p. 052. DOI: 10.1088/1126-6708/2000/08/052.

[20] F. W. Lawvere. "Unity and identity of opposites in calculus and physics". In: *Applied Categorical Structures* 4 (1996), p. 167.

[21] U. Schreiber. "Differential cohomology in a cohesive -topos". In: (2013). URL: https://arxiv.org/abs/1310.7930.

[22] J. Huerta and U. Schreiber. "M-theory from the superpoint". In: *Letters in Mathematical Physics* 108 (2018), p. 2695. DOI: 10.1007/s11005-018-1110-z.

[23] Yu. I. Manin. "Real multiplication and noncommutative geometry (ein Alterstraum)". In: *The Legacy of Niels Henrik Abel*. Berlin: Springer, 2004, pp. 685–727.

[24] A. Connes, M. R. Douglas, and A. Schwarz. "Noncommutative geometry and matrix theory". In: *Journal of High Energy Physics* 02 (1998), p. 003. DOI: 10.1088/1126-6708/1998/02/003.

[25] H. Elvang, A. Herderschee, and R. Morales. "String theory from maximal supersymmetry". In: (). URL: https://arxiv.org/abs/2601.11705.

[26] G. Veneziano. "Construction of a crossing-symmetric, Regge-behaved amplitude for linearly rising trajectories". In: *Il Nuovo Cimento A* 57 (1968), p. 190. DOI: 10.1007/BF02824451.

[27] Yu. I. Manin. "Numbers as functions". In: *p-Adic Numbers, Ultrametric Analysis, and Applications* 5 (2013), p. 313. DOI: 10.1134/S2070046613040055.

[28] A. Buium. *Arithmetic Differential Equations*. Vol. 118. Providence, RI, USA: American Mathematical Society, 2005.

[29] S. Laporta and E. Remiddi. "The analytical value of the electron (g-2) at order α^3 in QED". In: *Physics Letters B* 379 (1996), p. 283. DOI: 10.1016/0370-2693(96)00439-X.

[30] S. A. Hartnoll and M. Yang. "The conformal primon gas at the end of time". In: *Journal of High Energy Physics* 07 (2025), p. 281. URL: https://arxiv.org/abs/2502.02661.

[31] M. De Clerck, S. A. Hartnoll, and M. Yang. "Wheeler–DeWitt wavefunctions for 5d BKL dynamics, automorphic L-functions and complex primon gases". In: *Journal of High Energy Physics* 11 (2025), p. 160. URL: https://arxiv.org/abs/2507.08788.

[32] B. L. Julia. "Statistical theory of numbers". In: *Number Theory and Physics*. Berlin: Springer, 1990, p. 276.

[33] H. L. Montgomery. "The pair correlation of zeros of the zeta function". In: *Proceedings of Symposia in Pure Mathematics* 24 (1973), p. 181.

[34] A. M. Odlyzko. "On the distribution of spacings between zeros of the zeta function". In: *Mathematics of Computation* 48 (1987), p. 273.

[35] J. A. González García et al. "Simultaneous identities of the golden ratio and Euler's identity, formally verified in Lean 4 over the known Mersenne prime exponents". In: (2026). Companion paper (2026). DOI: 10.5281/zenodo.21681395.

[36] L. Pacioli. *De Divina Proportione*. 1509.

[37] V. Pletser. *Fibonacci numbers and the golden ratio in biology, physics, astrophysics, chemistry and technology: a non-exhaustive review*. Tech. rep. URL: https://arxiv.org/abs/1801.01369.

[38] A. Jagannathan. "The Fibonacci quasicrystal: case study of hidden dimensions and multifractality". In: *Reviews of Modern Physics* 93 (2021).

[39] O. Jedrkiewicz et al. "Coherent four-mode parametric generation in a hexagonally poled nonlinear crystal". In: *Scientific Reports* (2018).

[40] G. Sonnino et al. "The golden ratio family of extremal Kerr–Newman black holes and its implications for the cosmological constant". In: *Axioms* 13 (2024), p. 720.

[41] A. Hurwitz. "Über die angenäherte Darstellung der Irrationalzahlen durch rationale Brüche". In: *Mathematische Annalen* 39 (1891), p. 279.

[42] J. A. González García et al. "The Hilbert–Pólya Operator and the Primitive Structure of the Complex Plane". In: *Zenodo* (2026). Companion paper (2026). DOI: 10.5281/zenodo.17619486.

[43] J. Maldacena and L. Susskind. "Cool horizons for entangled black holes". In: *Fortschritte der Physik* 61 (2013), p. 781. URL: https://arxiv.org/abs/1306.0533.

[44] J. A. González García et al. "Odd Zeta Values from the PCF Torus: φ and π as Arithmetic-Geometric Sources". In: *Zenodo* (2026). Companion paper (2026). DOI: 10.5281/zenodo.19492791.

[45] J. Polchinski. *String Theory*. Vol. 1. Cambridge: Cambridge University Press, 1998.

[46] J. D. Bekenstein. "Black holes and entropy". In: *Physical Review D* 7 (1973), p. 2333. DOI: 10.1103/PhysRevD.7.2333.

[47] S. W. Hawking. "Particle creation by black holes". In: *Communications in Mathematical Physics* 43 (1975), p. 199. DOI: 10.1007/BF02345020.

[48] J. D. Brown and M. Henneaux. "Central charges in the canonical realization of asymptotic symmetries: an example from three-dimensional gravity". In: *Communications in Mathematical Physics* 104 (1986), p. 207. DOI: 10.1007/BF01211590.

[49] R. Apéry. "Irrationalité de $\zeta(2)$ et $\zeta(3)$". In: *Astérisque* 61 (1979), p. 11.

[50] S. S. Gubser, I. R. Klebanov, and A. M. Polyakov. "Gauge theory correlators from non-critical string theory". In: *Physics Letters B* 428 (1998), p. 105. DOI: 10.1016/S0370-2693(98)00377-3.

[51] E. Witten. "Anti-de Sitter space and holography". In: *Advances in Theoretical and Mathematical Physics* 2 (1998), p. 253. DOI: 10.4310/ATMP.1998.v2.n2.a2.

[52] S. Ryu and T. Takayanagi. "Holographic derivation of entanglement entropy from the AdS/CFT correspondence". In: *Physical Review Letters* 96 (2006).

[53] E. Witten. "String theory dynamics in various dimensions". In: *Nuclear Physics B* 443 (1995), p. 85. DOI: 10.1016/0550-3213(95)00158-O.

[54] A. Almheiri et al. "Black holes: complementarity or firewalls?" In: *Journal of High Energy Physics* 02 (2013), p. 062. URL: https://arxiv.org/abs/1207.3123.

[55] J. S. Bell. "On the Einstein Podolsky Rosen paradox". In: *Physics* 1 (1964), p. 195. DOI: 10.1103/PhysicsPhysiqueFizika.1.195.

[56] A. J. Leggett. "Nonlocal hidden-variable theories and quantum mechanics: an incompatibility theorem". In: *Foundations of Physics* 33 (2003), p. 1469. DOI: 10.1023/A:1026096313729.

[57] G. W. Gibbons and S. W. Hawking. "Cosmological event horizons, thermodynamics, and particle creation". In: *Physical Review D* 15 (1977), p. 2738. DOI: 10.1103/PhysRevD.15.2738.

[58] A. Kiely et al. "Metrological approach to the emergence of classical objectivity". In: *Physical Review A* 113 (2026).

[59] T. Jacobson. "Thermodynamics of spacetime: the Einstein equation of state". In: *Physical Review Letters* 75 (1995), p. 1260. URL: https://arxiv.org/abs/gr-qc/9504004.

[60] T. Jacobson. "Entanglement equilibrium and the Einstein equation". In: *Physical Review Letters* 116 (2016). URL: https://arxiv.org/abs/1505.04753.

[61] R. Barthes. *Le Degré zéro de l'écriture*. Paris: Seuil, 1953.

[62] J. Huerta. "How space-times emerge from the superpoint". In: *LMS/EPSRC Durham Symposium on Higher Structures in M-theory*. 2019. URL: https://arxiv.org/abs/1903.02822.

[63] D. Fiorenza, H. Sati, and U. Schreiber. "Super Lie n-algebra extensions, higher WZW models and super p-branes with tensor multiplet fields". In: *International Journal of Geometric Methods in Modern Physics* 12 (2015). URL: https://arxiv.org/abs/1308.5264.

[64] F. P. Oberarzbacher. "Los postulados básicos de la filosofía no-dualista del Shaivismo de Cachemira". In: *Estudios* 104 (2013), pp. 81–91.

[65] M. Foucault. *The Order of Things: An Archaeology of the Human Sciences (Les Mots et les Choses)*. Paris: Gallimard, 1966.

[66] G. C. dos Santos. "Arquitetura vitruviana e retórica antiga". In: *Archai* 28 (2019). DOI: 10.14195/1984-249X_28_4.

[67] G. Obied et al. "De Sitter space and the swampland". In: (2018). URL: https://arxiv.org/abs/1806.08362.

[68] R. Landauer. "Irreversibility and heat generation in the computing process". In: *IBM Journal of Research and Development* 5 (1961), p. 183. DOI: 10.1147/rd.53.0183.

[69] S. Gröblacher et al. "An experimental test of non-local realism". In: *Nature* 446 (2007), p. 871. DOI: 10.1038/nature05677.

[70] S. J. Freedman and J. F. Clauser. "Experimental test of local hidden-variable theories". In: *Physical Review Letters* 28 (1972), p. 938. DOI: 10.1103/PhysRevLett.28.938.

[71] A. Aspect, J. Dalibard, and G. Roger. "Experimental test of Bell's inequalities using time-varying analyzers". In: *Physical Review Letters* 49 (1982), p. 1804. DOI: 10.1103/PhysRevLett.49.1804.

[72] G. Weihs et al. "Violation of Bell's inequality under strict Einstein locality conditions". In: *Physical Review Letters* 81 (1998), p. 5039. DOI: 10.1103/PhysRevLett.81.5039.

[73] The Royal Swedish Academy of Sciences. "The Nobel Prize in Physics 2022: A. Aspect, J. F. Clauser, A. Zeilinger, for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science". In: (2022).

[74] C. Branciard et al. "Experimental falsification of Leggett's nonlocal variable model". In: *Physical Review Letters* 99 (2007). DOI: 10.1103/PhysRevLett.99.210407.

[75] E. Verlinde. "On the origin of gravity and the laws of Newton". In: *Journal of High Energy Physics* 04 (2011), p. 029. URL: https://arxiv.org/abs/1001.0785.

[76] E. Verlinde. "Emergent gravity and the dark universe". In: *SciPost Physics* 2 (2017). URL: https://arxiv.org/abs/1611.02269.

Email address: jorge@ttamayo.com

Email address: victor@ttamayo.com

Email address: luz@ttamayo.com

TTAMAYO PUNTO COM, S.A.P.I. DE C.V.; RESEARCH & DEVELOPMENT DIVISION; NAUCALPAN DE JUÁREZ, STATE OF MEXICO, MEXICO

Email address: itzel@omega-pcf.com

INDEPENDENT RESEARCHER